Weak existence and exact calibration of local stochastic volatility models

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Abstract

We construct weak solutions of local stochastic volatility models that retain the exact conditional-expectation quotient and have the marginals of a prescribed local diffusion. For a real-valued factor, the variance may depend on time, price and factor, be unbounded, and approach zero; the factor diffusion need not have a uniform positive lower bound. A second result covers positive square-root factors, including Heston–LSV when the Feller ratio exceeds one. Point initial conditions and joint initial laws satisfying explicit moment bounds are allowed. The price is a true martingale and the occupation density has positive-time Sobolev regularity. The proof combines scalar score estimates with a centred conditional entropy inequality, including the additional covariance caused by price-dependent variance. Finite Fourier systems preserve the prescribed marginal, and strong convergence identifies their calibrated fluxes. We also obtain flux stability and call and put price estimates from marginal defects. For denominator regularisation, a weighted entropy yields existence with a real-valued factor and price-independent variance under an explicit temporal integrability condition; the resulting vanilla price bias is linear in the regularisation parameter, uniformly in strike.

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1 Introduction

Let $a(t,x)$ be a prescribed local variance for the logarithm of a discounted price. The associated local model is $dZ_t = -a(t,Z_t)dt/2 + \sqrt{a(t,Z_t)}\,dW_t$. To introduce a stochastic factor Y while preserving the one-dimensional distributions of the price, we consider the effective variance

$$a(t,x)\,\frac{v(t,x,y)}{M(t,x)}, \qquad M(t,x) = \mathbb{E}[v(t,X_t,Y_t) \mid X_t = x].$$

If the joint law has a density p with marginal q, then

$$M(t,x) = \frac{\int v(t,x,y)p(t,x,y)\,dy}{q(t,x)}, \qquad \int \frac{v(t,x,y)}{M(t,x)}p(t,x,y)\,dy = q(t,x).$$

This identity gives calibration: the marginal equation in x closes to the equation of the local diffusion. It does not, however, establish existence of the joint law. The ratio depends on this law, its denominator may be small, and neither the variance nor its inverses need be bounded. We construct the law and the ratio simultaneously.

Theorems 2.1 and 2.2 are the main existence results. They concern continuous factors and the exact quotient v/M, with either a point initial condition or an initial joint law satisfying explicit moment bounds. The first allows an unbounded positive variance and a factor diffusion coefficient with no uniform lower bound, under (H1)–(H5). The second treats positive factors with diffusion coefficient $s(t,y)\sqrt{y}$ under polynomial bounds and a generator domination condition; it includes Heston–LSV when $2\kappa\theta/\sigma^2 > 1$. In both settings the exponential price is a true martingale, and the occupation density has Sobolev regularity at positive times.

We apply Fourier–Galerkin approximation to construct the periodic solutions used in the existence proof. The finite systems preserve positivity and the prescribed scalar marginal, and satisfy a conditional entropy inequality. Scalar score estimates and factor moments give compactness; conservation of mass gives strong convergence of the calibrated flux. Theorem 11.1 then gives a diagonal approximation on the whole space. Section 12 bounds changes in the calibrated flux caused by perturbing the density or variance, regularising the denominator, and smoothing in the price coordinate. Section 13 bounds call and put price errors in terms of a marginal defect and an integrable signed-measure residual. For solutions of the denominator-regularised equation, it gives a downward bias and rates under inverse-moment bounds. Theorem 13.5 constructs these solutions for fixed positive regularisation parameter when the factor is real-valued

and $v = v(t, y)$ satisfies the temporal condition in that theorem; time-independent variance is included. A weighted entropy gives existence, positive-time Sobolev regularity and a true martingale price; Corollary 13.6 then gives a price bias of order $\varepsilon\sqrt{t}$, uniformly in strike.

Earlier existence results address different coefficient classes or different quotients. Abergel and Tachet [1] prove short-time classical existence on a bounded domain, with positive smooth initial and boundary data and small spatial variation of the factor variance (Theorem 1 of the cited preprint). Jourdain and Zhou [6] construct weak solutions for a finite-state switching factor under their coercivity condition (C), allowing a general initial probability law; see Theorems 4.3–4.4 of the extended preprint. Lacker, Shkolnikov and Zhang [7, Theorems 1.3–1.4] treat stationary models with independent noises, bounded positive volatility factors and dissipative time-homogeneous coefficients. They obtain strong existence in stationarity and uniqueness under an additional structural identity. These results do not give the nonstationary continuous-factor construction considered here.

Djete [5, Proposition 2.8 and Remark 2.9] allows time- and price-dependent variance and correlated noises, with uniformly positive bounded coefficients and a regular initial density. His calibration application uses $(c + qv)/(c + qM)$, $c > 0$, and does not take $c = 0$. Mustapha [9, Theorems 1.2–1.3] proves strong well-posedness and propagation of chaos for this calibration-preserving form with a time-independent finite-valued factor, a small-range condition and sufficiently small smooth initial densities. Both quotients have conditional mean one. They differ from the exact v/M used in our main theorems and from $v/(M + \varepsilon)$ in our price-bias analysis.

For regularised numerical models, Bayer, Belomestny, Butkovsky and Schoenmakers [10, Theorems 2.2–2.3] prove well-posedness and propagation of chaos with a fixed reproducing-kernel regularisation parameter. Reisinger and Tsianni [17, 8] analyse kernel-regularised particle equations and time discretisations. Their Heston analysis includes an unbounded CIR factor, with fixed regularisation parameters and explicit restrictions on moments and time horizons. Wyns and in 't Hout [16] prove exact agreement between compatible spatial discretisations of the local and stochastic local volatility equations, when their discrete quotients are defined. Friz, Jourdain, Wagenhofer and Zhou [11] study weak errors of Euler and particle approximations. Theorem 3.3 of their preprint gives orders $h^{1/2}$, $h\log(1/h)$ and h according to payoff regularity, under bounded positive stochastic volatility and smooth uniformly elliptic local volatility; it does not assume well-posedness of the limiting LSV equation. Our flux and vanilla-price estimates concern different perturbations and are not convergence rates for a particular particle algorithm.

The contribution here is the conditional entropy construction for the exact quotient with a continuous factor, including the additional price covariance and its whole-space limit. It permits the unbounded coefficient classes and initial laws stated in the two existence theorems. The Fourier approximation, compactness criterion and superposition principle are established techniques; the argument uses them with a prescribed marginal and estimates uniform in the expanding domain. The weighted entropy for the additive denominator is separate: its time derivative explains the temporal hypothesis in Theorem 13.5.

Sections 3–8 prove Theorem 2.1. Sections 9–10 retain the whole-space entropy estimates needed to prove Theorem 2.2. The whole-space approximation and error estimates follow in Sections 11–13. Appendices A–C contain the extensions of the assumptions, the separate theorem for a Brownian factor with vanishing or singular variance, and changes of factor coordinate. Appendix D proves existence for the regularised denominator, including the controlled time-dependent case.

2 Assumptions and existence theorems

We state the coefficient and initial-law assumptions separately for real-valued and positive factors.

2.1 Real-valued factor

Growth bounds. A coefficient h belongs to $\mathscr{G}_0$ if

$$\forall e > 0 \quad \exists C_e < \infty \quad |h(t,x,y)| \le C_e \exp(ey^2). \tag{2.1}$$

The bound is uniform in all variables other than y. For a weak derivative it is understood almost everywhere. For coefficient values, we choose Borel versions satisfying the bound. We say that ℓ has subquadratic growth in the factor coordinate, uniformly in its other arguments, if

$$\forall e > 0 \quad \exists C_e < \infty \quad |\ell(t,x,y)| \le C_e + ey^2. \tag{2.2}$$

This condition is equivalent to $e^{\ell}, e^{-\ell} \in \mathscr{G}_0$.

Fix $T > 0$ and $(x_0, y_0) \in \mathbb{R}^2$. We impose the following assumptions, each on the entire indicated domain:

(H1) $a : [0,T] \times \mathbb{R} \to (0,\infty)$ is Borel measurable, $0 < a_- \le a \le a_+ < \infty$, and

$$K_a := \operatorname*{ess\,sup}_{(t,x)\in(0,T)\times\mathbb{R}} \left(\left| \frac{\partial a}{\partial x}(t,x) \right| + \left| \frac{\partial^2 a}{\partial x^2}(t,x) \right| \right) < \infty. \tag{2.3}$$

The spatial weak derivatives $\frac{\partial a}{\partial x}$, $\frac{\partial^2 a}{\partial x^2}$ are assumed to exist; the essential supremum is taken with respect to $\mathrm{d}t\,\mathrm{d}x$. By Fubini's theorem, for almost every t, $a(t,\cdot) \in W^{2,\infty}(\mathbb{R})$ and $\left|\frac{\partial a}{\partial x}(t,x)\right| + \left|\frac{\partial^2 a}{\partial x^2}(t,x)\right| \le K_a$ for almost every x. We choose the spatial $C^{1,1}$ representatives and set $a = a_-$ on the exceptional set of times. No time continuity is required.

(H2) $\eta : [0,T] \times \mathbb{R} \to (0,\infty)$ is Borel measurable, locally Lipschitz in y for almost every t, and

$$\eta \le \eta_+ < \infty, \qquad \eta^{-1},\ \frac{\partial \eta}{\partial y} \in \mathscr{G}_0. \tag{2.4}$$

(H3) $b : [0,T] \times \mathbb{R} \to \mathbb{R}$ is Borel measurable and, for some $C_d \ge 0$,

$$b \in \mathscr{G}_0, \qquad yb(t,y) \le C_d(1+y^2). \tag{2.5}$$

(H4) The prescribed variance $\mathsf{v} : [0,T] \times \mathbb{R}^2 \to (0,\infty)$ is Borel measurable and finite. There exists a finite Borel function $c_0 : [0,T] \to \mathbb{R}$ such that, on setting

$$\ell(t,x,y) := \log \mathsf{v}(t,x,y) - c_0(t), \qquad v(t,x,y) := e^{\ell(t,x,y)} = e^{-c_0(t)}\mathsf{v}(t,x,y),$$

the function $\ell(t,\cdot,\cdot)$ is locally Lipschitz for almost every t, with

$$\forall e > 0 \quad \exists C_e < \infty : \quad |\ell(t,x,y)| \le C_e + ey^2, \qquad \partial_x \ell,\ \partial_y \ell \in \mathscr{G}_0. \tag{2.6}$$

The bounds are uniform in (t,x); no time derivative of ℓ is required.

(H5) $\rho : [0,T] \times \mathbb{R}^2 \to \mathbb{R}$ is Borel measurable, locally Lipschitz in y for almost every (t,x), and

$$|\rho| \le \rho_* < 1, \qquad \frac{\partial \rho}{\partial y} \in \mathscr{G}_0, \qquad \alpha := 1 - \rho_* > 0. \tag{2.7}$$

All growth bounds are uniform in time and in the price coordinate x whenever present. Derivatives are weak derivatives and their bounds hold almost everywhere. We use the locally Lipschitz spatial representatives specified above. The coefficients may be redefined by admissible constants on the exceptional sections. We require no derivative of b, no time derivatives of the coefficients,

and no derivative of ρ in x. Thus a Borel correlation $\rho(t,x)$ independent of y satisfies the derivative condition with $\frac{\partial \rho}{\partial y} = 0$. The bound on yb is strictly one-sided.

The factor $e^{c_0(t)}$ is deterministic, positive and finite; it need not be uniformly bounded, and no time derivative is required. The estimates concern the normalised variance v. Section 2.3 justifies the return to the prescribed variance v. The choice $c_0 = 0$ recovers the assumptions without normalisation.

Initial laws for the real factor. The initial point in Theorem 2.1 may be replaced by a probability law μ_0 on $\mathbb{R}^2$ satisfying

$$\int e^{cy^2}\,\mu_0(dx\,dy) < \infty \quad \text{for some } c > 0, \qquad \int (|x|^m + e^{rx})\,\mu_0(dx\,dy) < \infty \quad (m, r > 0). \tag{2.8}$$

These sufficient conditions include every compactly supported law and impose neither independence of the coordinates nor existence of an initial density. For this alternative, q denotes the density of the scalar diffusion started from the first marginal μ_0^X, so that $q_0 = \mu_0^X$. The all-order exponential price moments are a convenient sufficient condition, not an optimal threshold.

2.2 Calibration with a real-valued factor

The scalar diffusion

$$\mathrm{d}Z_t = -\tfrac{1}{2}a(t, Z_t)\,\mathrm{d}t + \sqrt{a(t, Z_t)}\,\mathrm{d}W_t, \qquad Z_0 = x_0, \tag{2.9}$$

has a unique strong solution. At every time $t > 0$, its law has a positive density $q(t,x)$ satisfying

$$\frac{\partial q}{\partial t} = \tfrac{1}{2}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial}{\partial x}\right)(aq), \qquad q_0 = \delta_{x_0}. \tag{2.10}$$

These facts, together with the choice of a Borel version of q, are justified in Section 3.2. They concern only the scalar equation.

Theorem 2.1 (Calibrated weak existence)**.** *Under (H1)–(H5), there exist a filtered probability space, two Brownian motions W, B with $\langle W, B\rangle_t = \int_0^t \rho(s, X_s, Y_s)\,\mathrm{d}s$, and continuous adapted processes X, Y such that $X_0 = x_0$, $Y_0 = y_0$ and*

$$\begin{aligned} \mathrm{d}X_t &= -\frac{1}{2}a(t, X_t)\frac{\mathsf{v}(t, X_t, Y_t)}{\mathsf{M}(t, X_t)}\,\mathrm{d}t + \sqrt{a(t, X_t)\frac{\mathsf{v}(t, X_t, Y_t)}{\mathsf{M}(t, X_t)}}\,\mathrm{d}W_t, \\ \mathrm{d}Y_t &= b(t, Y_t)\,\mathrm{d}t + \eta(t, Y_t)\,\mathrm{d}B_t. \end{aligned} \tag{2.11}$$

Here M is a strictly positive finite Borel version of

$$\mathsf{M}(t,x) = \mathbb{E}[\mathsf{v}(t, X_t, Y_t) \mid X_t = x] \quad \textit{for } \mathrm{d}t\,\mathcal{L}(X_t)(\mathrm{d}x)\textit{-almost every } (t,x).$$

For every $t \in (0, T]$, $\mathcal{L}(X_t)(\mathrm{d}x) = q(t,x)\,\mathrm{d}x$. The measure $\mathrm{d}t\,\mathcal{L}(X_t, Y_t)(\mathrm{d}x\,\mathrm{d}y)$ has a density p, and, for all $0 < \delta < T$ and $1 < r < 2$,

$$p^{1/r} \in L^r(\delta, T; W^{1,r}(\mathbb{R}^2)), \qquad p \in L^1(\delta, T; W^{1,1}(\mathbb{R}^2)). \tag{2.12}$$

The process e^X is a true martingale. In this notation, a price-independent variance is viewed as a function constant in x. The same conclusions hold with $\mathcal{L}(X_0, Y_0) = \mu_0$ under (2.8), and with the scalar initial law μ_0^X. In particular $\mathbb{E}e^{X_t} = \int e^x\,\mu_0^X(dx)$ in that case. The proof in Sections 3–8 treats the stated price dependence and initial laws together. No uniqueness or strong existence is asserted. The processes W, B are individually Brownian motions; their pair is not asserted to be jointly Gaussian when the correlation is random.

2.3 Time-dependent normalisation of the variance

For any law constructed with v, suppose that $M(t,x) = \mathbb{E}[v(t,X_t,Y_t) \mid X_t = x]$ is positive and finite almost everywhere. At each fixed time, $e^{c_0(t)}$ is a finite positive constant. For every bounded nonnegative test function φ,

$$\mathbb{E}[\varphi(X_t)\mathsf{v}(t,X_t,Y_t)] = e^{c_0(t)}\mathbb{E}[\varphi(X_t)v(t,X_t,Y_t)] = \mathbb{E}[\varphi(X_t)e^{c_0(t)}M(t,X_t)].$$

Identification of the conditional expectation and cancellation of the factor give

$$\mathsf{M} = e^{c_0}M, \qquad \frac{\mathsf{v}}{\mathsf{M}} = \frac{v}{M}. \tag{2.13}$$

We therefore work with $v = e^{\ell}$. All subsequent moment bounds concern v. They do not imply corresponding uniform or time-integrated bounds for v without control of e^{c_0}. The exceptional values of the ratios can be chosen positive and finite in a Borel measurable way.

Additive denominator regularisation depends on this normalisation. For the same conditional law, (2.13) gives

$$\frac{\mathsf{v}}{\mathsf{M}+\varepsilon} = \frac{v}{M+\varepsilon e^{-c_0(t)}}, \qquad \frac{v}{M+\varepsilon} = \frac{\mathsf{v}}{\mathsf{M}+\varepsilon e^{c_0(t)}}. \tag{2.14}$$

The estimates in Sections 12.2 and 13 concern the variance denoted by v and the denominator $M+\varepsilon$. Under (H4), this is the normalised variance. Since c_0 is only assumed to be finite and Borel measurable, the same rates for $\mathsf{M}+\varepsilon$ require additional moment bounds, specified after (13.23).

2.4 Positive factor

Theorem 2.2 (Weak existence with a positive factor under generator domination)**.** *Fix $T > 0$, $x_0 \in \mathbb{R}$, $y_0 > 0$ and a coefficient a satisfying (H1). On $[0,T] \times (0,\infty)$, consider a Borel drift μ, a Borel function $s > 0$ and a finite Borel prescribed variance $\mathsf{v} > 0$. Suppose that there exist constants $A > 0$, $B \geq 0$, $C_\mu < \infty$, $0 < s_- \leq s_+ < \infty$, $c_v, C_v > 0$, $\gamma, \omega, m_v, m_-, \lambda, d \geq 0$, $L < \infty$, $0 \leq \rho_* < 1$, and a finite Borel function $c_0(t)$ with the following properties. Set $v = e^{-c_0}\mathsf{v}$.*

(P1) For all (t,y),

$$A - By \leq \mu(t,y), \qquad |\mu(t,y)| \leq C_\mu(1+y). \tag{2.15}$$

(P2) The function $s(t,\cdot)$ is locally Lipschitz for almost every t, and

$$\frac{s_-}{(1+y)^\lambda} \leq s(t,y) \leq s_+, \qquad \left| y\frac{\partial}{\partial y}s(t,y)\right| \leq L(1+y^d). \tag{2.16}$$

(P3) The function $\log v(t,\cdot)$ is locally Lipschitz for almost every t, and

$$\frac{c_v(1\wedge y^\gamma)}{(1+y)^{m_-}} \leq v(t,y) \leq C_v(1+y^{m_v}+y^{-\omega}), \qquad \left| y\frac{\partial}{\partial y}\log v(t,y)\right| \leq L(1+y^d). \tag{2.17}$$

(P4) The correlation $\rho(t,x,y)$ is Borel measurable, locally Lipschitz in y for almost every (t,x), and

$$|\rho(t,x,y)| \leq \rho_* < 1, \qquad \left| y\frac{\partial}{\partial y}\rho(t,x,y)\right| \leq L(1+y^d). \tag{2.18}$$

The bounds on weak derivatives hold almost everywhere, uniformly in the other variables. We use the conventions for representatives in (H1)–(H5). No derivative of μ, no time derivative, and no derivative of ρ in x is required. Define

$$\nu = \frac{2A}{s_+^2}, \qquad g_* = \max\{1, \gamma+\omega\}, \qquad \nu > g_*, \qquad r_{\nu,\gamma,\omega} = \frac{2\nu}{\nu+g_*}. \tag{2.19}$$

Then there exist a filtered probability space carrying two Brownian motions W, B and a continuous adapted weak solution, starting from (x_0, y_0), of the system

$$\begin{aligned} dX_t &= -\tfrac{1}{2}a(t,X_t)\frac{\mathsf{v}(t,Y_t)}{\mathsf{M}(t,X_t)}dt + \sqrt{a(t,X_t)\frac{\mathsf{v}(t,Y_t)}{\mathsf{M}(t,X_t)}}\,dW_t, \\ dY_t &= \mu(t,Y_t)dt + s(t,Y_t)\sqrt{Y_t}\,dB_t, \\ d\langle W,B\rangle_t &= \rho(t,X_t,Y_t)dt, \qquad \mathsf{M}(t,x) = \mathbb{E}[\mathsf{v}(t,Y_t) \mid X_t = x]. \end{aligned} \tag{2.20}$$

The last identity holds for $dt\,\mathcal{L}(X_t)(dx)$-almost every (t,x); M has a Borel representative that is positive and finite everywhere. Almost surely, $Y_t > 0$ for all $t \in [0,T]$. For every $t > 0$, $\mathcal{L}(X_t)(dx) = q(t,x)dx$, where q is defined by (2.10), and e^X is a true martingale.

The occupation measure has a density p_Y and, for all $0 < \delta < T$ and $1 < r < r_{\nu,\gamma,\omega}$,

$$p_Y^{1/r} \in L^r(\delta, T; W^{1,r}(\mathbb{R}\times(0,\infty))), \qquad p_Y \in L^1(\delta, T; W^{1,1}(\mathbb{R}\times(0,\infty))). \tag{2.21}$$

The Brownian motions are individually Brownian; their pair is not asserted to be jointly Gaussian when ρ is random. No uniqueness or strong existence is asserted for the coupled system.

The point initial condition may also be replaced by a joint probability law μ_0 on $\mathbb{R}\times(0,\infty)$ with

$$\begin{gathered} \int (y^m + y^{-p})\,\mu_0(dx\,dy) < \infty \quad (m > 0,\ 0 < p < \nu), \\ \int (|x|^m + e^{rx})\,\mu_0(dx\,dy) < \infty \quad (m, r > 0). \end{gathered} \tag{2.22}$$

Then the target scalar diffusion starts from μ_0^X, all the stated conclusions and the regularity range remain valid, and $\mathbb{E}e^{X_t} = \int e^x \mu_0^X(dx)$. No independence or initial density is required; the initial truncation is justified in Section 10.3.

The Heston case. Take $\mu(t,y) = \kappa(\theta - y)$, $s(t,y) = \sigma$, $v(t,y) = y$, $c_0 = 0$ and $\rho(t,x,y) = \rho_0 \in (-1,1)$. We may choose

$$A = \kappa\theta, \quad B = \kappa, \quad s_- = s_+ = \sigma, \quad \gamma = m_v = 1, \quad \omega = m_- = \lambda = d = 0.$$

The bounds on v follow from $1 \wedge y \le y \le 1 + y$, and $y\frac{\partial}{\partial y}\log v = 1$. Thus

$$\nu = 2\kappa\theta/\sigma^2 > 1, \qquad r_{\nu,1,0} = 2\nu/(\nu+1).$$

The critical case $\nu = 1$ is not covered: the proof requires an inverse moment of order strictly greater than one.

Interpretation of the conditions. The factor equation need not be affine, and its coefficients may be merely Borel measurable in time. The variance v need not equal y: its growth at infinity may be polynomial of any order, while γ measures the permitted degeneracy near zero and ω the permitted singularity in the upper bound. The variance may therefore also diverge as the factor tends to zero. The multiplier s and the variance v may decay polynomially at infinity; no constant lower bound is imposed there. The parameters λ and m_- do not restrict the regularity range, since the factor has positive moments of every order. The logarithmic derivatives may have polynomial growth at infinity. The correlation may depend on (t,x,y).

2.5 Conventions

We write $\mathbb{T}_L = \mathbb{R}/(L\mathbb{Z})$, equipped with Lebesgue measure on one period, rather than normalised measure. Periodic integration by parts has no boundary terms. Unless otherwise stated, identities between densities and weak derivatives hold almost everywhere; $0\log 0 = 0$. An

occupation density means a density of $dt\,\mathcal{L}(X_t, Y_t)$, with no assertion of a joint density at every time. The letter C denotes a constant whose value may change. In the construction on a fixed torus it may depend on the torus and the smoothed coefficients; in a uniform estimate it is independent of the parameters being sent to their limits. Constants denoted by $C_{\delta,r,T}$ may depend on $\delta > 0$, r and the time horizon T.

3 Approximation of the coefficients and the marginal

We work throughout with $v = v(t, x, y)$ under (H4) and an initial law satisfying (2.8); a point mass is included. The scalar point-start estimates below will also be used through their mixtures. The initial approximation is specified in Section 3.3.

3.1 Smoothing the local coefficient

Proposition 3.1 (Approximation of the local coefficient)**.** *Under (H1), there exists a sequence of smooth functions $\widetilde{a}_n$ such that*

$$a_- \le \widetilde{a}_n \le a_+, \qquad \left|\frac{\partial \widetilde{a}_n}{\partial x}\right| + \left|\frac{\partial^2 \widetilde{a}_n}{\partial x^2}\right| \le K_a.$$

All derivatives are bounded for each fixed n. Moreover,

$$\widetilde{a}_n \longrightarrow a \quad \text{in } L^1_{\mathrm{loc}}([0,T]\times\mathbb{R}),$$

and, for every $L > 0$,

$$\int_0^T \sup_{|x|\le L} |\widetilde{a}_n(t,x) - a(t,x)|^2 \,\mathrm{d}t \longrightarrow 0.$$

Proof. Extend a by a_- outside $[0,T]$, calling the extension $\widetilde{a}$, and set $\widetilde{a}_n = \widetilde{a} * \varphi_n$, where $\varphi_n(t,x) = n^2\varphi(nt, nx)$ and φ is a nonnegative smooth kernel of mass one supported in the unit ball. Positivity of the kernel and the identities $\partial_x^k \widetilde{a}_n = (\mathbf{1}_{(0,T)}\partial_x^k a) * \varphi_n$, $k = 1, 2$, give

$$a_- \le \widetilde{a}_n \le a_+, \qquad |\partial_x \widetilde{a}_n| + |\partial_x^2 \widetilde{a}_n| \le K_a. \tag{3.1}$$

Every derivative is bounded for fixed n, since $\|\partial_t^j \partial_x^k \widetilde{a}_n\|_\infty \le \|\widetilde{a}\|_\infty \|\partial_t^j \partial_x^k \varphi_n\|_1$. The approximate-identity property gives

$$\widetilde{a}_n \to \widetilde{a} \quad \text{in } L^1_{\mathrm{loc}}(\mathbb{R}^2). \tag{3.2}$$

For almost every t, $f_n(t,\cdot) = \widetilde{a}_n(t,\cdot) - a(t,\cdot)$ is $2K_a$-Lipschitz and bounded by $2a_+$. If $A = \max_{|x|\le L} |f_n(t,x)| > 0$ is attained at x_*, then $|f_n| \ge A/2$ on the interval of radius $A/(4K_a + 4a_+) \le 1/2$ around x_*. Consequently,

$$\sup_{|x|\le L} |f_n(t,x)|^2 \le (4K_a + 4a_+) \int_{-L-1}^{L+1} |f_n(t,x)| \,\mathrm{d}x.$$

This also holds for $A = 0$. Integrating and using (3.2) proves

$$\int_0^T \sup_{|x|\le L} |\widetilde{a}_n(t,x) - a(t,x)|^2 \,\mathrm{d}t \to 0. \tag{3.3}$$

□

3.2 The scalar diffusion exists independently of the joint model

Proposition 3.2 (Local diffusion and stability)**.** *Under (H1), equation* (2.9) *has a unique strong solution Z. For every $t > 0$, the law of Z_t has a density $q(t, \cdot)$ which is positive almost everywhere. The function q satisfies* (2.10) *with initial trace δ_{x_0}.*

Let $\widetilde{Z}_n$ be the diffusions associated with the coefficients $\widetilde{a}_n$ from Proposition 3.1, driven by the same Brownian motion as Z. If $\widetilde{Z}_n(0) \to Z_0$ in L^2, then

$$\mathbb{E}\left[\sup_{0\le t\le T} |\widetilde{Z}_n(t) - Z_t|^2\right] \longrightarrow 0.$$

Proof. The coefficients $-a/2$ and $\sqrt{a}$ are bounded, Borel measurable in time and uniformly Lipschitz in space; the latter has Lipschitz constant $K_a/(2\sqrt{a_-})$. The Lipschitz SDE theorem [12, Theorem 2.20 and Corollary 2.19, without a jump term] gives strong existence and uniqueness. Bounded coefficients also give $\mathbb{E}\sup_{t\le T}|Z_t|^r < \infty$ for every finite r when $Z_0 = x_0$.

For the common-noise coupling, write $\Delta = \widetilde{Z}_n - Z$, $D_n = \widetilde{a}_n(t, \widetilde{Z}_n) - a(t, Z)$ and $S_n = \sqrt{\widetilde{a}_n(t, \widetilde{Z}_n)} - \sqrt{a(t, Z)}$. Then

$$\Delta_t = \Delta_0 - \tfrac{1}{2}\int_0^t D_n \, \mathrm{d}s + \int_0^t S_n \, \mathrm{d}W, \qquad |D_n| \le K_a|\Delta| + e_n, \quad |S_n| \le |D_n|/(2\sqrt{a_-}),$$

where $e_n(t) = |\widetilde{a}_n(t, Z_t) - a(t, Z_t)|$. Cauchy–Schwarz for the drift and Doob's L^2 inequality and Itô's isometry for the martingale give, for $F_n(t) = \mathbb{E}\sup_{u\le t}|\Delta_u|^2$,

$$F_n(t) \le 3\mathbb{E}|\Delta_0|^2 + B_T\int_0^t (K_a^2 F_n(s) + \mathbb{E}e_n(s)^2)\, \mathrm{d}s, \qquad B_T = 3T/2 + 6/a_-.$$

Gronwall therefore yields, with $C_T = \max\{3, B_T\}e^{B_T K_a^2 T}$,

$$\mathbb{E}\sup_{t\le T}|\widetilde{Z}_n(t) - Z_t|^2 \le C_T\left(\mathbb{E}|\widetilde{Z}_n(0) - Z_0|^2 + \int_0^T \mathbb{E}e_n(t)^2 \, \mathrm{d}t\right). \tag{3.4}$$

Bounded coefficients and $Z_0 \in L^2$ give a finite second moment of the time supremum, so

$$\int_0^T \mathbb{E}e_n(t)^2 \, \mathrm{d}t \le \int_0^T \sup_{|x|\le L} |\widetilde{a}_n(t, x) - a(t, x)|^2 \, \mathrm{d}t + C/L^2.$$

Here C is independent of n, L. First use (3.3), then let $L \to \infty$; this proves stability, also for the square-integrable initial laws used below.

For the density assertion take $\widetilde{Z}_n(0) = x_0$. The forward operator has divergence form

$$\partial_t q = \partial_x\big((a/2)\partial_x q + (\partial_x a + a)q/2\big).$$

For its smooth approximations the leading coefficient lies in $[a_-/2, a_+/2]$ and the lower-order coefficient is bounded by $(K_a + a_+)/2$. Aronson's bounds [2, 3] thus give $g_- \le q_n \le g_+$ with constants independent of n, where

$$g_-(t, x) = \frac{c_1}{\sqrt{t}}e^{-c_2|x-x_0|^2/t}, \qquad g_+(t, x) = \frac{c_3}{\sqrt{t}}e^{-c_4|x-x_0|^2/t}.$$

At each $t > 0$, coupling convergence passes these inequalities against nonnegative continuous compactly supported tests to $g_- \, \mathrm{d}x \le \mathcal{L}(Z_t) \le g_+ \, \mathrm{d}x$. Hence $\mathcal{L}(Z_t)$ has a density r_t between these two positive bounds.

For a jointly Borel version, let φ_j be smooth nonnegative approximate identities on $\mathbb{R}$ and put $Q_j(t,x) = \mathbb{E}\varphi_j(x - Z_t)$. Path continuity and dominated convergence make Q_j continuous; Lebesgue differentiation gives $Q_j(t,\cdot) \to r_t$ almost everywhere for each $t > 0$. Thus

$$q(t,x) = \min\{g_+(t,x), \max\{g_-(t,x), \limsup_j Q_j(t,x)\}\}$$

is Borel and is a density of Z_t at every positive time. Finally, Itô's formula gives, for $\psi \in C_c^\infty(\mathbb{R})$,

$$\int \psi q(t) - \psi(x_0) = \tfrac{1}{2}\int_0^t \int a(s,x)(\psi'' - \psi')q(s,x)\ \mathrm{d}x\ \mathrm{d}s.$$

This is (2.10), and path continuity gives the initial trace δ_{x_0}. □

3.3 Initial law and its approximation

Fix a joint law μ_0 satisfying (2.8); for the point-start case take $\mu_0 = \delta_{(x_0,y_0)}$. Let (X_0, Y_0) have law μ_0, and let G_1, G_2 be independent standard Gaussians, independent of this pair. Choose $\lambda_0 > 0$ with $2\lambda_0 < c$ and $0 < \varepsilon_n \le \min\{1/(4n), 1/(8\lambda_0)\}$ tending to zero. The initial density p_n^0 is the periodisation of the joint law of

$$(X_0 + \sqrt{\varepsilon_n}G_1, Y_0 + \sqrt{\varepsilon_n}G_2).$$

Convolution with the periodised Gaussian kernel is smooth and strictly positive on each fixed torus and preserves total mass. It preserves the dependence of the initial coordinates in the limit. The centred projections converge almost surely to (X_0, Y_0). They decrease absolute values, and

$$\mathbb{E}e^{\lambda_0(Y_0+\sqrt{\varepsilon_n}G_2)^2} \le \frac{\mathbb{E}e^{2\lambda_0 Y_0^2}}{\sqrt{1 - 4\lambda_0\varepsilon_n}} \le \sqrt{2}\mathbb{E}e^{2\lambda_0 Y_0^2} < \infty. \tag{3.5}$$

Polynomial price moments are uniform by $|u+v|^j \le 2^{j-1}(|u|^j + |v|^j)$, $j \ge 1$. No initial entropy bound is required.

Start the scalar diffusion from X_0, independent of its Brownian motion, and denote its density by q. Proposition 3.2 extends to this initial value: the coupling proof uses only $\mathbb{E}|X_0|^2 < \infty$ for the localisation tails, and the initial error of $X_0 + \sqrt{\varepsilon_n}G_1$ is ε_n in squared L^2 norm. For completeness, conditional on $X_0 = z$, the density bounds proved there have the same constants and centre z. Integration gives the bounds

$$g_\pm^{\mu_0}(t,x) = \int g_\pm(t,x;z)\,\mu_0^X(\,\mathrm{d}z).$$

The upper bound is finite; a bounded set of positive μ_0^X mass gives a strictly positive lower bound on every compact positive-time cylinder. The Borel version is obtained by the same explicit formula using $Q_j(t,x) = \mathbb{E}\varphi_j(x - Z_t)$, clamped between these two bounds. Itô's formula gives the scalar weak equation with initial term $\int \psi\ \mathrm{d}\mu_0^X$. Continuity of Z gives its initial trace. The periodic score estimates below depend neither on this initial law nor on the approximating initial densities.

3.4 Periodic coefficients and identification of the marginal

Lemma 3.3 (Scalar equation on a circle)**.** *Let a be smooth and strictly positive on $[0,T] \times \mathbb{T}_L$. For any smooth and strictly positive initial function q_0 on $\mathbb{T}_L$, the equation*

$$\frac{\partial q}{\partial t} = \frac{1}{2}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial}{\partial x}\right)(aq), \qquad q(0,\cdot) = q_0,$$

has a smooth and strictly positive solution with constant mass. The curve $t \mapsto q(t,x)\,\mathrm{d}x$ is unique among narrowly continuous curves of finite positive measures satisfying the weak formulation with the same initial data.

Proof. Write

$$L_t q = \tfrac{a}{2}\partial_x^2 q + (\partial_x a + a/2)\partial_x q + \tfrac{1}{2}(\partial_x^2 a + \partial_x a)q.$$

On the compact circle its coefficients are smooth and its leading coefficient is bounded below by $\min a/2 > 0$. Proposition 2.3 and Remark 2.4 of [13] give the smooth solution. With $m_0 = \min q_0 > 0$ and $K = \|\partial_x^2 a + \partial_x a\|_\infty/2$, the function $m_0 e^{-Kt}$ is a subsolution; comparison yields $q \geq m_0 e^{-Kt} > 0$. Integrating the divergence-form equation over the circle proves conservation of mass.

For measure-valued uniqueness, fix $t > 0$ and $\phi \in C^\infty(\mathbb{T}_L)$. The same linear existence theorem, after reversing time, gives a smooth solution of

$$\partial_s \psi + L_s^* \psi = 0, \qquad \psi(t) = \phi, \qquad L_s^* \psi = \tfrac{a(s,\cdot)}{2}(\partial_x^2 \psi - \partial_x \psi).$$

For a narrowly continuous measure solution μ, compactness gives $M_T = \sup_s \mu_s(\mathbb{T}_L) < \infty$. For each smooth spatial test β, the weak formulation says that $m_\beta(s) = \int \beta \, \mathrm{d}\mu_s$ is absolutely continuous with derivative $\int L_s^* \beta \, \mathrm{d}\mu_s$. The product rule therefore extends that formulation to finite sums of smooth time functions times spatial tests. For a general smooth ψ, its spatial Fourier coefficients satisfy $\sup_s |\partial_s^j c_k(s)| \leq C_m |k|^{-m}$, $j = 0, 1$, by m periodic integrations by parts. Taking $m = 4$ shows that the Fourier sums converge uniformly together with one time and two spatial derivatives. The bound M_T and bounded coefficients then pass the finite-sum identity to

$$\int \psi(t) \, \mathrm{d}\mu_t - \int \psi(0) \, \mathrm{d}\mu_0 = \int_0^t \int (\partial_s \psi + L_s^* \psi) \, \mathrm{d}\mu_s \, \mathrm{d}s.$$

For the backward solution the right side vanishes, determining $\int \phi \, \mathrm{d}\mu_t$ from μ_0. Density of smooth functions in $C(\mathbb{T}_L)$ proves uniqueness at every time. □

Proposition 3.4 (Periodic approximation)**.** *Under (H1)–(H5), after the normalisation* (2.13)*, one can construct the smooth periodic coefficients* $(a_n, b_n, \eta_n, V_n, \rho_n)$ *of* (3.7) *on* $\Omega_n = \mathbb{T}^2_{2R_n}$*. They satisfy the uniform growth bounds* (3.8)*, the one-sided bound* (3.9)*, and the local convergences* (3.10)*. On each fixed torus,* η_n, V_n *are bounded away from zero. The scalar marginals* q_n *constructed below are smooth, positive and of mass one.*

Let Z *solve the scalar equation of Proposition 3.2 with initial law* μ_0^X*, as constructed in Section 3.3. For every bounded Lipschitz function* $\phi : \mathbb{R} \to \mathbb{R}$*,*

$$\sup_{0 \leq t \leq T} \left| \int_{-R_n}^{R_n} \phi(x) q_n(t,x) \, \mathrm{d}x - \mathbb{E}\phi(Z_t) \right| \longrightarrow 0.$$

Proof. We first construct the periodic coefficients, then identify their scalar marginals by coupling.

3.4.1 Construction of the coefficients and preservation of the bounds

Extend b, ℓ, ρ by zero outside $[0, T]$, and η by η_+. Choose $\chi \in C_c^\infty(\mathbb{R})$, $0 \leq \chi \leq 1$, equal to one on $[-1, 1]$ and zero outside $[-2, 2]$. Let ϑ_h and $\widehat{\vartheta}_h$ be dilations of smooth positive mollifiers of mass one, supported in $[-h, h]^2$ and $[-h, h]^3$, respectively. Set

$$\begin{aligned} h_n &= 1/n, \qquad R_n = 2n + 2, \qquad \Omega_n = \mathbb{T}^2_{2R_n}, \\ b_n^\circ &= (\chi(\cdot/n) b) * \vartheta_{h_n}, \\ \ell_n^\circ(t,x,y) &= [\chi(x/n)\chi(y/n)\ell(t,x,y)] * \widehat{\vartheta}_{h_n}, \\ \eta_n^\circ &= \eta * \vartheta_{h_n}, \qquad V_n^\circ = \exp(\ell_n^\circ), \qquad \rho_n^\circ = \rho * \widehat{\vartheta}_{h_n}. \end{aligned} \tag{3.6}$$

The convolution defining ℓ_n° is in (t, x, y). When ℓ is independent of x, we instead take $\ell_n^\circ(t, y) = [\chi(y/n)\ell(t, y)] * \vartheta_{h_n}$. This preserves price independence for the constructions that use

it; the same bounds below hold, with $\partial_x \ell_n^\circ = 0$. Let $\zeta \in C_c^\infty((-1,1))$ take values in $[0,1]$ and equal one on $[-1/2, 1/2]$. On $[-R_n, R_n]^2$, define

$$\begin{aligned} a_n(t,x) &= a_- + \zeta(x/R_n)(\widetilde{a}_n(t,x) - a_-), \\ \eta_n(t,y) &= \eta_+ + \zeta(y/R_n)(\eta_n^\circ(t,y) - \eta_+), \\ b_n &= b_n^\circ, \qquad V_n = V_n^\circ, \\ \rho_n(t,x,y) &= \zeta(x/R_n)\zeta(y/R_n)\rho_n^\circ(t,x,y). \end{aligned} \tag{3.7}$$

Since $2n + h_n < R_n$, the function b_n° vanishes near the endpoints in y. In the general construction, ℓ_n° vanishes near every spatial join, where $V_n = 1$; in the price-independent construction it vanishes near the joins in y and is constant in x. The functions a_n and η_n are constant near the endpoints of their intervals, and ρ_n vanishes near the boundary of the square. Their periodic extensions are therefore smooth.

For each $e > 0$, we prove the following bounds for all $t \in [0,T]$ and all $x, y \in [-R_n, R_n]$, with constants independent of n:

$$\begin{aligned} a_- \le a_n \le a_+, \qquad \left|\frac{\partial a_n}{\partial x}\right| + \left|\frac{\partial^2 a_n}{\partial x^2}\right| &\le C_a, \\ 0 < \eta_n \le \eta_+, \qquad |\rho_n| &\le \rho_*, \\ \eta_n^{-1} + \left|\frac{\partial \eta_n}{\partial y}\right| + |b_n| + \left|\frac{\partial \rho_n}{\partial y}\right| + |\partial_x \log V_n| + \left|\frac{\partial \log V_n}{\partial y}\right| &\le C_e \exp(ey^2), \\ |\log V_n| &\le C_e + ey^2. \end{aligned} \tag{3.8}$$

The bounds on the values of a_n, η_n, ρ_n follow from positivity and unit mass of the kernels and the inequalities $0 \le \zeta \le 1$.

For every $e > 0$, convolution preserves a common $\mathscr{G}_0$ envelope, uniformly in the other variables: on its support $|z| \le 1$,

$$|f(t-r, x-u, y-z)| \le C_{e/2} e^{e(y-z)^2/2} \le C_{e/2} e^e e^{ey^2}.$$

Multiplication by a cutoff in $[0,1]$ preserves this bound. For the inverse diffusion, Jensen and $\eta_n \ge \eta_n^\circ$ give $\eta_n^{-1} \le (\eta * \vartheta_{h_n})^{-1} \le \eta^{-1} * \vartheta_{h_n}$. The spatial derivatives of η_n, ρ_n are their mollified weak derivatives multiplied by the cutoffs, plus cutoff terms bounded respectively by $\|\zeta'\|_\infty \eta_+ / R_n$ and $\|\zeta'\|_\infty \rho_* / R_n$. These satisfy the same envelopes. For a_n, the product rule and (3.1) give the uniform bound

$$|\partial_x a_n| + |\partial_x^2 a_n| \le K_a + \frac{\|\zeta'\|_\infty (a_+ - a_- + 2K_a)}{R_n} + \frac{\|\zeta''\|_\infty (a_+ - a_-)}{R_n^2} =: C_{a,n} \le C_a.$$

For V_n, put $c_n(x,y) = \chi(x/n)\chi(y/n)$ in the general case. The spatial product rules give

$$\partial_i(c_n \ell) = c_n \partial_i \ell + (\partial_i c_n)\ell, \qquad i = x, y, \qquad \partial_i \log V_n = [\partial_i(c_n \ell)] * \widehat{\vartheta}_{h_n}.$$

The first terms satisfy common $\mathscr{G}_0$ bounds by (H4). Since $n \ge 1$ and $|\ell(t,x,y)| \le C_1 + y^2$,

$$|(\partial_i c_n)\ell| \le \|\chi'\|_\infty (C_1 + y^2), \qquad i = x, y.$$

This polynomial envelope belongs to $\mathscr{G}_0$; the positive convolution estimate therefore bounds both logarithmic derivatives uniformly in n, t, x. For the price-independent construction the same argument applies with $c_n = \chi(y/n)$, the two-variable kernel, and zero x-derivative. Positivity and unit mass of either kernel, together with $|z| \le 1$ in its factor coordinate, also give

$$|\log V_n(t,x,y)| \le C_{e/2} + e + ey^2, \qquad V_n(t,x,y)^{\pm 1} \le C_e \exp(ey^2).$$

Here the constants are adjusted by using the subquadratic bound with a smaller exponent before exponentiation. These estimates prove (3.8). On each fixed torus, continuity and positivity give positive minima for η_n on $[0,T]\times\mathbb{T}_{2R_n}$ and for V_n on $[0,T]\times\Omega_n$.

We check the one-sided drift condition:

$$\operatorname{sgn}(y)b_n(t,y)\le D_0(1+|y|),\qquad D_0=2C_d+\sup_{t\in[0,T],\,|y|\le 3}|b(t,y)|+1. \tag{3.9}$$

For $|y|\ge 2$ and $|z|\le 1$, $u=y-z$ has the sign of y and $|u|\ge 1$, so $\operatorname{sgn}(y)b(t,u)\le C_d(|u|+|u|^{-1})\le 2C_d(1+|y|)$. Multiplication by $0\le\chi\le 1$ preserves this positive upper bound. For $|y|\le 2$, use the supremum of $|b|$ over $|u|\le 3$. Both estimates hold for the zero temporal extension; integration against the positive kernel proves (3.9). On any fixed compact set, the factors $\chi(x/n)$, $\chi(y/n)$, $\zeta(x/R_n)$ and $\zeta(y/R_n)$ equal one for sufficiently large n, as they do on the neighbourhoods needed for convolution. For $\widetilde{a}_n$, we use the explicit convolution construction of Proposition 3.1. Convergence of mollifications at Lebesgue points, together with continuity of the exponential for $V_n=\exp(\ell_n^\circ)$, gives

$$(a_n,b_n,\eta_n,V_n,\rho_n)\longrightarrow(a,b,\eta,v,\rho)\quad\text{locally almost everywhere.} \tag{3.10}$$

The estimates (3.8) also show that these coefficients are bounded uniformly in n on each compact set. The spatial derivatives $a_{n,x}, a_{n,xx}, \eta_{n,y}, \rho_{n,y}$ and $\partial_i\log V_n$, $i=x,y$, converge locally almost everywhere as well, after extraction if needed. On each compact interior cylinder the cutoff derivatives eventually vanish and these are mollifications of the corresponding locally integrable weak derivatives. This convergence is used for the whole-space entropy sources; identification of the Fokker–Planck fluxes uses only coefficient values.

3.4.2 Identification of the limiting periodic marginals

Use p_n^0 from Section 3.3. Its marginal is

$$q_{n,0}(x)=\int\sum_{k\in\mathbb{Z}}\gamma^z_{\varepsilon_n}(x+2kR_n)\,\mu_0^X(\,\mathrm{d}z).$$

The periodised Gaussian kernel and all its derivatives are bounded on the fixed circle, so differentiation under this integral is valid. Positivity and Tonelli's theorem give a smooth positive density of mass one. Lemma 3.3, applied to a_n and $q_{n,0}$, gives a smooth, strictly positive periodic scalar solution q_n of mass one. In particular,

$$\min_{[0,T]\times\mathbb{T}_{2R_n}} q_n>0,$$

where the lower bound may depend on n.

Let $\xi_n=X_0+\sqrt{\varepsilon_n}G$, with G standard normal and independent of X_0,W. Couple Z, $\widetilde{Z}_n$ and Z_n^{lift} using W, coefficients $a,\widetilde{a}_n,a_n$, and initial values X_0,ξ_n,ξ_n, respectively; here a_n is extended periodically to the line. All these SDEs have bounded Lipschitz coefficients. Itô's formula for periodic tests and Lemma 3.3 identify the law of $\pi_n(Z_n^{\mathrm{lift}}(t))$ with $q_n(t,x)\,\mathrm{d}x$, where π_n is the centred representative modulo $2R_n$.

Because $a_n=\widetilde{a}_n$ on $[-R_n/2,R_n/2]$, pathwise uniqueness makes the last two processes agree up to exit from this interval. Their bounded coefficients and uniformly square-integrable initial values give

$$\mathbb{E}\sup_{t\le T}|Z_n^{\mathrm{lift}}(t)|^2+\mathbb{E}\sup_{t\le T}|\widetilde{Z}_n(t)|^2\le C_T.$$

The probability of either exit is at most C_T/R_n^2; on its complement π_n changes neither process. Thus, for bounded Lipschitz ϕ,

$$\sup_{t\le T}\left|\int_{-R_n}^{R_n}\phi q_n(t)-\mathbb{E}\phi(Z_t)\right|\le\frac{C_T\|\phi\|_\infty}{R_n^2}+\operatorname{Lip}(\phi)\left(\mathbb{E}\sup_{t\le T}|\widetilde{Z}_n(t)-Z_t|^2\right)^{1/2}.$$

Since $\mathbb{E}|\xi_n - X_0|^2 = \varepsilon_n \to 0$, (3.4) gives

$$\sup_{t\le T}\left|\int_{-R_n}^{R_n}\phi(x)q_n(t,x)\,\mathrm{d}x - \mathbb{E}\phi(Z_t)\right| \to 0. \tag{3.11}$$

This identification uses neither the joint density nor derivatives of q_n. □

3.5 Scalar score estimate

Proposition 3.5 (Score moments)**.** *Let $q > 0$ be a smooth density of mass one on $[0,T]\times\mathbb{T}_L$ solving*

$$\frac{\partial q}{\partial t} = \tfrac{1}{2}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial}{\partial x}\right)(aq), \qquad a_- \le a \le a_+, \qquad \left|\frac{\partial a}{\partial x}\right| + \left|\frac{\partial^2 a}{\partial x^2}\right| \le K_a, \tag{3.12}$$

where a is smooth. For every $p > 0$, there exists $C_p = C_p(a_-, a_+, K_a)$, independent of L and the initial data, such that, with $S = \frac{\partial q}{\partial x}/q$,

$$\int_{\mathbb{T}_L} q|S|^p \le C_p(1 + t^{-p/2}), \qquad 0 < t \le T. \tag{3.13}$$

Proof. The calculations take place on the compact cylinder, where q is bounded away from zero. Set $A = a/2$, $d = \left(\frac{\partial a}{\partial x} + a\right)/2$, $F = AS + d$. Then

$$\frac{\partial q}{\partial t} = \frac{\partial(qF)}{\partial x}, \qquad \frac{\partial \log q}{\partial t} = \frac{\partial F}{\partial x} + SF, \qquad \frac{\partial S}{\partial t} = \frac{\partial^2 F}{\partial x^2} + \frac{\partial S}{\partial x}F + S\frac{\partial F}{\partial x}. \tag{3.14}$$

For $p \ge 2$, write $I_p = \int q|S|^p$ and $J_p = \int q|S|^{p-2}\left(\frac{\partial S}{\partial x}\right)^2$, with $|S|^0 = 1$ when $p = 2$. The function $\psi(s) = |s|^p$ is C^2. Integrating by parts twice gives the cancellation

$$\begin{aligned} I_p' &= \int \frac{\partial(qF)}{\partial x}\psi(S) + \int q\psi'(S)\left(\frac{\partial^2 F}{\partial x^2} + F\frac{\partial S}{\partial x} + S\frac{\partial F}{\partial x}\right) \\ &= \int q\psi'(S)\frac{\partial^2 F}{\partial x^2} + \int qS\psi'(S)\frac{\partial F}{\partial x} \\ &= -\int\left(qS\psi'(S) + q\psi''(S)\frac{\partial S}{\partial x}\right)\frac{\partial F}{\partial x} + \int qS\psi'(S)\frac{\partial F}{\partial x} \\ &= -p(p-1)\int q|S|^{p-2}\frac{\partial S}{\partial x}\left(A\frac{\partial S}{\partial x} + \frac{\partial A}{\partial x}S + \frac{\partial d}{\partial x}\right). \end{aligned}$$

Only a, $\frac{\partial a}{\partial x}$ and $\frac{\partial^2 a}{\partial x^2}$ remain. Young's inequality, $A \ge a_-/2$ and $|S|^{p-2} \le 1 + |S|^p$ imply

$$I_p' \le -\frac{p(p-1)a_-}{4}J_p + C_p(1 + I_p).$$

Since $\frac{\partial q}{\partial x} = qS$, a further integration by parts and Hölder's inequality give

$$\begin{aligned} I_p &= \int \frac{\partial q}{\partial x}S|S|^{p-2} = -(p-1)\int q|S|^{p-2}\frac{\partial S}{\partial x}, \\ I_p^2 &\le (p-1)^2 J_p \int q|S|^{p-2} \le (p-1)^2 J_p I_p^{(p-2)/p}. \end{aligned}$$

For $p = 2$, the last factor equals 1. Dividing when $I_p > 0$, and treating $I_p = 0$ trivially, gives

$$I_p' \le -c_p I_p^{1+2/p} + C_p(1 + I_p), \qquad c_p = \frac{pa_-}{4(p-1)} > 0. \tag{3.15}$$

With $B_p = \max\{1, (4C_p/c_p)^{p/2}\}$, on $\{I_p \geq B_p\}$ we have $I_p' \leq -c_p I_p^{1+2/p}/2$. The level B_p cannot be crossed upwards: between the last contact with this level and a point above it, the derivative would be strictly negative. If $I_p(t) > B_p$, it must therefore have remained above this level on $[0,t]$, whence

$$(I_p^{-2/p})' \geq c_p/p, \qquad I_p(t) \leq \max\{B_p, (p/(c_p t))^{p/2}\}.$$

For $0 < p < 2$, Hölder's inequality under the probability measure $q\,dx$ gives $I_p \leq I_2^{p/2}$. This proves (3.13). □

In particular,

$$\int q|S|^4 \leq C(1 + t^{-2}). \tag{3.16}$$

Applied to a_n, q_n with the common constant C_a from (3.8), the estimate gives, for every $r > 0$,

$$\int q_n |S_n|^r \leq C_r(1 + t^{-r/2}), \qquad S_n = \frac{\partial q_n}{\partial x}/q_n. \tag{3.17}$$

4 Calibration and conditional entropy

For a fixed marginal q, we seek an inequality controlling derivatives of the joint density despite the ratio V/M. The identities in Section 4.1 isolate the centred derivative $f - m$; the moments in Section 4.2 control the weights in Young's inequality. Section 4.3 combines the three fluxes and proves the inequality. Section 4.4 reformulates it for passage to weak limits.

4.1 Auxiliary setting and centering identity

In this section, fix a torus $\Omega = \mathbb{T}_{L_x} \times \mathbb{T}_{L_y}$, smooth coefficients $a(t,x)$, $b(t,y)$, $\eta(t,y)$, $V(t,x,y)$, $\rho(t,x,y)$, with $a, \eta, V > 0$ and $|\rho| \leq \rho_* < 1$, and a smooth marginal $q > 0$ of mass one. The static calculations use only these properties; they do not assume existence of a joint solution.

For a regular positive density p with marginal q, set

$$\begin{aligned} u = p/q, \quad M = \int V u \,\mathrm{d}y, \quad w = V/M, \quad E(p,q) = \iint p \log u, \quad S = \frac{\partial q}{\partial x}/q, \\ f = \frac{\partial \log u}{\partial x}, \quad g = \frac{\partial \log u}{\partial y}, \quad m = \int w f u \,\mathrm{d}y, \quad K = \int u/w \,\mathrm{d}y - 1, \\ D_x = \int w(f-m)^2 u \,\mathrm{d}y, \quad D_y = \int \eta^2 g^2 u \,\mathrm{d}y, \quad \mathcal{D} = \int q(aD_x + D_y)\,\mathrm{d}x. \end{aligned} \tag{4.1}$$

On the fixed torus, $0 < V_- \leq V \leq V_+$, so

$$V_- \leq M \leq V_+, \quad V_-/V_+ \leq w \leq V_+/V_-, \qquad \int u \,\mathrm{d}y = \int wu \,\mathrm{d}y = 1. \tag{4.2}$$

The ratio is therefore defined for every conditional density, without modifying its denominator. Put $L = \partial_x \log V$ and $n = \int wLu \,\mathrm{d}y$. Differentiation of the conditional integrals gives

$$\partial_x \log M = m + n, \qquad \partial_x \log w = L - m - n, \qquad \partial_y \log w = \partial_y \log V, \qquad \int f u \,\mathrm{d}y = 0. \tag{4.3}$$

In particular, m is the conditional mean of f under $wu\,\mathrm{d}y$; it is not in general $\partial_x \log M$.

The identities $\int fu = 0$, $\int wfu = m$ and $\int wu = 1$ give

$$\int (f-m)(w^{-1}-1)wu = -m, \qquad \int (w^{-1}-1)^2 wu = K \geq 0.$$

The Cauchy–Schwarz inequality under the probability measure $wu\,dy$ yields

$$m^2 \le K D_x, \qquad K \ge 0. \tag{4.4}$$

This estimate bounds the mean m by the centred dissipation.

The nonlinear operator associated with the fixed coefficients is

$$\mathcal{F}_t(p) = \tfrac{1}{2}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial}{\partial x}\right)(awp) + \frac{\partial^2}{\partial x \partial y}(\rho\eta\sqrt{aw}\,p) + \tfrac{1}{2}\frac{\partial^2}{\partial y^2}(\eta^2 p) - \frac{\partial}{\partial y}(bp). \tag{4.5}$$

Periodic integration in y and (4.2) give

$$\int \mathcal{F}_t(p)\,\mathrm{d}y = \tfrac{1}{2}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial}{\partial x}\right)(aq). \tag{4.6}$$

If $\frac{\partial p}{\partial t} = \mathcal{F}_t(p)$ and q solves the scalar equation, then

$$E' = \iint \frac{\partial p}{\partial t}\log u + \iint \frac{\partial p}{\partial t} - \int \frac{\partial q}{\partial t}\left(\int u\,\mathrm{d}y\right)\mathrm{d}x = \iint \frac{\partial p}{\partial t}\log u.$$

No time derivative of V appears in this identity.

4.2 Moments of the conditional weights

Proposition 4.1 (Conditional moments)**.** *Let $p = qu$ be a probability density, $V > 0$, $M = \int Vu$ positive and finite, $w = V/M$ and $K = M\int V^{-1}u - 1$. For $h \ge 1$ and conjugate exponents $A, B > 1$, each of $\int q(1+K)^h$, $\iint w^h p$ and $\iint w^{-h}p$ is bounded by*

$$\left(\iint V^{hA}p\right)^{1/A}\left(\iint V^{-hB}p\right)^{1/B}. \tag{4.7}$$

The inequality remains valid when the right-hand side is infinite.

Proof. Put $U_h = \int V^h u\,\mathrm{d}y$ and $V_h = \int V^{-h}u\,\mathrm{d}y$. Conditional Jensen gives $M^h \le U_h$, $M^{-h} \le V_h$ and $(\int V^{-1}u)^h \le V_h$; hence

$$(1+K)^h \le U_h V_h, \qquad \int w^h u = M^{-h}U_h \le U_h V_h, \qquad \int w^{-h}u = M^h V_h \le U_h V_h.$$

Hölder under $q\,\mathrm{d}x$, followed by conditional Jensen, yields

$$\int qU_hV_h \le \left(\int qU_h^A\right)^{1/A}\left(\int qV_h^B\right)^{1/B} \le \left(\iint V^{hA}p\right)^{1/A}\left(\iint V^{-hB}p\right)^{1/B}.$$

Tonelli justifies all integrations, including when the right side is infinite. □

For $A = B = 2$, we obtain the two forms used in the construction:

$$\int q(1+K)^r \le \left(\iint V^{2r}p\right)^{1/2}\left(\iint V^{-2r}p\right)^{1/2}, \tag{4.8}$$

$$\iint (w^r + w^{-r})p \le 2\left(\iint V^{2r}p\right)^{1/2}\left(\iint V^{-2r}p\right)^{1/2}. \tag{4.9}$$

4.3 Computation of the three fluxes

In the fixed smooth setting of Section 4.1, define

$$
\begin{aligned}
k_\rho &= \rho\frac{\partial \eta}{\partial y} + \eta\frac{\partial \rho}{\partial y} + \tfrac{1}{2}\rho\eta\,\frac{\partial}{\partial y}\log V, \qquad d = b/\eta - \frac{\partial \eta}{\partial y},\\
H(p) &= \iint p(k_\rho^2 + d^2), \qquad J(p) = \int q\frac{\left(\frac{\partial a}{\partial x} + a + aS\right)^2}{a}K\,\mathrm{d}x,\\
H_x(p) &= \iint awp(L-n)^2, \qquad \alpha = 1-\rho_* > 0,\\
G(p) &= \frac{2H(p) + J(p)/2 + H_x(p)/2}{\alpha}.
\end{aligned}
\tag{4.10}
$$

Proposition 4.2 (Conditional entropy inequality)**.** *For every regular positive density p with marginal q on the fixed torus, the quantities in* (4.1)–(4.10) *satisfy*

$$
\langle \mathcal{F}_t(p), \log(p/q)\rangle + c_\alpha \mathcal{D}(p) \le G(p). \tag{4.11}
$$

Here $c_\alpha = \alpha/8$. If V is independent of x, then $H_x = 0$ and one may take $c_\alpha = \alpha/4$.

Proof. For the pure price flux, (4.3) gives

$$
\begin{aligned}
I_x &= -\tfrac{1}{2}\iint [\partial_x(awp) + awp]f\\
&= -\tfrac{1}{2}\int aqD_x\,\mathrm{d}x - \tfrac{1}{2}\int \mathcal{B}m\,\mathrm{d}x - \tfrac{1}{2}\iint awp(f-m)(L-n),\\
\mathcal{B} &= q(a_x + a + aS).
\end{aligned}
$$

Indeed, $\int wfu = m$, $\int wLu = n$ and $\int wu = 1$ imply

$$
\int wuf(f + L - m - n) = \int wu(f-m)^2 + \int wu(f-m)(L-n).
$$

The extra price term is therefore a conditional covariance.

For the pure factor flux, $\frac{\partial p}{\partial y} = pg$ gives

$$
I_y = -\tfrac{1}{2}\iint \frac{\partial(\eta^2 p)}{\partial y}g + \iint bpg = -\tfrac{1}{2}\int qD_y\,\mathrm{d}x + \iint p\eta g\,d.
$$

Finally, integration in x and cancellation of an integrated periodic derivative in y give

$$
\begin{aligned}
I_{xy} &= -\int q\sqrt{a}\int f\,\frac{\partial}{\partial y}(\rho\eta\sqrt{w}\,u)\,\mathrm{d}y\,\mathrm{d}x\\
&= -\int q\sqrt{a}\int (f-m)\frac{\partial}{\partial y}(\rho\eta\sqrt{w}\,u)\,\mathrm{d}y\,\mathrm{d}x\\
&= -\iint p\sqrt{aw}(f-m)(\rho\eta g + k_\rho).
\end{aligned}
$$

To justify the second equality, $m(t,x)$ is independent of y and periodicity gives

$$
m(t,x)\int \frac{\partial}{\partial y}(\rho\eta\sqrt{w}\,u)\,\mathrm{d}y = 0.
$$

For the last equality, we expand only the derivative in y:

$$
\frac{\partial}{\partial y}(\rho\eta\sqrt{w}\,u) = \sqrt{w}\,u\left(\rho\eta g + \rho\frac{\partial \eta}{\partial y} + \eta\frac{\partial \rho}{\partial y} + \frac{\rho\eta}{2}\frac{\partial \log V}{\partial y}\right).
$$

This is $\sqrt{w}\,u(\rho\eta g + k_\rho)$ by (4.10); no derivative of ρ in x is needed.

Write $A = \sqrt{aw}(f - m)$ and $B = \eta g$. The quadratic part is

$$-\tfrac{1}{2}\iint p(A^2 + 2\rho AB + B^2) \le -\tfrac{\alpha}{2}\iint p(A^2 + B^2) = -\tfrac{\alpha}{2}\mathcal{D},$$

since $2|AB| \le A^2 + B^2$. The first remainder is a scalar product of the vector fields (A, B) and $(-k_\rho, d)$ in $L^2(p)$. For the other remainder, (4.4) gives

$$|\mathcal{B}m| \le \sqrt{aqD_x}\sqrt{q\frac{(\partial a/\partial x + a + aS)^2}{a}K}.$$

Cauchy–Schwarz in these two spaces therefore yields

$$\left|\iint p(-Ak_\rho + Bd)\right| \le \sqrt{\mathcal{D}H}, \qquad \left|\int \mathcal{B}m\,\mathrm{d}x\right| \le \sqrt{\mathcal{D}J}.$$

Young's inequality gives $\sqrt{\mathcal{D}H} \le \alpha\mathcal{D}/8 + 2H/\alpha$ and $\sqrt{\mathcal{D}J}/2 \le \alpha\mathcal{D}/8 + J/(2\alpha)$. The covariance is bounded by $\sqrt{\mathcal{D}H_x}/2 \le \alpha\mathcal{D}/8 + H_x/(2\alpha)$. Subtracting these three costs from $\alpha\mathcal{D}/2$ leaves $\alpha\mathcal{D}/8$. If $L = 0$, the third term vanishes and $\alpha\mathcal{D}/4$ remains. This proves both assertions. Also, conditional variance and Cauchy–Schwarz give

$$H_x = \iint awpL^2 - \int aqn^2 \le a_+\left(\iint w^2 p\right)^{1/2}\left(\iint |L|^4 p\right)^{1/2}. \tag{4.12}$$

□

4.4 A form of the dissipation suitable for passage to the limit

For $\widehat{p} = wp$, calibration gives $\int \widehat{p}\,\mathrm{d}y = q$. Since $\frac{\partial \log \widehat{p}}{\partial x} = S + f - m + L - n$ and $\frac{\partial \log p}{\partial y} = g$,

$$\mathcal{D}(p) = \iint a\left|2\frac{\partial}{\partial x}\sqrt{\widehat{p}} - (S + L - n)\sqrt{\widehat{p}}\right|^2 + 4\iint \eta^2\left|\frac{\partial}{\partial y}\sqrt{p}\right|^2. \tag{4.13}$$

For weak densities on a fixed torus, the formula is justified as follows.

Write $h = \sqrt{p} \in H^1(\Omega)$. By the Cauchy–Schwarz inequality, $2h\nabla h \in L^1$, and the chain rule obtained by truncation gives $p = h^2 \in W^{1,1}$. Define the weighted scores by $\sqrt{p}\,f = 2\frac{\partial h}{\partial x} - Sh$ and $\sqrt{p}\,g = 2\frac{\partial h}{\partial y}$, with value zero on $\{p = 0\}$; the derivatives of h vanish almost everywhere on this set. The section properties of Sobolev functions imply, for almost every x, the identities in y used above. Since q is smooth and bounded away from zero, $u = p/q \in W^{1,1}$. Since V and V_x are bounded, Fubini's theorem applied against a spatial test function gives

$$M \in W^{1,1}(\mathbb{T}_{L_x}), \qquad \frac{\partial u}{\partial x} = fu \quad \text{almost everywhere,}$$
$$\frac{\partial M}{\partial x} = \int (V_x u + V u_x)\,\mathrm{d}y = M(m + n).$$

Here $fu = \left(2h\frac{\partial h}{\partial x} - Sp\right)/q$ is integrable; we do not divide by p on its zero set. Thus $\int fu = 0$, $m = \int wfu$ and $M_x = M(m + n)$ hold as weak identities; $|n| \le \|L\|_\infty$.

The quantity $\iint pf^2$ is finite, since $pf^2 = \left|2\frac{\partial h}{\partial x} - Sh\right|^2 \le 8\left|\frac{\partial h}{\partial x}\right|^2 + 2S^2h^2$. By the Cauchy–Schwarz inequality under $wu\,\mathrm{d}y$,

$$\int qm^2\,\mathrm{d}x \le \iint wpf^2 \le (V_+/V_-)\iint pf^2 < \infty.$$

Since $q \geq q_* > 0$, we have $m \in L^2_x$; since $M \leq V_+$, we also have $\frac{\partial M}{\partial x} = M(m+n) \in L^2_x$. Thus $M \in H^1(\mathbb{T}_{L_x})$. The function $z \mapsto z^{-1/2}$ has bounded derivative on $[V_-, V_+]$; its composition with M belongs to H^1 and

$$\frac{\partial M^{-1/2}}{\partial x} = -\tfrac{1}{2} M^{-1/2}(m+n).$$

For $\widehat{h} = \sqrt{V}\, M^{-1/2} h$, the derivative in x is

$$\frac{\partial \widehat{h}}{\partial x} = \sqrt{w}\, \frac{\partial h}{\partial x} + \tfrac{1}{2}(L - m - n)\sqrt{w}\, h.$$

Both terms belong to L^2: w, L, n are bounded and $\iint m^2 w p = \int q m^2 < \infty$. This product rule can first be checked in distributions in L^1; its two L^2 terms then identify the weak derivative in L^2. We thus recover the square-root formula without a lower bound on p. The centering identities follow by the same expansions of the integrals, all of which are finite by these estimates. This justification applies to the weak solutions constructed below.

4.5 From dissipation to gradient estimates

Proposition 4.3 (A common estimate for the roots)**.** *Fix $1 < r < 2$ and $\beta = r/(2-r)$. Let $p = qu$ be a density, $a \geq a_- > 0$, $w, \eta > 0$, and consider the quantities in (4.1). Suppose that the weak centering identities hold, with $\sqrt{p} \in H^1_{\mathrm{loc}}$ and $\frac{\partial}{\partial x} p = p(S+f)$, $\frac{\partial}{\partial y} p = pg$ in the weak sense. If $\mathcal{D}$ and the three moments*

$$A_\beta = \iint p w^{-\beta}, \qquad B_\beta = \iint p \eta^{-2\beta}, \qquad C_\beta = \int q K^\beta$$

are finite, then

$$\iint |\nabla p^{1/r}|^r \leq C_{r,a_-} \mathcal{D}^{r/2} (A_\beta^{1-r/2} + B_\beta^{1-r/2} + C_\beta^{1-r/2}) + C_r \int q |S|^r. \tag{4.14}$$

On any interval (δ, T) on which these moments are essentially bounded, $\mathcal{D} \in L^1$ and $\int q|S|^r \in L^1_t$, we have $p^{1/r} \in L^r_t W^{1,r}$ and $p \in L^r_t W^{1,1}$.

Proof. Hölder's inequality with exponents $2/r$ and $2/(2-r)$, followed by $m^2 \leq K D_x$, gives

$$\begin{aligned}
\iint p|g|^r &\leq \left(\iint p \eta^2 g^2 \right)^{r/2} B_\beta^{1-r/2}, \\
\iint p|f-m|^r &\leq \left(\iint p w (f-m)^2 \right)^{r/2} A_\beta^{1-r/2}, \\
\int q|m|^r &\leq \left(\int q D_x \right)^{r/2} C_\beta^{1-r/2}.
\end{aligned} \tag{4.15}$$

For example, $p|g|^r = (p\eta^2 g^2)^{r/2} (p\eta^{-2r/(2-r)})^{1-r/2}$, which determines the required power. On $\{p > 0\}$, the derivative identities give

$$\left| \frac{\partial p^{1/r}}{\partial x} \right|^r = r^{-r} p |(f-m) + m + S|^r, \qquad \left| \frac{\partial p^{1/r}}{\partial y} \right|^r = r^{-r} p |g|^r.$$

We use $|A+B+C|^r \leq 3^{r-1}(|A|^r + |B|^r + |C|^r)$ in the first identity. Since m and S depend only on x, their integrals against p equal their integrals against q. Finally, $\int q D_x \leq a_-^{-1} \mathcal{D}$ and $\int q D_y \leq \mathcal{D}$. Substituting (4.15) proves (4.14), once the following chain-rule justification is made. To justify this rule, use $\Phi_\epsilon(p) = (p+\epsilon)^{1/r} - \epsilon^{1/r}$: away from the zero set, its gradients are dominated by $r^{-1} p^{1/r - 1} |\nabla p|$, whose rth power was just estimated. Weak gradients vanish on the zero set. Dominated convergence and distributional tests therefore identify $\nabla p^{1/r}$. Finally,

$$\int_\delta^T \mathcal{D}^{r/2} \leq (T-\delta)^{1-r/2} \Big(\int_\delta^T \mathcal{D} \Big)^{r/2}, \qquad \|\nabla p(t)\|_1 \leq r \|\nabla p^{1/r}(t)\|_r,$$

since $\int p(t) = 1$. These inequalities give the time-integrated conclusions. □

5 Periodic existence by finite Fourier systems

Fix a torus $\Omega = \mathbb{T}_{L_x} \times \mathbb{T}_{L_y}$ and smooth coefficients a, b, η, V, ρ as in Section 4.1, with $a, \eta, V > 0$ and $|\rho| \le \rho_* < 1$. Let p^0 be a smooth strictly positive probability density, and let q solve the periodic scalar equation with coefficient a and initial marginal $\int p^0 \,\mathrm{d}y$. Lemma 3.3 gives a smooth, strictly positive q of mass one. The torus, coefficients and initial data are fixed throughout this section. The constants may depend on their bounds and positive minima. We use the operator $\mathcal{F}_t$ in (4.5), the entropy E, dissipation $\mathcal{D}$, and source G of Section 4, and write $D = \mathcal{D}$. Proposition 4.2 gives

$$\langle \mathcal{F}_t(p), \log(p/q) \rangle + c_\alpha D(p) \le G_t(p), \qquad \alpha = 1 - \rho_*. \tag{5.1}$$

The bounds in (4.2) and smoothness on the compact cylinder give $G_t(p) \le G_*$ uniformly over densities with marginal q. The finite systems below retain the exact scalar marginal; all their integrals are evaluated exactly.

5.1 Discrete coefficients and the finite-dimensional system

For $N \ge 1$, set $\tau = T/N$, $t_k = k\tau$, $q^k = q(t_k)$, and

$$\bar{a}^k(x) = \frac{1}{\tau q^k(x)} \int_{t_{k-1}}^{t_k} a(s,x) q(s,x) \,\mathrm{d}s. \tag{5.2}$$

Then

$$(q^k - q^{k-1})/\tau = \tfrac{1}{2} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial}{\partial x} \right) (\bar{a}^k q^k) \tag{5.3}$$

holds exactly. If $q_* \le q \le q^*$, then $a_- q_*/q^* \le \bar{a}^k \le a_+ q^*/q_*$. The required derivatives are uniformly bounded in N, k. Uniform continuity of aq, q with values in C^2, together with $q \ge q_* > 0$, gives

$$\sup_k \sup_{t \in [t_{k-1}, t_k]} \| \bar{a}^k - a(t) \|_{C^2(\mathbb{T}_{L_x})} \longrightarrow 0. \tag{5.4}$$

Let P_m be the Fourier projection onto frequencies $|j_x|, |j_y| \le m$. We use real trigonometric polynomials and set

$$X_m = \left\{ z \in \operatorname{Ran} P_m : \int_{\mathbb{T}_{L_y}} z(x,y) \,\mathrm{d}y = 0 \right\}.$$

Equip $H^4(\Omega)$ with the inner product $\sum_{|\gamma| \le 4} \int \partial^\gamma z \, \partial^\gamma \varphi$. The Fourier projections are contractions for this norm. For $z \in X_m$, define

$$p[z](x,y) = q^k(x) \frac{\exp z(x,y)}{\int_{\mathbb{T}_{L_y}} \exp z(x,y') \,\mathrm{d}y'}. \tag{5.5}$$

Let $\mathcal{F}^k$ be the operator with $\bar{a}^k$ in both price fluxes and with b, η, V, ρ evaluated at t_k. For every density p with marginal q^k, calibration and (5.3) give

$$\int \mathcal{F}^k(p) \,\mathrm{d}y = (q^k - q^{k-1})/\tau. \tag{5.6}$$

For $0 < \kappa \le 1$, each step consists in finding $z^k \in X_m$ such that

$$\begin{aligned} \kappa (z^k, \varphi)_{H^4} + \left\langle \frac{p[z^k] - p^{k-1}}{\tau} - \mathcal{F}^k(p[z^k]), \varphi \right\rangle = 0 \quad (\varphi \in X_m), \\ p^k = p[z^k]. \end{aligned} \tag{5.7}$$

This is a finite system of nonlinear equations. The density p^k is not itself a trigonometric polynomial: the discretised quantity is the conditional logarithm, up to a function of x.

Proposition 5.1 (Periodic existence and Fourier approximation)**.** *Under the assumptions above, every step of* (5.7) *has at least one solution. All resulting densities are strictly positive, have mass one, and have marginal q^k. For any sequence*

$$m_j \longrightarrow \infty, \qquad \tau_j = T/N_j \longrightarrow 0, \qquad \kappa_j \longrightarrow 0,$$

the piecewise constant interpolants $\bar{p}_j = p_j^k$ on $(t_{k-1}, t_k]$ have a subsequence converging strongly in $L^1((0,T)\times\Omega)$ to a weak solution $p \geq 0$ of $\frac{\partial p}{\partial t} = \mathcal{F}_t(p)$. It has initial datum p^0, marginal q, and satisfies

$$\sqrt{p} \in L^2(0,T;H^1(\Omega)), \qquad p \in L^2((0,T)\times\Omega). \tag{5.8}$$

It has a narrowly continuous probability-measure representative, with marginal $q(t,x)\,\mathrm{d}x$ at every time, and satisfies

$$E' + c_\alpha D \leq G(p) \quad \text{in } \mathcal{D}'(0,T). \tag{5.9}$$

No relation between m_j, τ_j, κ_j is required on the fixed torus.

5.2 Existence of one step and the entropy estimate

Proof of Proposition 5.1, first part. The parametrisation gives positivity and the prescribed marginal. Set $R(z) = (p[z] - p^{k-1})/\tau - \mathcal{F}^k(p[z])$. Integration in y, calibration and (5.3) give $\int R(z)\,\mathrm{d}y = 0$. Thus the residual annihilates every function of x alone.

Joint convexity of $(P,Q) \mapsto P\log(P/Q)$ gives

$$E(p[z],q^k) - E(p^{k-1},q^{k-1}) \leq \iint \log(p[z]/q^k)(p[z]-p^{k-1}). \tag{5.10}$$

Indeed, the additional terms vanish:

$$\iint (p[z]-p^{k-1}) = 0, \qquad -\int (q^k - q^{k-1}) \left(\int p[z]/q^k \,\mathrm{d}y \right) \mathrm{d}x = 0.$$

Write $\bar{E} = E + \log L_y \geq 0$. Since $z - \log(p[z]/q^k)$ depends only on x,

$$\langle R(z), z\rangle \geq (\bar{E}(p[z],q^k) - \bar{E}_{k-1})/\tau + c_\alpha D_k(p[z]) - G_*.$$

Here G_* is uniform in m, N, k, κ on the fixed torus, since the bounds on $\bar{a}^k, q^k$ and their derivatives are uniform.

Identify X_m' with X_m through the H^4 inner product. The left-hand side of (5.7) defines a continuous map $\Phi : X_m \to X_m$. Continuity follows from finite dimensionality and strict positivity of the denominators on bounded sets. We have

$$(\Phi(z), z)_{H^4} \geq \kappa \|z\|_{H^4}^2 - \bar{E}_{k-1}/\tau - G_*.$$

The right-hand side is positive on a sufficiently large sphere. Brouwer's lemma gives a zero of Φ in the ball. Indeed, if there were no zero, the continuous map $z \mapsto -R\Phi(z)/\|\Phi(z)\|_{H^4}$ from the ball into itself would have a fixed point on its boundary. At that point $(\Phi(z), z)_{H^4} < 0$, a contradiction.

Testing (5.7) with z^k gives

$$\bar{E}_k - \bar{E}_{k-1} + \tau c_\alpha D_k + \tau\kappa \left\| z^k \right\|_{H^4}^2 \leq \tau G_k(p^k) \leq \tau G_*. \tag{5.11}$$

Induction constructs all the steps. Summing up to any index l gives

$$\sup_k \bar{E}_k + \sum_k \tau D_k + \kappa \sum_k \tau \left\| z^k \right\|_{H^4}^2 \leq C. \tag{5.12}$$

Every branch of solutions of the systems satisfies this estimate. □

5.3 Spatial compactness and untested modes

Second part. Set $w_* = V_-/V_+$, $K^* = V_+/V_- - 1$, $\eta_* = \min \eta > 0$, and $a_* = \inf_{k,x} \bar{a}^k > 0$. With the notation of Section 4.1, the identity $\frac{\partial}{\partial x} \log p = (f - m) + m + S$ and (4.4) give

$$\iint p(f-m)^2 \le (w_* a_*)^{-1} D, \qquad \iint p m^2 \le K^* a_*^{-1} D, \qquad \iint p g^2 \le \eta_*^{-2} D.$$

Since $\iint pS^2 = \int qS^2 \le C$, it follows that

$$\iint p |\nabla \log p|^2 \le C(D(p) + 1). \tag{5.13}$$

Together with (5.12) and unit mass, this gives

$$\sum_k \tau \iint p^k |\nabla \log p^k|^2 \le C, \qquad \sum_k \tau \left\| \sqrt{p^k} \right\|_{H^1}^2 \le C. \tag{5.14}$$

Unit mass and Cauchy–Schwarz give $\left\|\nabla p^k\right\|_1 \le 2 \left\|\nabla \sqrt{p^k}\right\|_2$. The Gagliardo–Nirenberg inequality on the two-dimensional torus [4] gives $\|h\|_4^4 \le C \|h\|_2^2 \|\nabla h\|_2^2 + C \|h\|_2^4$. Hence

$$\sum_k \tau \left(\left\|p^k\right\|_{W^{1,1}}^2 + \left\|p^k\right\|_2^2 \right) \le C. \tag{5.15}$$

Let $\widetilde{p}$ be the affine interpolant of the p^k, including p^0. The scheme controls $P_m \frac{\partial}{\partial t} \widetilde{p}$, rather than $\frac{\partial}{\partial t} \widetilde{p}$ itself, in H^{-4}. Indeed, $P_m \varphi$ is the sum of an element of X_m and a function of x alone. For the latter, the residual vanishes and $(z^k, \varphi(x))_{H^4} = 0$. Consequently,

$$\left\langle P_m \frac{\partial}{\partial t} \widetilde{p}, \varphi \right\rangle = \langle \mathcal{F}^k(p^k), P_m \varphi \rangle - \kappa (z^k, P_m \varphi)_{H^4}.$$

The fluxes have uniformly bounded masses, since $w, \bar{a}^k, \eta, b, \rho$ are bounded on the fixed torus. The embedding $H^4 \hookrightarrow C^2$ and (5.12) imply

$$\left\| \frac{\partial}{\partial t} P_m \widetilde{p} \right\|_{L^2(0,T;H^{-4})} \le C, \qquad \|P_m(\widetilde{p} - \bar{p})\|_{L^2(0,T;H^{-4})} \le C\tau. \tag{5.16}$$

It remains to control the high frequencies. The Fourier coefficients of any positive measure μ of mass one on the torus are bounded in absolute value by a constant depending on the torus. Equivalence of Sobolev and spectral norms gives

$$\|(I - P_m)\mu\|_{H^{-4}}^2 \le C \sum_{\max(|j_x|,|j_y|) > m} (1 + |j|^2)^{-4} \le C(1+m)^{-6}.$$

Thus

$$\|(I - P_m)\mu\|_{H^{-4}} \le C(1+m)^{-3}. \tag{5.17}$$

Both interpolants are positive and have mass one, so this bound holds for each of them at every time.

Combining (5.16), (5.17), and the time-translation estimate for the affine interpolant gives, for $0 < h < T$,

$$\int_0^{T-h} \|\bar{p}(t+h) - \bar{p}(t)\|_{H^{-4}} \, \mathrm{d}t \le C\big(h + \tau + (1+m)^{-3}\big).$$

We use the compact embedding $W^{1,1}(\Omega) \Subset L^1(\Omega)$ together with the continuous injective embedding $L^1(\Omega) \hookrightarrow H^{-4}(\Omega)$. These embeddings imply that, for every $\epsilon > 0$, there is a constant C_ϵ such that

$$\|f\|_1 \le \epsilon \|f\|_{W^{1,1}} + C_\epsilon \|f\|_{H^{-4}}. \tag{5.18}$$

For completeness, if (5.18) failed for some $\epsilon > 0$, normalisation would give f_j with

$$\|f_j\|_1 = 1, \qquad \epsilon\|f_j\|_{W^{1,1}} + j\|f_j\|_{H^{-4}} < 1.$$

A subsequence would converge in L^1 to a function f of norm one, while converging to zero in H^{-4}. The injective embedding identifies these limits, a contradiction. Applying (5.18) to the time increments and using (5.15), we obtain

$$\int_0^{T-h} \|\bar{p}(t+h) - \bar{p}(t)\|_1 \,\mathrm{d}t \le C\epsilon + C_\epsilon\big(h + \tau + (1+m)^{-3}\big).$$

Letting the parameters tend to their limits, and then h, ϵ tend to zero, makes the time translations uniformly small on tails of the sequence. Estimate (5.15) controls the spatial translations. Extension by zero in time adds at most $2h$ to the translation integrals, by unit mass. The Fréchet–Kolmogorov criterion therefore gives a subsequence such that

$$\bar{p}_j \longrightarrow p \quad \text{in } L^1((0,T)\times\Omega) \quad \text{and almost everywhere.} \tag{5.19}$$

The L^2 bound passes to the limit. □

5.4 Identification of the equation, entropy and initial trace

Third part. Write $q_j = q(t_k)$, $V_j = V(t_k)$, $\bar{a}_j = \bar{a}^k$, and similarly for the other coefficients on each time interval. The discrete coefficients converge uniformly to their continuous values; for $\bar{a}^k$, convergence holds in C^2. Hence

$$\int p\,\mathrm{d}y = q, \qquad \int V_j\bar{p}_j\,\mathrm{d}y \longrightarrow \int Vp\,\mathrm{d}y \quad \text{in } L^1.$$

The bounds $q_j \ge q_* > 0$ and $\int V_j\bar{p}_j\,\mathrm{d}y \ge V_- q_j$ give, after extraction, $M_j \to M$ and $w_j \to w$ almost everywhere. Since $V_-/V_+ \le w_j \le V_+/V_-$,

$$w_j\bar{p}_j \to wp, \qquad \sqrt{w_j}\bar{p}_j \to \sqrt{w}\,p \quad \text{in } L^1.$$

The other fluxes also converge strongly in L^1, by the uniform coefficient bounds.

For smooth periodic ψ vanishing near T, test each step with $P_{m_j}\psi(t_{k-1})$ and multiply by τ_j. Summation by parts gives

$$\begin{aligned}\sum_k \langle p_j^k - p_j^{k-1}, P_{m_j}\psi(t_{k-1})\rangle = &-\langle p^0, P_{m_j}\psi(0)\rangle \\ &- \sum_{k=1}^{N_j-1} \langle p_j^k, P_{m_j}(\psi(t_k) - \psi(t_{k-1}))\rangle.\end{aligned}$$

Convergence of the Fourier sums, together with the required time and space derivatives, and L^1 convergence of the densities give the left-hand side of the weak formulation with trace p^0. The fluxes give the right-hand side. The penalisation term is bounded by

$$\left(\kappa_j \sum_k \tau_j \left\|z_j^k\right\|_{H^4}^2\right)^{1/2} \left(\kappa_j \sum_k \tau_j \left\|P_{m_j}\psi(t_{k-1})\right\|_{H^4}^2\right)^{1/2} \le C_\psi\sqrt{\kappa_j} \longrightarrow 0.$$

We obtain the weak formulation

$$\begin{aligned}-\int_0^T \iint p\frac{\partial\psi}{\partial t} - \iint p^0\psi(0) = \int_0^T \iint p\Big[&\tfrac{1}{2}aw\left(\frac{\partial^2\psi}{\partial x^2} - \frac{\partial\psi}{\partial x}\right) + \rho\eta\sqrt{aw}\frac{\partial^2\psi}{\partial x\partial y} \\ &+ \tfrac{1}{2}\eta^2\frac{\partial^2\psi}{\partial y^2} + b\frac{\partial\psi}{\partial y}\Big].\end{aligned} \tag{5.20}$$

Strong L^1 convergence implies $\sqrt{\bar{p}_j} \to \sqrt{p}$ and $\sqrt{w_j \bar{p}_j} \to \sqrt{wp}$ in L^2. The bound (5.14) identifies the weak limit of $\nabla\sqrt{\bar{p}_j}$ as $\nabla\sqrt{p}$, which proves the first assertion in (5.8). The dissipation can be written as

$$D_j = \iint \bar{a}_j \left| 2\frac{\partial}{\partial x}\sqrt{w_j \bar{p}_j} - (S_j + L_j - n_j)\sqrt{w_j \bar{p}_j} \right|^2 + 4\eta_j^2 \left| \frac{\partial}{\partial y}\sqrt{\bar{p}_j} \right|^2 .$$

For the limiting density, this expression in square roots defines $D(p)$ through weak derivatives; no definition of $\log p$ is needed where $p = 0$. The lower bounds on the coefficients give the required gradients in L^2. Their weak limits are identified in distributions. On this fixed cylinder V_j, L_j, S_j converge uniformly and

$$n_j = \frac{\int V_j L_j \bar{p}_j \, \mathrm{d}y}{\int V_j \bar{p}_j \, \mathrm{d}y} \longrightarrow n = \frac{\int VLp \, \mathrm{d}y}{\int Vp \, \mathrm{d}y}$$

almost everywhere after extraction. The denominators are at least $V_- q_* > 0$ and $|n_j| \le \|L_j\|_\infty$. Strong convergence of $w_j \bar{p}_j$ therefore gives strong L^2 convergence of $(S_j + L_j - n_j)\sqrt{w_j \bar{p}_j}$. Lower semicontinuity gives, for $\xi \in C_c^\infty(0,T)$, $\xi \ge 0$,

$$\int \xi D(p) \le \liminf_j \int \xi D_j. \tag{5.21}$$

We check uniform integrability before passing to the entropy. For $z \ge R > 1$, $z \log z \le c_R z^2$, where $c_R = \sup_{z \ge R} \log z / z \to 0$. The L^2 bound on $\bar{p}_j$ therefore makes the integrals of its positive entropy tail uniformly small. On bounded values of the density the integrand is bounded, and $|z \log z| \le 1/e$ for $0 \le z \le 1$. Since the cylinder has finite measure, these observations prove uniform integrability of $\bar{p}_j \log \bar{p}_j$. Vitali's theorem and uniform convergence of $\log q_j$ to $\log q$ give $E_j \to E$ in $L^1(0,T)$. The conditional moments of $V_j^{\pm 1}$ converge strongly and are bounded. The variance identity $H_{x,j} = \int \bar{a}_j w_j \bar{p}_j L_j^2 - \int \bar{a}_j q_j n_j^2$ gives convergence of the additional source. The explicit formula for G_j therefore gives $G_j \to G(p)$ in $L^1(0,T)$. Summing (5.11) against $\xi(t_{k-1})$, followed by summation by parts, yields $-\int E\xi' + c_\alpha \int D\xi \le \int G\xi$ in the limit. The errors from replacing time-dependent tests by their endpoint values are $O(\tau_j)$, by (5.12).

Finally, for every smooth spatial test function φ, the projected estimates give

$$|\langle \widetilde{p}_j(t) - \widetilde{p}_j(s), \varphi \rangle| \le C \, \|\varphi\|_{H^4} \, |t-s|^{1/2} + 2 \left\| (I - P_{m_j})\varphi \right\|_\infty .$$

The second term tends to zero. Compactness of the torus, extraction on a countable family of test functions, and this asymptotic equicontinuity give a narrowly continuous limiting curve, with convergence uniform in time against each continuous test function. By (5.16) and (5.17), the difference between the affine and piecewise constant interpolants tends to zero in $L^1(0,T;H^{-4})$. Their occupation measures therefore have the same limiting density p. At $t = 0$, the affine interpolant equals p^0. Its marginal is the affine interpolant of the q^k, which converges uniformly to q. This identifies the initial trace and the marginal at every time, and completes the proof. □

5.5 Nonlinear systems solved up to a residual

Proposition 5.2. *The scheme* (5.7) *may be replaced by*

$$\kappa_j (z_j^k, \varphi)_{H^4} + \left\langle (p_j^k - p_j^{k-1})/\tau_j - \mathcal{F}^k(p_j^k), \varphi \right\rangle = r_j^k[\varphi] \quad (\varphi \in X_{m_j}),$$

provided that the exponential parametrisation is retained and

$$\frac{1}{\kappa_j} \sum_k \tau_j \left\| r_j^k \right\|_{X'_{m_j}}^2 \longrightarrow 0. \tag{5.22}$$

The dual norm is induced by the H^4 inner product restricted to X_{m_j}. The conclusions of Proposition 5.1 then remain valid.

Proof. Testing with z_j^k introduces the term

$$|r_j^k[z_j^k]| \le \frac{\kappa_j}{2}\left\|z_j^k\right\|_{H^4}^2 + \frac{1}{2\kappa_j}\left\|r_j^k\right\|_{X'_{m_j}}^2 .$$

Multiplying by τ_j and summing, we obtain a total error bounded by

$$\frac{1}{2\kappa_j}\sum_k \tau_j \|r_j^k\|_{X'_{m_j}}^2 \longrightarrow 0.$$

Thus half the penalisation remains on the left of the discrete entropy inequality, and the bounds (5.12) hold with constants independent of j. The residual does not act on functions of x alone, for which the marginal already imposes the exact identity; extend it by zero on these functions. On the other test functions, its L^2 norm in time tends to zero, since $\sum \tau_j \left\|r_j^k\right\|^2 = \kappa_j\, o(1)$. The projected time estimates and passage to the limit in the weak formulation therefore remain valid. The preceding proof applies, including the entropy inequality. □

This proposition gives a sufficient condition on the computed residual. It does not guarantee that Newton's method, or another iteration, attains the tolerance from an arbitrary initial iterate.

Proposition 5.3 (Measure bounds for the projected residual)**.** *On the fixed torus of Proposition 5.2, extend each functional r_j^k by*

$$\overline{r}_j^k[\varphi] = r_j^k[Q_{m_j}\varphi],$$

where Q_m is the Fourier projection onto X_m. This extension is represented by a real trigonometric polynomial R_j^k. Set $\mathfrak{e}_j = \sum_k \tau_j \|r_j^k\|_{X'_{m_j}}^2$. Then

$$\sum_k \tau_j \|R_j^k\|_{L^1(\Omega)} \le C_\Omega \sqrt{T}(1+m_j)^4 \mathfrak{e}_j^{1/2}. \tag{5.23}$$

For solutions satisfying the tolerance in Proposition 5.2, the full projected discrete residual also satisfies

$$\int_0^T \left\| P_{m_j}\left(\frac{\partial \widetilde{p}_j}{\partial t} - \mathcal{F}^k(p_j^k)\right)\right\|_{L^1(\Omega)} \mathrm{d}t \le C(1+m_j)^4\big(\mathfrak{e}_j^{1/2} + \sqrt{\kappa_j}\big). \tag{5.24}$$

Here k is the time-cell index, and C may depend on the fixed torus, coefficients and initial data, but not on j. In particular, $(1+m_j)^4\sqrt{\kappa_j} \to 0$, together with the original residual tolerance, implies convergence of this projected residual to zero in time-integrated total variation.

Proof. In an orthonormal Fourier basis, let $\lambda_\ell \asymp_\Omega (1+|\ell|^2)^4$ be the weights of the squared H^4 norm. Represent r_j^k by its Fourier coefficients on X_{m_j} and zero coefficients elsewhere; these define the real polynomial R_j^k. Orthogonality gives

$$\|R_j^k\|_2^2 \le C_\Omega(1+m_j)^8 \sum_{\ell \in X_{m_j}} \frac{|\widehat{R}_j^k(\ell)|^2}{\lambda_\ell} = C_\Omega(1+m_j)^8 \|r_j^k\|_{X'_{m_j}}^2 .$$

Here the sum is over the frequencies of X_{m_j}. The inequalities $\|R\|_1 \le |\Omega|^{1/2}\|R\|_2$ and Cauchy–Schwarz in discrete time prove (5.23). For the multiplier $\mathcal{H}_4$ with weights λ_ℓ, the same spectral bound gives $\|\mathcal{H}_4 z\|_2 \le C_\Omega(1+m)^4\|z\|_{H^4}$, $z \in X_m$. The scheme and exact marginal identity imply

$$P_{m_j}(\partial_t \widetilde{p}_j - \mathcal{F}^k(p_j^k)) = R_j^k - \kappa_j \mathcal{H}_4 z_j^k.$$

Using $\kappa_j \sum_k \tau_j \|z_j^k\|_{H^4}^2 \le C$ and Cauchy–Schwarz once more proves (5.24). Finally, (5.22) says $\mathfrak{e}_j = o(\kappa_j)$, giving the asserted sufficient condition. □

For example, the choice $\kappa_j = (1+m_j)^{-10}$ satisfies the additional condition on a fixed torus. It is sufficient for the projected measure bound, not necessary for the convergence result of Proposition 5.1. On the centred fundamental domain, the same bound with weight $\exp(x)$ follows by multiplication by $\exp(L_x/2)$. These constants are not uniform as the torus grows. Both R_j^k and $\mathcal{H}_4 z_j^k$ have zero integral in y, since their modes belong to X_{m_j}. They therefore introduce no scalar marginal residual. The result does not control the unprojected part of the equation, errors from replacing the discrete coefficients by continuous ones in total variation, or boundary terms after extension to the line. It is consequently not, by itself, a bound on the whole-space residual in Section 13.

The additional frequency factor cannot be omitted on the basis of a negative Sobolev norm alone. On $\mathbb{T}_{2\pi}$, the measures $\sigma_N = N\sin(Nx)\,\mathrm{d}x$ satisfy

$$\|\sigma_N\|_{H^{-4}} \le CN^{-3} \longrightarrow 0, \qquad \|\sigma_N\|_{\mathrm{TV}} = 4N \longrightarrow \infty.$$

This gives a direct obstruction to inferring total-variation convergence from distributional convergence without further control.

For each n, apply Proposition 5.1 to the coefficients of (3.7) and the joint initial densities of Section 3.3. Denote the resulting density and curve by p_n and μ_t^n. The internal limits $m \to \infty$ and $\tau, \kappa \to 0$ are taken on each fixed torus before the limit in n.

6 Uniform estimates on expanding tori

Proposition 6.1 (Uniform estimates)**.** *Under (H1)–(H5), apply Proposition 5.1 to the coefficients and initial data of Proposition 3.4. The resulting solutions p_n and their curves of laws μ^n satisfy the Gaussian moment bound* (6.2)*, the position moment bounds* (6.4)*, and the coefficient and weight moment bounds* (6.5)–(6.6)*. For every $0 < \delta < T$ and $1 < r < 2$, they also satisfy the entropy and dissipation bounds* (6.14)*, and the gradient bounds* (6.15) *and* (6.16)*. All constants are independent of n; they may depend on δ, r where these parameters occur.*

Proof. We derive moment bounds before estimating the entropy sources and dissipation.

6.1 Gaussian moment of the factor

Proposition 5.1 has established existence of the solutions p_n and curves μ^n. In this section, integrals against μ_t^n are defined at every time. The density expressions $p_n, u_n, M_n, w_n, E_n, \mathcal{D}_n$ are understood almost everywhere in time. The coordinates x, y denote their centred representatives in $[-R_n, R_n]$. Constants are independent of n.

Set $c_* = 4/\pi^2$, $C_* = 2\pi D_0 + \eta_+^2 + 1$, where D_0 is defined in (3.9), and

$$\lambda(t) = \frac{\lambda_0 e^{-C_* t}}{1 + (2\eta_+^2\lambda_0/C_*)(1 - e^{-C_* t})}, \qquad \kappa_T = c_*\lambda(T) > 0. \tag{6.1}$$

We prove that

$$\sup_n \sup_{0\le t\le T} \int e^{\kappa_T y^2}\,\mathrm{d}\mu_t^n \le C_{\mu_0} := \sqrt{2}e^{\lambda_0}\mathbb{E}e^{2\lambda_0 Y_0^2}. \tag{6.2}$$

With $R = R_n$, $\omega = \pi/(2R)$, set $F_R(y) = \sin(\omega y)/\omega$ and $U_R = 1 + F_R^2$. This function is smooth and $2R$-periodic. On $[-R, R]$,

$$\begin{gathered} c_*(1+y^2) \le U_R \le 1 + y^2, \qquad U_R' = 2F_R\cos(\omega y), \qquad U_R'' = 2\cos(2\omega y) \le 2, \\ |U_R'|^2 \le 4U_R, \qquad \operatorname{sgn}(U_R') = \operatorname{sgn}(y) \quad (0 < |y| < R). \end{gathered} \tag{6.3}$$

For $\mathcal{L}_Y^n = b_n\partial/\partial y + \eta_n^2\partial^2/(2\partial y^2)$, bound (3.9) implies

$$\mathcal{L}_Y^n U_R \le C_* U_R, \qquad \eta_n^2|U_R'|^2 \le 4\eta_+^2 U_R.$$

Since $\lambda' = -C_*\lambda - 2\eta_+^2\lambda^2$, we obtain

$$(\partial/\partial t + \mathcal{L}_Y^n)e^{\lambda U_R} = e^{\lambda U_R}\left[\lambda' U_R + \lambda \mathcal{L}_Y^n U_R + \tfrac{1}{2}\lambda^2\eta_n^2|U_R'|^2\right] \le 0.$$

This test is admissible in the marginal equation on each fixed torus. With $G_n = Y_0 + \sqrt{\varepsilon_n}G_2$, (3.5) gives

$$\mathbb{E}e^{\lambda_0 U_R(\pi_R G_n)} \le C_{\mu_0}, \qquad \lambda(t)U_R \ge \kappa_T y^2.$$

This proves (6.2). Only the one-sided bound on yb is used, since U_R' has the sign of y.

6.2 Moments of the positions, coefficients and weights

The marginal q_n is the projection of the scalar diffusion on the line with periodic coefficient a_n. Uniform boundedness of the scalar drift and diffusion coefficients, the martingale inequalities, and $|\pi_R x| \le |x|$ give, for every $r \ge 1$,

$$\sup_n \sup_{t\le T} \int |x|^r \,\mathrm{d}\mu_t^n \le C_{r,T}. \tag{6.4}$$

For every approximating coefficient h_n satisfying the growth bound $|h_n| \le C_e e^{ey^2}$ in (3.8), and every power $r > 0$, choose $e < \kappa_T/r$. Then

$$\operatorname*{ess\,sup}_{t\le T} \iint |h_n|^r p_n \le C_e^r \sup_{t\le T}\int e^{\kappa_T y^2}\,\mathrm{d}\mu_t^n \le C_{r,T}. \tag{6.5}$$

This applies to $V_n^{\pm 1}, \eta_n^{-1}, b_n, \frac{\partial}{\partial y}\eta_n, \frac{\partial}{\partial y}\rho_n, \partial_x \log V_n, \frac{\partial}{\partial y}\log V_n$ and products of their fixed powers. Inequalities (4.8)–(4.9) therefore give

$$\operatorname*{ess\,sup}_{t\le T}\left[\int q_n K_n^r \,\mathrm{d}x + \iint (w_n^r + w_n^{-r})p_n\right] \le C_{r,T} \quad (r \ge 1). \tag{6.6}$$

In particular, the L^1 norms of the fluxes are uniformly bounded at almost every time:

$$\iint \left(a_n w_n + \eta_n^2 + |b_n| + |\rho_n|\eta_n\sqrt{a_n w_n}\right) p_n \le C_T. \tag{6.7}$$

By calibration, the first term equals $\int a_n q_n \le a_+$; the second is at most η_+^2; the third is controlled by (6.5); and the last is at most $\rho_*\eta_+\sqrt{a_+}$ by the Cauchy–Schwarz inequality.

For every test function $\phi \in C_c^\infty(\mathbb{R}^2)$ and all sufficiently large n, its support lies inside the centred domain, so it can be periodised by zero. The weak equation and (6.7) then give, for $s, t \in [0, T]$,

$$\left|\int \phi \,\mathrm{d}\mu_t^n - \int \phi \,\mathrm{d}\mu_s^n\right| \le C_T\|\phi\|_{C^2}|t - s|. \tag{6.8}$$

6.3 The entropy source and weighted coercivity

The coefficient moments in (6.5) give

$$H_n \le C\iint p_n(|\partial_y\eta_n|^2 + |\partial_y\rho_n|^2 + |\partial_y \log V_n|^2 + |b_n|^2\eta_n^{-2}) \le C_T.$$

For the product use Hölder with the fourth moments of b_n and η_n^{-1}. The score bound of order four and weight bound of order two give

$$J_n \le C\left(\int q_n K_n^2\right)^{1/2}\left(\int q_n(1 + S_n^2)^2\right)^{1/2} \le C_T(1 + t^{-1}). \tag{6.9}$$

By (4.12), the moments of w_n^2 and $|\partial_x \log V_n|^4$ also give $H_{x,n} \le C_T$. The solutions of Proposition 5.1 therefore satisfy

$$E_n' + c_\alpha \mathcal{D}_n \le C_T(1 + t^{-1}) \quad \text{in } \mathcal{D}'(0,T). \tag{6.10}$$

The coercivity estimate that removes dependence on the initial entropy is obtained directly on the circle. Fix an admissible time, and set $L = 2R_n$ and $Q_x = \int \eta_n^{-2} u_n \,\mathrm{d}y$. For almost every x, $u_n(x,\cdot) \in W^{1,1}(\mathbb{T}_L)$ by (5.8) and the lower bound on q_n for fixed n. Its continuous representative takes a value at most $1/L$, since its integral is one. The fundamental theorem of calculus and the Cauchy–Schwarz inequality give

$$\|u_n(x,\cdot)\|_\infty \le L^{-1} + \int \left|\frac{\partial u_n}{\partial y}\right| \mathrm{d}y \le L^{-1} + \sqrt{D_{y,n}(x) Q_x}.$$

On the zero set, weak derivatives vanish and the ratios are understood in the weighted sense. Set $e_x = \int u_n \log u_n \,\mathrm{d}y$. Since $e_x \le \log \|u_n(x,\cdot)\|_\infty$, Jensen's inequality followed by the Cauchy–Schwarz inequality under $q_n \,\mathrm{d}x$ gives

$$\begin{aligned} e^{E_n} &\le \int q_n e^{e_x} \,\mathrm{d}x \le L^{-1} + \int q_n \sqrt{D_{y,n} Q_x} \,\mathrm{d}x \\ &\le L^{-1} + \sqrt{\mathcal{D}_n Q_n}, \qquad Q_n = \iint \eta_n^{-2} p_n \le Q_* < \infty. \end{aligned} \tag{6.11}$$

The bound Q_* follows from (6.5), not from a uniform lower bound on η_n. Since $L \ge 1$, for $E_n \ge \log 2$,

$$\mathcal{D}_n \ge \frac{e^{2E_n}}{4Q_*}. \tag{6.12}$$

For the lower bound, relative entropy with respect to e^{-y^2}/Z_R, $Z_R = \int_{-R}^{R} e^{-y^2} \,\mathrm{d}y \le \sqrt{\pi}$, gives

$$\int u_n \log u_n \,\mathrm{d}y \ge -\int y^2 u_n \,\mathrm{d}y - \log Z_R.$$

Together with (6.2), this yields $E_n \ge -C_T$.

6.4 Entropy bounds at positive times

We detail the passage from a distributional inequality to a pointwise bound. Suppose that a locally integrable function E, bounded below, satisfies

$$E' + \lambda e^{2E} \mathbf{1}_{\{E \ge A\}} \le G, \qquad G \ge 0,$$

with a locally integrable dissipative term. Then its right-continuous representative satisfies, for $0 < s < t < T$,

$$E(t) \le \int_s^t G(r) \,\mathrm{d}r + \max\{A, -\tfrac{1}{2}\log(2\lambda(t-s))\}. \tag{6.13}$$

Indeed, $F(r) = E(r) - \int_s^r G$ has nonpositive distributional derivative, and hence a nonincreasing representative of locally bounded variation. As long as $F > A$, we have $E \ge F > A$ and $F' \le -\lambda e^{2F}$ almost everywhere. For $U = e^{-2F}$, the chain rule for monotone functions gives $U' \ge 2\lambda$ almost everywhere; the jumps and the singular part of $\mathrm{d}U$ are nonnegative. Thus $U(t) \ge 2\lambda(t-s)$ if F has remained above A. After its first passage below A, monotonicity gives $F \le A$. These two cases prove (6.13).

The representative is the same for all pairs of times: $G - E'$ is a positive measure, so $E - \int G$ has a right-continuous nonincreasing representative. The integrated inequality and the comparison hold for every $s < t$ for this representative. We may therefore choose $s = t/2$ and then return to Lebesgue values outside a single null set.

Apply this calculation to (6.10) and (6.12), with $\lambda = c_\alpha/(4Q_*)$, $A = \log 2$ and $s = t/2$. The right-hand side integrates over this interval as follows:

$$\int_{t/2}^{t} C_T(1 + r^{-1})\,\mathrm{d}r = C_T(t/2 + \log 2).$$

For almost every $t > 0$, we obtain

$$-C_T \le E_n(t) \le C_T - \tfrac{1}{2}\log t, \qquad \int_\delta^T \mathcal{D}_n(t)\,\mathrm{d}t \le C_T(1 + |\log\delta|) \quad (0 < \delta < T). \tag{6.14}$$

For the second bound, integrate (6.10) between a Lebesgue time $s \in (\delta/2, \delta)$ and $t < T$, use the upper bound at s and the lower bound at t, and let $t \uparrow T$. All constants are independent of n. The entropy of the initial Gaussian may diverge with n; it does not enter this estimate.

6.5 Uniform gradient bounds

Fix $\delta > 0$, $1 < r < 2$ and $\beta = r/(2 - r)$. For fixed n, (5.8) and Section 4.4 give the weak identities required by Proposition 4.3. The three moments in that proposition are uniformly bounded by (6.5)–(6.6). Its right-hand side is bounded by $C_r\mathcal{D}_n^{r/2} + C_r\int q_n|S_n|^r$. Integrating, Hölder in time, (6.14) and (3.17) give

$$\int_\delta^T \iint_{\Omega_n} |\nabla p_n^{1/r}|^r \le C_{r,\delta,T}, \tag{6.15}$$

and, since the spatial mass is one,

$$\int_\delta^T \|p_n(t)\|_{W^{1,1}(\Omega_n)}^r\,dt \le C_{r,\delta,T}. \tag{6.16}$$

These estimates give both compactness of the densities and regularity of the limit. □

7 Compactness and identification of the limiting law

Lemma 7.1 (Convergence of weighted products)**.** *On a measure space, suppose that $p_n \ge 0$, $p_n \to p$ in L^1 and almost everywhere, $h_n \to h$ almost everywhere, and $\sup_n \int |h_n|^s p_n \le C_s$ for some $s > 1$. Then $h_n p_n \to hp$ in L^1.*

Proof. We separate a bounded part of the weight from its tail. For $R > 0$, set $T_R(z) = \max(-R, \min(z, R))$. The almost-everywhere convergences imply $|h_n|^s p_n \to |h|^s p$, so Fatou's lemma gives $\int |h|^s p \le C_s$. Moreover, on $\{|z| > R\}$ we have $|z| \le R^{1-s}|z|^s$. Therefore

$$\int |h|^s p \le C_s, \qquad \int |h_n - T_R(h_n)|p_n \le C_s R^{1-s}, \qquad \int |h - T_R(h)|p \le C_s R^{1-s}.$$

For fixed R, dominated convergence against p and strong convergence give

$$\|T_R(h_n)p_n - T_R(h)p\|_1 \le R\|p_n - p\|_1 + \int |T_R(h_n) - T_R(h)|p \longrightarrow 0.$$

Thus $\limsup_n \|h_n p_n - hp\|_1 \le 2C_s R^{1-s}$, and letting $R \to \infty$ concludes the proof. The proof also applies on a domain of infinite measure. □

Proposition 7.2 (Limiting density and fluxes)**.** *For the sequence of solutions of Proposition 6.1, there exist a common subsequence and a density $p \ge 0$ such that $p_n \to p$ in $L^1((\delta, T) \times \mathbb{R}^2)$ for every $0 < \delta < T$, where p_n is extended by zero outside the centred domain. The marginal of p is q almost everywhere in time. The moment $\mathcal{V}(t, x) = \int v(t, x, y)p(t, x, y)\,\mathrm{d}y$ is positive and finite almost everywhere; $M = \mathcal{V}/q$ defines the ratio $w = v/M$. The fluxes converge as shown below, and p satisfies* (7.11) *with regularity* (2.12)*. There exists a narrowly continuous curve of probability measures μ_t, with trace μ_0, occupation density p, and marginal $\mathcal{L}(Z_t)$ at every time.*

Proof. We first extract the density, then identify the quotient and each flux along the same subsequence.

7.1 Compactness of the densities and identification of the marginal

In this section only, regard p_n and q_n as functions on $\mathbb{R}^2$ and $\mathbb{R}$, extended by zero outside the centred domains. The coefficients remain periodic. No global Sobolev regularity is asserted for these extensions, which may have a jump at the boundary.

Fix $\delta > 0$ and $\chi_0 \in C_c^\infty(\mathbb{R}^2)$. For sufficiently large n, its support lies inside the torus. Estimates (6.16) and (6.7) give

$$\|\chi_0 p_n\|_{L^r(\delta,T;W^{1,1})} \le C_{\chi_0,r,\delta}, \qquad \left\|\frac{\partial(\chi_0 p_n)}{\partial t}\right\|_{L^\infty(\delta,T;H^{-4})} \le C_{\chi_0}.$$

We take $r = 3/2$ here; the argument applies to any available $r > 1$. The embeddings $W^{1,1} \Subset L^1 \hookrightarrow H^{-4}$ on a fixed bounded domain allow us to apply [14, Theorem 5 and Lemma 8]. We obtain local strong convergence in L^1 along a subsequence.

The tails are uniformly small. For $L > 0$, (6.2) and (6.4) give

$$\int_\delta^T \iint_{\{|x|>L\}\cup\{|y|>L\}} p_n \le C_T \left(L^{-2} + e^{-\kappa_T L^2}\right). \tag{7.1}$$

To pass from local to global convergence, let $Q_L = [-L, L]^2$. Fatou's lemma gives the same tail bound for the local limit p. Hence

$$\|p_n - p\|_{L^1((\delta,T)\times\mathbb{R}^2)} \le \|p_n - p\|_{L^1((\delta,T)\times Q_L)} + 2C_T(L^{-2} + e^{-\kappa_T L^2}).$$

For fixed L the first term tends to zero; we then let $L \to \infty$. This proves global convergence and prevents loss of mass. A diagonal extraction over increasing compact sets and $\delta \downarrow 0$ gives a common function $p \ge 0$, of mass one at almost every time, such that

$$p_n \to p \quad \text{in } L^1((\delta, T) \times \mathbb{R}^2) \quad \text{for every } \delta > 0. \tag{7.2}$$

On $(0, \delta)$, the L^1 distance is at most 2δ, since both masses are one. Convergence therefore also holds on $(0, T)$. The position moments pass to p by Fatou's lemma, after almost-everywhere extraction and integration against nonnegative time test functions. They are thus bounded in essential supremum in time. The moments of the original coefficients follow from their growth bounds and the Gaussian moment.

Integration in y contracts the L^1 norm, so $q_n \to \bar{q} := \int p \, \mathrm{d}y$ in $L^1((0, T) \times \mathbb{R})$. The scalar identification (3.11), integrated against a time test function and a bounded Lipschitz spatial test function, implies $\bar{q} = q$. Consequently,

$$\int p(t, x, y) \, \mathrm{d}y = q(t, x) \quad \text{almost everywhere}, \qquad q_n \to q \quad \text{in } L^1((0, T) \times \mathbb{R}). \tag{7.3}$$

If necessary, extract a subsequence along which the densities and their marginals converge almost everywhere. No limiting score is used in this identification.

7.2 The variance moment and the limiting ratio

Write

$$\mathcal{V}_n(t, x) = \int V_n(t, x, y) p_n(t, x, y) \, \mathrm{d}y, \qquad \mathcal{V}(t, x) = \int v(t, x, y) p(t, x, y) \, \mathrm{d}y.$$

The uniform moments of V_n of order greater than one, (7.2), and (3.10) allow us to apply Lemma 7.1. Integration in y contracts the L^1 norm. Hence, for every $\delta > 0$,

$$V_n p_n \to v p \quad \text{in } L^1((\delta,T)\times\mathbb{R}^2), \qquad \mathcal{V}_n \to \mathcal{V} \quad \text{in } L^1((\delta,T)\times\mathbb{R}). \tag{7.4}$$

Finiteness of the integrated moment implies $\mathcal{V} < \infty$ almost everywhere. If $\mathcal{V}(t,x) = 0$ while $v(t,x,y) > 0$, nonnegativity of p forces $p(t,x,\cdot) = 0$ almost everywhere, contradicting (7.3) and $q(t,x) > 0$. Thus

$$0 < \mathcal{V} < \infty \quad \text{almost everywhere}, \qquad M = \mathcal{V}/q, \qquad w = v/M = qv/\mathcal{V}. \tag{7.5}$$

The almost-everywhere convergences extracted from (7.3) and (7.4) give

$$w_n \to w \quad \text{almost everywhere in space-time.} \tag{7.6}$$

Outside the centred domain in x, we may set $M_n = 1$ and $w_n = V_n$ arbitrarily, since $p_n = 0$; this affects neither the integrals nor eventual local convergence. No uniform lower bound on M_n is used in the external limit.

7.3 Strong convergence of all fluxes

Set $F_n = w_n p_n$ and $F = wp$ on $D_\delta = (\delta,T)\times\mathbb{R}^2$. Convergences (7.4)–(7.6) give $F_n \to F$ almost everywhere. Calibration gives $\int F_n(t) = \int F(t) = 1$ for almost every time. Dominated convergence of $\min(F_n, F)$ against F yields

$$\|F_n - F\|_1 = 2(T-\delta) - 2\int_{D_\delta} \min(F_n, F) \longrightarrow 0.$$

This mass-conservation argument applies on any interval $J \subset (0,T)$, including $J = (0,T)$, once almost-everywhere convergence of the ratio is established. It requires no additional moment of the calibrated flux. Moreover,

$$\|\sqrt{F_n}\sqrt{p_n} - \sqrt{F}\sqrt{p}\|_1 \le \|F_n - F\|_1^{1/2}\|p_n\|_1^{1/2} + \|F\|_1^{1/2}\|p_n - p\|_1^{1/2} \longrightarrow 0,$$

by the Cauchy–Schwarz inequality and $|\sqrt{u} - \sqrt{v}|^2 \le |u - v|$. Thus

$$w_n p_n \to wp, \qquad \sqrt{w_n}p_n \to \sqrt{w}\,p \quad \text{in } L^1(D_\delta). \tag{7.7}$$

Lemma 7.1, with a moment of b_n of order strictly greater than one, gives separately

$$b_n p_n \to bp \quad \text{in } L^1(D_\delta). \tag{7.8}$$

For uniformly bounded c_n converging almost everywhere to c, and $H_n \to H$ in L^1, dominated convergence gives

$$\|c_n H_n - cH\|_1 \le C\|H_n - H\|_1 + \int |c_n - c||H| \longrightarrow 0. \tag{7.9}$$

With $G_n = \eta_n^2 p_n \to G = \eta^2 p$ in L^1, we also have

$$\|\sqrt{G_n F_n} - \sqrt{GF}\|_1 \le \|G_n - G\|_1^{1/2}\|F_n\|_1^{1/2} + \|G\|_1^{1/2}\|F_n - F\|_1^{1/2} \longrightarrow 0.$$

This last argument remains valid when η_n is not uniformly bounded, provided convergence of G_n has been established. The bounded multipliers $a_n, \rho_n\sqrt{a_n}$ then give

$$\begin{aligned} a_n w_n p_n \to awp, \qquad \eta_n^2 p_n \to \eta^2 p, \\ \rho_n\eta_n\sqrt{a_n w_n}\,p_n \to \rho\eta\sqrt{aw}\,p \quad \text{in } L^1(D_\delta). \end{aligned} \tag{7.10}$$

The limiting calibration $\int wp\,dy = q$ and the moments of p give the limiting version of (6.7). The flux integrals over $(0,\delta)$ are $O(\delta)$ uniformly in n. The preceding convergences therefore hold on $(0,T)\times\mathbb{R}^2$.

We can now pass to the limit in the periodic weak formulation against any test function in $C_c^\infty([0,T)\times\mathbb{R}^2)$, which is admissible for sufficiently large n. The joint initial approximations converge narrowly to μ_0 by Section 3.3. We obtain

$$\frac{\partial p}{\partial t} = \tfrac{1}{2}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial}{\partial x}\right)(awp) + \frac{\partial^2}{\partial x\partial y}(\rho\eta\sqrt{aw}\,p) + \tfrac{1}{2}\frac{\partial^2}{\partial y^2}(\eta^2 p) - \frac{\partial}{\partial y}(bp), \tag{7.11}$$

with this weak initial trace. For every $\psi \in C_c^\infty([0,T)\times\mathbb{R}^2)$, its weak formulation is

$$\begin{aligned} -\int_0^T \iint p\frac{\partial\psi}{\partial t} - \int \psi(0,x,y)\,\mu_0(\,\mathrm{d}x\,\mathrm{d}y) \\ = \int_0^T \iint p\Big[\tfrac{1}{2}aw\left(\frac{\partial^2\psi}{\partial x^2} - \frac{\partial\psi}{\partial x}\right) + \rho\eta\sqrt{aw}\,\frac{\partial^2\psi}{\partial x\partial y} \\ + \tfrac{1}{2}\eta^2\frac{\partial^2\psi}{\partial y^2} + b\frac{\partial\psi}{\partial y}\Big]. \end{aligned} \tag{7.12}$$

7.4 Regularity of the same limiting density

Fix $\delta > 0$ and $1 < r < 2$. The inequality $|A^{1/r} - B^{1/r}|^r \le |A - B|$ for $A, B \ge 0$, together with (7.2), gives $p_n^{1/r} \to p^{1/r}$ in $L^r((\delta,T)\times\mathbb{R}^2)$.

On every compact spatial set and for sufficiently large n, bound (6.15) controls the derivatives of these roots without reaching the boundary of the torus. Reflexivity of L^r and distributional testing identify their weak limits as the derivatives of $p^{1/r}$. Lower semicontinuity on each compact set, followed by exhaustion of $\mathbb{R}^2$, gives

$$\int_\delta^T \iint_{\mathbb{R}^2} |\nabla p^{1/r}|^r \le C_{r,\delta,T}.$$

Since $\int |p^{1/r}|^r = 1$ almost everywhere in time, we obtain

$$p^{1/r} \in L^r(\delta, T; W^{1,r}(\mathbb{R}^2)).$$

Truncating $h = p^{1/r}$ in the chain rule for h^r and applying Hölder's inequality gives

$$\nabla p = r h^{r-1}\nabla h \in L^1((\delta,T)\times\mathbb{R}^2).$$

This proves (2.12). For every other exponent r, the argument applies to the same strong limit p; there is no need to extract a common subsequence for an uncountable set of exponents.

7.5 A law at every time

The moment bounds (6.2) and (6.4) give uniform tightness, and (6.8) gives equicontinuity against every smooth compactly supported test function. For sufficiently large n, these tests are periodic by zero extension; no boundary join is differentiated. Extraction on a countable separating family, followed by control of the tails, gives a narrowly continuous curve μ and

$$\mu_t(dx\,dy) = p(t,x,y)dx\,dy \quad \text{for almost every } t, \qquad \mu|_{t=0} = \mu_0. \tag{7.13}$$

Strong convergence identifies the occupation density; narrow continuity extends the marginal identity to every time. Position moments pass to the limit at every time by lower semicontinuity; moments of Borel coefficients follow from their growth bounds. The weak formulation already obtained, with its trace, and the flux bounds give

$$\int_0^T \iint (aw + \eta^2 + |b| + |\rho|\eta\sqrt{aw})p < \infty. \tag{7.14}$$

□

8 Stochastic reconstruction

The following result gives the passage from the Fokker–Planck equation to the calibrated model. It will also be applied in the logarithmic coordinate of the positive factor.

Proposition 8.1 (Reconstruction from the fluxes)**.** *Let a satisfy (H1), and let b, η, v, ρ be finite Borel coefficients on the corresponding domains, with $\eta, v > 0$ and $|\rho| \le 1$. Let $(\mu_t)_{0\le t\le T}$ be a narrowly continuous curve of probability measures on $\mathbb{R}^2$, with trace μ_0 and first marginal $\mathcal{L}(Z_t)$, where Z is the scalar diffusion with initial law μ_0^X and $\int e^x \, \mu_0^X(\,\mathrm{d}x) < \infty$. Suppose that its occupation measure has a density p, that*

$$0 < \mathcal{V}(t,x) := \int v(t,x,y)p(t,x,y)\,dy < \infty \quad \text{for almost every } (t,x),$$
$$M = \mathcal{V}/q, \qquad w = v/M,$$

and that μ solves (7.11) *with*

$$\int_0^T \iint (aw + \eta^2 + |b| + |\rho|\eta\sqrt{aw})p < \infty. \tag{8.1}$$

Then μ is realised by a continuous weak solution of (2.11)*, with v in place of v, and $M(t,x) = \mathbb{E}[v(t,X_t,Y_t) \mid X_t = x]$ for $dt\,q(t,x)dx$-almost every (t,x). The process e^X is a true martingale. A normalisation $\mathsf{v} = e^{c_0(t)}v$ is restored by* (2.13)*.*

Proof. We apply superposition to the limiting equation and use the scalar marginal to prove the martingale property.

8.1 Superposition and Brownian motions

Choose Borel representatives of $p, q, \mathcal{V}$ and complete M by 1 on the exceptional set where $\mathcal{V}/q$ is not positive and finite. Tonelli's theorem makes $\mathcal{V}$ Borel measurable; this exceptional set is null for $dt\,q\,dx$, hence also for $dt\,\mu_t$. Set

$$\beta = \begin{pmatrix} -aw/2 \\ b \end{pmatrix}, \qquad A = \begin{pmatrix} aw & \rho\eta\sqrt{aw} \\ \rho\eta\sqrt{aw} & \eta^2 \end{pmatrix}. \tag{8.2}$$

The matrix is positive semidefinite, since $\xi^T A\xi = (\sqrt{aw}\xi_1 + \rho\eta\xi_2)^2 + (1-\rho^2)\eta^2\xi_2^2$, and $|\beta| + |A| \le C(aw + \eta^2 + |b| + |\rho|\eta\sqrt{aw})$. The curve therefore solves $\frac{\partial}{\partial t}\mu = \mathcal{L}^*\mu$ for

$$\mathcal{L}\phi = -\tfrac{1}{2}aw\frac{\partial\phi}{\partial x} + b\frac{\partial\phi}{\partial y} + \tfrac{1}{2}aw\frac{\partial^2\phi}{\partial x^2} + \rho\eta\sqrt{aw}\frac{\partial^2\phi}{\partial x\partial y} + \tfrac{1}{2}\eta^2\frac{\partial^2\phi}{\partial y^2},$$

with characteristics integrable against $dt\,\mu_t$. Trevisan's superposition principle [15, Definition 2.2 and Theorem 2.5] gives a probability measure on $C([0,T];\mathbb{R}^2)$ with marginals μ_t solving the martingale problem for $\mathcal{L}$.

For the canonical process, the compensated coordinates

$$N^x = X - X_0 + \tfrac{1}{2}\int_0^{\cdot} aw\,dt, \qquad N^y = Y - Y_0 - \int_0^{\cdot} b\,dt$$

are continuous local martingales, and

$$d\langle N^x\rangle = aw\,dt, \quad d\langle N^y\rangle = \eta^2 dt, \quad d\langle N^x, N^y\rangle = \rho\eta\sqrt{aw}\,dt.$$

The bracket identities follow by testing with x, y, x^2, y^2, xy, localised before exit from a ball and before the integral of the characteristics exceeds a threshold. Continuity and (8.1) allow these stopping times to be removed.

Define the local stochastic integrals

$$W = \int_0^{\cdot} (aw)^{-1/2}\, dN^x, \qquad B = \int_0^{\cdot} \eta^{-1}\, dN^y.$$

Positivity of the coefficients and the bracket identities justify these integrals without a uniform lower bound: their brackets equal t. Lévy's characterisation [12, Theorem 1.4] gives two Brownian motions with $d\langle W, B\rangle_t = \rho(t, X_t, Y_t)dt$. The stochastic equations follow from the definitions of N^x, N^y.

The disintegration $p = qu$ gives $M(t,x) = \mathbb{E}[v(t, X_t, Y_t) \mid X_t = x]$ in the stated sense. Compositions of Borel coefficients with continuous adapted processes are progressively measurable. By Tonelli's theorem, a $dt\,\mu_t$-null set is visited for zero Lebesgue time almost surely; the completion of M and modifications on exceptional sets therefore have no effect.

8.2 The martingale property of the price

Conditional on $Z_0 = z$, $e^{Z_t - z}$ is a stochastic exponential with bounded volatility. Stopping at level N, Itô's formula and Gronwall give

$$\mathbb{E}[e^{2(Z_{t\wedge\tau_N} - z)} \mid Z_0 = z] \le e^{a+t}.$$

Uniform integrability removes the stopping, giving $\mathbb{E}[e^{Z_t} \mid Z_0 = z] = e^z$. Integration against μ_0^X and equality of the scalar marginals yield

$$\mathbb{E}e^{X_t} = \mathbb{E}e^{Z_t} = \int e^z\, \mu_0^X(\,\mathrm{d}z).$$

Itô's formula for the reconstructed price equation shows that e^X is a positive local martingale, hence a supermartingale. For $s \le t$, the random variable

$$e^{X_s} - \mathbb{E}[e^{X_t} \mid \mathcal{F}_s]$$

is nonnegative and has expectation $\mathbb{E}e^{X_s} - \mathbb{E}e^{X_t} = 0$. It therefore vanishes almost surely, which is the martingale property.

□

8.3 Proof of Theorem 2.1

Proof of Theorem 2.1. Use the coefficient and joint initial-law approximation of Section 3. Normalisation (2.13) allows us to work with v. Under (H1)–(H5), Propositions 3.1, 3.2 and 3.4 provide the approximating coefficients and the target marginal. Proposition 5.1, using the identity of Proposition 4.2, constructs the periodic solutions. The moments and dissipation in Proposition 6.1 are independent of the period; Proposition 7.2 therefore gives a density p, its limiting fluxes, and a continuous curve μ with marginal $\mathcal{L}(Z_t)$ and initial trace μ_0. The conditional moment is positive and finite by (7.5), and (7.14) verifies (8.1). Proposition 8.1 gives the paths and calibration. The regularity is that obtained in Section 7.4 for the same density p. Finally, (2.13) restores v. □

9 Entropy on the whole space

We retain the whole-space entropy inequality for the solution already constructed, writing z for the factor coordinate. The price-dependent source is explicit, and the remaining constants are independent of the factor coefficients; this dependence is needed in Section 10.

Proposition 9.1 (Passage of the entropy inequality to the limit)**.** *Under (H1)–(H5), the solution of Theorem 2.1 can be chosen to have the following property. Writing z for the factor coordinate, set*

$$u = p/q, \quad M = \int vu\,dz, \quad w = v/M, \quad \widehat{p} = wp,$$

$$K = M\int v^{-1}u\,dz - 1, \qquad E = \iint p\log(p/q),$$

$$k = \rho\frac{\partial\eta}{\partial z} + \eta\frac{\partial\rho}{\partial z} + \tfrac{1}{2}\rho\eta\frac{\partial\ell}{\partial z}, \qquad d = b/\eta - \frac{\partial\eta}{\partial z},$$

$$H = \iint p(k^2+d^2), \qquad \mathcal{B} = \int qK^2\,dx.$$

Define also

$$L = \partial_x \log v, \qquad \bar{L} = \frac{\int vLp\,dz}{\int vp\,dz}, \qquad H_x = \iint awp(L-\bar{L})^2. \tag{9.1}$$

Conditional means are defined arbitrarily on the null exceptional set where their denominators are not positive and finite. Then, for almost every $t>0$, these quantities are finite, and

$$\mathcal{A}(p) = 2a_- \iint \left|\frac{\partial}{\partial x}\sqrt{wp}\right|^2 + 4\iint \eta^2\left|\frac{\partial}{\partial z}\sqrt{p}\right|^2 \tag{9.2}$$

is locally integrable in time. In $\mathcal{D}'(0,T)$,

$$E' + \frac{\alpha}{8}\mathcal{A}(p) \le \frac{2}{\alpha}H + C_\alpha H_x + C_{a,\alpha,T}(1+t^{-1})(1+\sqrt{\mathcal{B}}). \tag{9.3}$$

One may take $C_\alpha = 1/(2\alpha)+\alpha/4$. The constant $C_{a,\alpha,T}$ depends only on a_-, a_+, K_a, α, T and the fixed scalar cutoffs. It is independent of the factor coefficients. More generally, for every $h>1$, set $\mathcal{B}_h = \int qK^h\,dx$. The same solution satisfies

$$E' + \frac{\alpha}{8}\mathcal{A}(p) \le \frac{2}{\alpha}H + C_\alpha H_x + C_{a,\alpha,T,h}(1+t^{-1})(1+\mathcal{B}_h^{1/h}). \tag{9.4}$$

The choice $h=2$ recovers (9.3)*. If v is independent of x, then $H_x = 0$ and both inequalities hold with $\alpha/4$ in place of $\alpha/8$, for the same selected solution. On an open interval $I \Subset (0,T)$, one may use a bound $|\rho| \le 1-\alpha_I$ valid in a neighbourhood of I and replace α by α_I in this conclusion on I.*

Proof. Fix the coefficients. Use Proposition 3.4 and the periodic construction of Section 5. Their uniform estimates and Proposition 7.2 give strong convergence $p_n \to p$. For price-independent variance, choose the price-independent approximation branch, so that $L_n = 0$ throughout. On every $D_\delta = (\delta,T)\times\mathbb{R}^2$, the densities are extended by zero; no derivative is taken across the boundary of the torus.

We first derive the inequality satisfied by the approximations. Set $L_n = \partial_x \log V_n$ and $\bar{L}_n = \int V_n L_n p_n\,dz / \int V_n p_n\,dz$. With the centred dissipation D_n of Section 4,

$$D_n = \iint a_n|2\partial_x\sqrt{\widehat{p}_n} - (S_n + L_n - \bar{L}_n)\sqrt{\widehat{p}_n}|^2 + 4\iint \eta_n^2|\partial_z\sqrt{p_n}|^2.$$

Since $|A-B|^2 \ge |A|^2/2 - |B|^2$ and $\int \widehat{p}_n\,dz = q_n$, this gives

$$D_n \ge \mathcal{A}_n - 2a_+\int q_n S_n^2 - 2H_{x,n}. \tag{9.5}$$

Moreover,

$$H_{x,n} \le a_+\iint w_n p_n L_n^2 \le a_+\left(\iint w_n^2 p_n\right)^{1/2}\left(\iint |L_n|^4 p_n\right)^{1/2} \le C_T.$$

The last bound follows from the conditional-weight and Gaussian factor moments. It is used for integrability, whereas the source H_x is kept explicit in the asserted inequality. For $h > 1$ and $h' = h/(h-1)$, Proposition 3.5 yields

$$J_n \le C \left(\int q_n K_n^h \right)^{1/h} \left(\int q_n (1 + S_n^2)^{h'} \right)^{1/h'} \le C_{a,h} (1 + t^{-1}) \mathcal{B}_{n,h}^{1/h}.$$

With $h = 2$, the periodic inequality $E_n' + \alpha D_n/8 \le 2H_n/\alpha + J_n/(2\alpha) + H_{x,n}/(2\alpha)$ implies

$$E_n' + \frac{\alpha}{8} \mathcal{A}_n \le \frac{2}{\alpha} H_n + C_\alpha H_{x,n} + C_{a,\alpha,T} (1 + t^{-1})(1 + \sqrt{\mathcal{B}_n}). \tag{9.6}$$

The displayed constant is independent of the factor coefficients.

We next prove strong convergence of the entropy. Bound (6.16) with $r = 3/2$ and the Sobolev inequality on tori of side length $L_n \ge 1$,

$$\|f\|_2 \le C(\|\nabla f\|_1 + L_n^{-1} \|f\|_1),$$

give, by interpolation and $\|p_n(t)\|_1 = 1$,

$$\int_{D_\delta} p_n^{7/4} \le \int_\delta^T \|p_n(t)\|_2^{3/2} dt \le C_\delta. \tag{9.7}$$

For $m = 7/4$, $0 < d < m - 1$ and any set A,

$$\int_A p_n \log^+ p_n \le C_d \left(\int_A p_n \right)^{1 - d/(m-1)} \left(\int_A p_n^m \right)^{d/(m-1)}. \tag{9.8}$$

Strong convergence of the densities therefore controls small sets and the tails of the positive part. For the negative part, set $W = 1 + x^2 + z^2$ and use $-s \log s \le W s + e^{-W-1}$ for $0 < s \le 1$. Fourth-order position moments control the tails of $W p_n$, and $s \log^- s \le 1/e$ on compact sets. Vitali's theorem gives

$$p_n \log p_n \longrightarrow p \log p \quad \text{in } L^1(D_\delta). \tag{9.9}$$

Similarly, $q_n \to q$ strongly, and the score bound gives $\|q_n(t)\|_\infty \le L_n^{-1} + \int |\partial q_n/\partial x| \le C_\delta$. The scalar moments control the negative part. Hence $q_n \log q_n \to q \log q$ in L^1 and

$$E_n \longrightarrow E \quad \text{in } L^1(\delta, T). \tag{9.10}$$

This strong convergence justifies passage to the limit in $-\int E_n \xi'$.

We now identify the limiting sources and energy. The smoothed coefficients and their required spatial derivatives converge almost everywhere along a subsequence. Their growth bounds and (6.5) allow us to apply Lemma 7.1 to $k_n^2 + d_n^2$, giving

$$H_n \longrightarrow H \quad \text{in } L^1(\delta, T). \tag{9.11}$$

Applied to $V_n^{\pm 1}$, this lemma also gives convergence of the two moments defining K_n, hence $K_n \to K$ almost everywhere on $\{q > 0\}$. For every $h > 1$, conditional moments of order $h + \zeta$, $\zeta > 0$, are available for fixed coefficients. The same lemma, with density q_n and weight K_n^h, implies

$$q_n K_n^h \to q K^h \text{ in } L^1(dt\, dx), \qquad \mathcal{B}_{n,h} \to \mathcal{B}_h \text{ in } L^1(dt). \tag{9.12}$$

The inequality $|u^{1/h} - v^{1/h}|^h \le |u - v|$ gives convergence of the source term $\mathcal{B}_{n,h}^{1/h}$ in $L^1(\delta, T)$.

It remains to identify the additional source caused by price dependence. The convolution construction gives $L_n \to L$ locally almost everywhere after extraction, with common $\mathscr{G}_0$ bounds.

Write $F_n = w_n p_n$ and $F = wp$; calibrated-flux convergence gives $F_n \to F$ in $L^1(D_\delta)$. For every finite $r > 1$,

$$\int_{D_\delta} F_n |L_n|^r \le \left(\int_{D_\delta} w_n^2 p_n \right)^{1/2} \left(\int_{D_\delta} |L_n|^{2r} p_n \right)^{1/2} \le C_{\delta,r}.$$

Lemma 7.1, applied to the strongly convergent nonnegative densities F_n and the weights $a_n L_n^2$, therefore gives $a_n F_n L_n^2 \to a F L^2$ in L^1. The same lemma applied under p_n to $V_n L_n$ and V_n gives convergence of their z-integrals. Hence $\bar{L}_n \to \bar{L}$ almost everywhere in (t, x), since $q > 0$ and $\int vp \, dz > 0$ almost everywhere. Conditional Jensen under F_n/q_n yields, for every $r > 1$,

$$\int q_n |\bar{L}_n|^{2r} \, dt \, dx \le \int F_n |L_n|^{2r} \, dt \, dx \le C_{\delta,r}.$$

Applying the weighted-products lemma under $q_n \to q$ now gives $a_n q_n \bar{L}_n^2 \to a q \bar{L}^2$ in L^1. The conditional-variance identity

$$H_{x,n} = \iint a_n F_n L_n^2 - \int a_n q_n \bar{L}_n^2$$

and its limiting version show that

$$H_{x,n} \longrightarrow H_x \quad \text{in } L^1(\delta, T). \tag{9.13}$$

All quantities involving weak derivatives are understood almost everywhere; no convergence of a differentiated conditional moment is required.

Finally, (7.7) gives $\sqrt{\widehat{p_n}} \to \sqrt{wp}$ and $\sqrt{p_n} \to \sqrt{p}$ in L^2. The energy bound identifies $\partial\sqrt{wp}/\partial x$ weakly. On compact sets, the growth bounds on η_n^{-1} give a lower bound on η_n; almost-everywhere convergence of the coefficients also identifies the limit of $\eta_n \partial\sqrt{p_n}/\partial z$. For every time test function $\xi \ge 0$, lower semicontinuity followed by spatial exhaustion gives

$$\int \xi \mathcal{A}(p) \le \liminf_n \int \xi \mathcal{A}_n.$$

Together with (9.10), (9.11), (9.12) and (9.13), this passes (9.6) and its version for $h > 1$ to the limit. Extractions may be made for rational exponents; for arbitrary $h > 1$, choose rational $1 < h_0 < h$ and use $\mathcal{B}_{h_0}^{1/h_0} \le \mathcal{B}_h^{1/h}$ under $q \, dx$. When v is independent of x, use the sharper periodic inequality with coefficient $\alpha/4$ and the sharper bound $D_n \ge \mathcal{A}_n - a_+ \int q_n S_n^2$. The identical limit argument gives the stronger asserted coefficient. The same solution therefore satisfies all the assertions. If $|\rho| \le 1 - \alpha_I$ in a neighbourhood of I, sufficiently small kernels preserve this bound on the support of a test function in I. The same limit argument gives the local version.

□

Lemma 9.2 (Coercivity and positive-time entropy bounds on the line)**.** *Let $p = qu$ be a density of mass one with marginal q, and let $\eta > 0$. Suppose that the entropy is finite and*

$$Q = \iint \eta^{-2} p < \infty, \qquad A_y = 4 \iint \eta^2 \left| \frac{\partial}{\partial z} \sqrt{p} \right|^2 < \infty.$$

Then $e^{2E} \le Q A_y$. For a family of such densities, if $Q \le Q_*$, $E \ge -C_*$ and, in distributions, $E' + cA \le G$ with $A \ge A_y$, $c > 0$ and $G \ge 0$ locally integrable, then, for almost every $0 < s < t < T$,

$$E(t) \le \int_s^t G(u) \, du - \tfrac{1}{2} \log\bigl(2c(t-s)/Q_*\bigr). \tag{9.14}$$

In particular, $G(t) \le C(1 + t^{-1})$ implies $\operatorname{ess\,sup}_{[\delta,T]} E + \int_\delta^T A \le C_\delta$.

Proof. For $q\,\mathrm{d}x$-almost every x, set $Q_x = \int \eta^{-2} u\,\mathrm{d}z$ and $A_x = 4\int \eta^2 |\partial_z \sqrt{u}|^2\,\mathrm{d}z$. The Sobolev chain rule and Cauchy–Schwarz give $\|\partial_z u\|_1 \le \sqrt{Q_x A_x}$. Since $u \in W^{1,1}(\mathbb{R}) \cap L^1(\mathbb{R})$ and $\int u = 1$, $e^{\int u \log u} \le \|u\|_\infty \le \sqrt{Q_x A_x}$. Jensen under $q\,\mathrm{d}x$, then Cauchy–Schwarz, yields

$$e^E \le \int q e^{\int u \log u}\,\mathrm{d}x \le \int q\sqrt{Q_x A_x}\,\mathrm{d}x \le \sqrt{Q A_y}.$$

For the family in the second assertion, apply the comparison in Section 6.4 with $\lambda = c/Q_*$ and no threshold. It gives (9.14) for the right-continuous representative. Choosing $s = t/2$ when $G \le C(1 + t^{-1})$ bounds E on $[\delta, T]$; integration of the entropy inequality, using $E \ge -C_*$, bounds $\int_\delta^T A$. □

Lemma 9.3 (Score of the limiting marginal)**.** *For the scalar density q in* (2.10)*, the weak score $S = \frac{\partial q}{\partial x}/q$ satisfies, for almost every $t > 0$,*

$$\int q(t,x)|S(t,x)|^4\,dx \le C_a(1 + t^{-2}).$$

More generally, for every $p \ge 2$,

$$\int q(t,x)|S(t,x)|^p\,dx \le C_{a,p}(1 + t^{-p/2}) \quad \text{for almost every } t > 0. \tag{9.15}$$

Proof. For $p \ge 2$, convergence $q_n \to q$ in L^1 implies convergence of $q_n^{1/p}$ in L^p. By Proposition 3.5,

$$p^p \int \left|\frac{\partial q_n^{1/p}}{\partial x}\right|^p = \int q_n |S_n|^p \le C_{a,p}(1 + t^{-p/2}).$$

Weak convergence of the derivatives, lower semicontinuity tested against a nonnegative time function, and spatial exhaustion pass this bound to the limit. The chain rule $\partial q/\partial x = pq^{1-1/p}\partial q^{1/p}/\partial x$ and $q > 0$ almost everywhere identify the score. Exponents below two follow by Hölder's inequality. This argument can be made for each fixed exponent without changing the limiting density q. A common full-measure set is obtained by retaining all integer exponents and, for real $p \ge 2$, using $k = \lceil p \rceil$ and $\int q|S|^p \le (\int q|S|^k)^{p/k} \le C_{a,p}(1 + t^{-p/2})$. □

10 Proof for positive factors

We first take a joint initial law with compact support in $\mathbb{R} \times (0, \infty)$; point masses are included. Its image under $(x, y) \mapsto (x, \log y)$ satisfies (2.8). The final step removes compact support under (2.22). We work with the normalised variance v. In the coordinate $z = \log y$, replace the diffusion coefficient $s(t, e^z)e^{-z/2}$ by $s(t, e^z)/\sqrt{e^z + \varepsilon}$. Theorem 2.1 constructs the approximations, and Proposition 9.1 supplies their entropy inequality. It remains to establish the uniform moment and gradient estimates needed to let ε tend to zero.

10.1 Approximations and uniform moments

Proposition 10.1 (Construction of the approximations and uniform moments)**.** *Under the assumptions of Theorem 2.2, for $0 < \varepsilon \le 1$ there exists a calibrated weak solution with factor $Y^\varepsilon > 0$ satisfying*

$$dY_t^\varepsilon = \mu(t, Y_t^\varepsilon)dt + h_\varepsilon(t, Y_t^\varepsilon)dB_t, \qquad h_\varepsilon(t,y) = \frac{s(t,y)y}{\sqrt{y + \varepsilon}}. \tag{10.1}$$

The price uses exactly $v(t, Y^\varepsilon)/\mathbb{E}[v(t, Y^\varepsilon) \mid X^\varepsilon]$, and the correlation is $\rho(t, X^\varepsilon, Y^\varepsilon)$. Its marginal is the same q for every ε. Its density in (x, z), denoted by P_ε, satisfies the price-independent case of Proposition 9.1, with $H_x = 0$ and dissipation coefficient $\alpha/4$.

These same solutions satisfy the uniform bounds

$$\sup_{\varepsilon\in(0,1]}\sup_{0\le t\le T}\mathbb{E}[(Y_t^\varepsilon)^m]<\infty\quad(m>0),\qquad \sup_{\varepsilon\in(0,1]}\sup_{0\le t\le T}\mathbb{E}[(Y_t^\varepsilon)^{-p}]<\infty\quad(0<p<\nu).\tag{10.2}$$

This is a supremum of expectations; no expectation of the time supremum of inverse powers is asserted. Moreover, for every $0<\theta<\min\{1,\nu-1\}$,

$$\varepsilon\int_0^T\mathbb{E}[(Y_t^\varepsilon)^{-2}]\,dt\le C_\theta\varepsilon^\theta.\tag{10.3}$$

This estimate does not assert that inverse moments of order two are bounded uniformly in ε *and time.*

Proof. We first construct the approximations in logarithmic coordinates and then derive uniform moment bounds.

10.1.1 Construction without regularising the ratio

Fix ε before taking any internal limit, and set

$$\begin{aligned}\eta_\varepsilon(t,z)&=\frac{s(t,e^z)}{\sqrt{e^z+\varepsilon}},\\ b_\varepsilon(t,z)&=e^{-z}\mu(t,e^z)-\frac{s(t,e^z)^2}{2(e^z+\varepsilon)},\\ V(t,z)&=v(t,e^z),\qquad \widehat\rho(t,x,z)=\rho(t,x,e^z).\end{aligned}\tag{10.4}$$

We have $0<\eta_\varepsilon\le s_+/\sqrt{\varepsilon}$ and, with $y=e^z$,

$$\frac{\partial}{\partial z}\eta_\varepsilon=\frac{y\frac{\partial}{\partial y}s}{\sqrt{y+\varepsilon}}-\frac{sy}{2(y+\varepsilon)^{3/2}}.\tag{10.5}$$

The functions b_ε, η_ε^{-1} and $\frac{\partial}{\partial z}\eta_\varepsilon$ grow at most exponentially in $|z|$ for fixed ε, and hence belong to $\mathscr{G}_0$. By (2.17),

$$|\log V(t,z)|\le C+(m_v+m_-+\gamma+\omega)|z|,\qquad \left|\frac{\partial}{\partial z}\log V\right|+\left|\frac{\partial}{\partial z}\widehat\rho\right|\le 2L(1+e^{dz}).$$

Thus (H4) and (H5) hold, without time derivatives.

To check (H3), note that $\nu>g_*\ge 1$ implies $A>s_+^2/2$. If $z<0$, then

$$b_\varepsilon(t,z)\ge(A-s_+^2/2)e^{-z}-B\ge -B,\qquad zb_\varepsilon(t,z)\le B|z|.$$

If $z\ge 0$, the upper bound on μ gives $b_\varepsilon(t,z)\le C_\mu(1+e^{-z})\le 2C_\mu$, hence $zb_\varepsilon\le 2C_\mu z$. In either case, $zb_\varepsilon\le C(1+z^2)$. All the assumptions of Theorem 2.1 are satisfied.

That theorem, with the selection in Proposition 9.1, gives $(X^\varepsilon,Z^\varepsilon)$ and P_ε. Itô's formula applied to $Y^\varepsilon=e^{Z^\varepsilon}$ gives

$$Y^\varepsilon(b_\varepsilon+\eta_\varepsilon^2/2)=\mu(t,Y^\varepsilon),\qquad Y^\varepsilon\eta_\varepsilon=h_\varepsilon(t,Y^\varepsilon).$$

The correction terms cancel exactly, yielding (10.1). The ratio and correlation are those stated, by the definitions of V and $\widehat\rho$.

10.1.2 Uniform moments by comparison with a CIR generator

We first estimate the positive moments of the factor. The generator of the approximating factor is

$$L_{\varepsilon,t}f(y) = \mu(t,y)f'(y) + \frac{s(t,y)^2y^2}{2(y+\varepsilon)}f''(y).$$

For an integer $m \geq 2$, (2.15)–(2.16) give

$$\begin{aligned} L_{\varepsilon,t}y^m &= m\mu(t,y)y^{m-1} + \tfrac{1}{2}m(m-1)s(t,y)^2\frac{y^m}{y+\varepsilon} \\ &\leq mC_\mu(y^{m-1}+y^m) + \tfrac{1}{2}m(m-1)s_+^2y^{m-1} \leq C_m(1+y^m). \end{aligned}$$

Itô's formula stopped on $[1/N, N]$, Gronwall's lemma and Fatou's lemma give the first bound. The preceding construction already ensures that the processes are positive and nonexplosive, so the stopping times exhaust the time horizon. The other powers follow from $y^m \leq 1 + y^k$ for an integer $k \geq \max\{2, m\}$.

For the inverse moments, we introduce a comparison function. Define the auxiliary operator $L_0 = (A - By)\frac{\partial}{\partial y} + (s_+^2 y/2)\frac{\partial^2}{\partial y^2}$ and, for $\lambda \geq 0$,

$$\Phi(t,y;\lambda) = (1+c_t\lambda)^{-\nu}\exp\left(-\frac{\lambda e^{-Bt}}{1+c_t\lambda}y\right), \qquad c_t = \frac{s_+^2}{2}\int_0^t e^{-Bu}\,du. \tag{10.6}$$

This formula includes $B = 0$, in which case $c_t = s_+^2 t/2$. To check the equation, write $\Phi = C(t)e^{-d(t)y}$, where

$$C(t) = (1+c_t\lambda)^{-\nu}, \qquad d(t) = \frac{\lambda e^{-Bt}}{1+c_t\lambda}.$$

Since $c_t' = s_+^2 e^{-Bt}/2$ and $\nu = 2A/s_+^2$, differentiation gives

$$\frac{C'}{C} = -Ad, \qquad d' = -Bd - \frac{s_+^2}{2}d^2.$$

Consequently

$$\frac{\partial_t\Phi}{\Phi} = -Ad + \left(Bd + \frac{s_+^2}{2}d^2\right)y = \frac{L_0\Phi}{\Phi}.$$

Also, $C(0) = 1$, $d(0) = \lambda$, $\partial_y\Phi = -d\Phi$ and $\partial_y^2\Phi = d^2\Phi$. These formulas prove the initial condition, $0 \leq \Phi \leq 1$, and the two derivative signs. These are the only properties of the CIR transform used in the comparison.

We compare the two generators using the derivative signs just established. At every time u, any twice differentiable function satisfies

$$(L_{\varepsilon,u} - L_0)f = [\mu(u,y) - (A-By)]f' + \frac{1}{2}\left[\frac{s(u,y)^2y^2}{y+\varepsilon} - s_+^2 y\right]f''. \tag{10.7}$$

The first bracket is nonnegative by (2.15); the second is nonpositive since $s(u,y)^2y^2/(y+\varepsilon) \leq s_+^2 y$. Applied to $f(y) = \Phi(t-u, y;\lambda)$, they give

$$(L_{\varepsilon,u} - L_0)\Phi(t-u,\cdot;\lambda) \leq 0.$$

Itô's formula stopped on $[1/N, N]$, followed by dominated convergence ($0 \leq \Phi \leq 1$ and the approximating paths are positive and nonexplosive), gives

$$\mathbb{E}e^{-\lambda Y_t^\varepsilon} \leq \mathbb{E}\Phi(t, Y_0;\lambda). \tag{10.8}$$

This compares expectations through the generators, without a pathwise coupling with a CIR process or a uniqueness assumption.

We now integrate the Laplace-transform bound to obtain inverse moments. The Gamma identity

$$y^{-p} = \frac{1}{\Gamma(p)} \int_0^\infty \lambda^{p-1} e^{-\lambda y}\, d\lambda \qquad (p > 0)$$

and Tonelli's theorem give

$$\mathbb{E}(Y_t^\varepsilon)^{-p} \le \frac{1}{\Gamma(p)} \int_0^\infty \lambda^{p-1} \mathbb{E}\Phi(t, Y_0; \lambda)\, d\lambda.$$

For each fixed initial value $y_0 > 0$ and $t > 0$, set $d_t = y_0 e^{-Bt}$ and $u = c_t\lambda/(1 + c_t\lambda)$. Then $\lambda = u/[c_t(1-u)]$, $d\lambda = du/[c_t(1-u)^2]$, and the right-hand side becomes

$$\frac{c_t^{-p}}{\Gamma(p)} \int_0^1 u^{p-1}(1-u)^{\nu-p-1} e^{-d_t u/c_t}\, du. \tag{10.9}$$

If $p < \nu$, then on $0 < u < 1/2$ the factor $(1-u)^{\nu-p-1}$ is bounded, while $d_t \ge d_* = y_0 e^{-BT} > 0$. The contribution is therefore bounded by

$$C c_t^{-p} \int_0^\infty u^{p-1} e^{-d_* u/c_t}\, du = C\Gamma(p) d_*^{-p}.$$

On $1/2 < u < 1$, integrability of $(1-u)^{\nu-p-1}$ gives a bound $C c_t^{-p} e^{-d_*/(2c_t)}$, uniform for $0 < c_t \le c_T$. Both parts are bounded by $C_{p,T} y_0^{-p}$: for the second use $c_t^{-p} e^{-y_0 e^{-BT}/(2c_t)} \le C_{p,T} y_0^{-p}$. Taking expectation over Y_0 gives $\mathbb{E}(Y_t^\varepsilon)^{-p} \le C_{p,T} \mathbb{E} Y_0^{-p}$, also at $t = 0$. This proves the second bound in (10.2).

We next estimate the regularisation defect integrated in time. Fix $0 < \theta < \min\{1, \nu - 1\}$. Such a choice is possible since $\nu > 1$, and the inverse moment of order $1 + \theta < \nu$ has just been established. For $f_\varepsilon(y) = \varepsilon/y$, the generator satisfies

$$\begin{aligned} L_{\varepsilon,t} f_\varepsilon(y) &= -\frac{\varepsilon \mu(t,y)}{y^2} + \frac{s(t,y)^2 \varepsilon}{y(y+\varepsilon)} \\ &\le -A\varepsilon y^{-2} + B\varepsilon y^{-1} + s_+^2 \frac{\varepsilon}{y(y+\varepsilon)} \\ &\le -A\varepsilon y^{-2} + B\varepsilon y^{-1} + s_+^2 \varepsilon^\theta y^{-1-\theta}. \end{aligned}$$

The last line follows from

$$\frac{\varepsilon}{y+\varepsilon} \le \min\{1, \varepsilon/y\} \le (\varepsilon/y)^\theta.$$

Apply Itô's formula up to $T \wedge \tau_N$, where τ_N is the first exit time from $[1/N, N]$, taking N large enough to contain the initial factor support. The stopped martingale is integrable. Since $f_\varepsilon \ge 0$, its terminal expectation may be dropped:

$$\begin{aligned} A\varepsilon \mathbb{E} \int_0^{T\wedge\tau_N} (Y_t^\varepsilon)^{-2} dt &\le \varepsilon \mathbb{E} Y_0^{-1} + B\varepsilon \int_0^T \mathbb{E}(Y_t^\varepsilon)^{-1} dt + s_+^2 \varepsilon^\theta \int_0^T \mathbb{E}(Y_t^\varepsilon)^{-1-\theta} dt \\ &\le C_\theta(\varepsilon + \varepsilon^\theta) \le 2C_\theta \varepsilon^\theta. \end{aligned}$$

Both moments on the right are uniform by (10.2). The positivity and nonexplosion already established imply $T \wedge \tau_N \uparrow T$ almost surely. Monotone convergence proves (10.3). In particular, the defect vanishes in L^1 in time.

The positive and inverse moments also control the logarithmic coordinate and the variance. For $0 < \zeta < \nu$, the inequality $e^{\zeta|\log y|} \le y^\zeta + y^{-\zeta}$ gives

$$\sup_{\varepsilon, t \le T} \mathbb{E} e^{\zeta |Z_t^\varepsilon|} < \infty, \qquad Z^\varepsilon = \log Y^\varepsilon. \tag{10.10}$$

All polynomial moments of $|Z^\varepsilon|$ follow. By (2.17), for every $h > 0$,

$$v(t,y)^h \le C_h(1 + y^{m_v h} + y^{-\omega h}), \qquad v(t,y)^{-h} \le C_h(1 + y^{-\gamma h} + y^{m_- h}).$$

Thus

$$\sup_{\varepsilon,t} \mathbb{E} v(t, Y_t^\varepsilon)^h < \infty \quad (\omega h < \nu), \qquad \sup_{\varepsilon,t} \mathbb{E} v(t, Y_t^\varepsilon)^{-h} < \infty \quad (\gamma h < \nu). \tag{10.11}$$

When $\omega = 0$, the first assertion holds for all $h > 0$; when $\gamma = 0$, the second does so. These bounds follow from the construction; they are not additional assumptions on a law yet to be constructed. □

10.2 Uniform entropy and gradient estimates

Proposition 10.2 (Uniform entropy, dissipation and gradient estimates)**.** *For the solutions of Proposition 10.1, set*

$$u_\varepsilon = P_\varepsilon/q, \quad M_\varepsilon = \int V u_\varepsilon \, dz, \quad w_\varepsilon = V/M_\varepsilon, \quad K_\varepsilon = M_\varepsilon \int V^{-1} u_\varepsilon \, dz - 1,$$

where $V(t,z) = v(t,e^z)$. The entropy $E_\varepsilon = \int P_\varepsilon \log(P_\varepsilon/q)$ and the energy $\mathcal{A}_\varepsilon$ in (9.2) satisfy

$$\begin{aligned} &E_\varepsilon' + \frac{\alpha}{4}\mathcal{A}_\varepsilon \le C(1 + t^{-1}) + C\mathcal{R}_\varepsilon(t), \qquad \alpha = 1 - \rho_* > 0, \\ &\mathcal{R}_\varepsilon(t) = \varepsilon \mathbb{E}[(Y_t^\varepsilon)^{-2}], \qquad \int_0^T \mathcal{R}_\varepsilon(t) dt \le C_\theta \varepsilon^\theta \quad (0 < \theta < \min\{1, \nu - 1\}). \end{aligned} \tag{10.12}$$

and, for every $0 < \delta < T$,

$$\operatorname*{ess\,sup}_{t \in [\delta, T]} E_\varepsilon(t) + \int_\delta^T \mathcal{A}_\varepsilon(t) \, dt \le C_\delta. \tag{10.13}$$

The constants are independent of ε.

For every $\delta > 0$ and $1 < r < r_{\nu,\gamma,\omega}$,

$$\sup_{\varepsilon \in (0,1]} \left\{ \int_\delta^T \iint |\nabla P_\varepsilon^{1/r}|^r + \int_\delta^T \|P_\varepsilon(t)\|_{W^{1,1}}^r \, dt \right\} < \infty. \tag{10.14}$$

Proof. We compute the entropy sources and derive the weighted estimates needed for compactness.

10.2.1 Exact computation of the sources and positive-time entropy bounds

We first compute each source in the logarithmic coordinate. Write $y = e^z$, $s = s(t,y)$, $\mu = \mu(t,y)$, $\rho = \rho(t,x,y)$ and $\ell(t,z) = \log V(t,z)$. The derivatives in z are $\frac{\partial \ell}{\partial z} = y\frac{\partial}{\partial y}\log v$ and $\frac{\partial \widehat{\rho}}{\partial z} = y\frac{\partial}{\partial y}\rho$. By (10.4)–(10.5), the two sources in the entropy inequality are exactly

$$\begin{aligned} k_\varepsilon &= \widehat{\rho}\,\frac{\partial}{\partial z}\eta_\varepsilon + \eta_\varepsilon \frac{\partial}{\partial z}\widehat{\rho} + \tfrac{1}{2}\widehat{\rho}\eta_\varepsilon \frac{\partial \ell}{\partial z} \\ &= \frac{\rho y \frac{\partial s}{\partial y} + s y \frac{\partial \rho}{\partial y} + \frac{1}{2}\rho s\left(\frac{\partial \ell}{\partial z} - 1\right)}{\sqrt{y+\varepsilon}} + \frac{\rho s \varepsilon}{2(y+\varepsilon)^{3/2}}, \\ d_\varepsilon &= b_\varepsilon/\eta_\varepsilon - \frac{\partial}{\partial z}\eta_\varepsilon = \frac{\mu\sqrt{y+\varepsilon}}{sy} - \frac{y\frac{\partial s}{\partial y}}{\sqrt{y+\varepsilon}} - \frac{s\varepsilon}{2(y+\varepsilon)^{3/2}}. \end{aligned} \tag{10.15}$$

For the last identity, the two terms involving s combine as

$$-\frac{s}{2\sqrt{y+\varepsilon}} + \frac{sy}{2(y+\varepsilon)^{3/2}} = -\frac{s\varepsilon}{2(y+\varepsilon)^{3/2}}.$$

The numerator of the first fraction in k_ε is bounded in absolute value by $C(1+y^d)$, by (P2)–(P4). For $0 < \varepsilon \le 1$,

$$\begin{aligned}
\frac{(1+y^d)^2}{y+\varepsilon} &\le C(y^{-1}+y^{2d-1}) \le C(1+y^{-1}+y^{2d}),\\
\frac{\varepsilon^2}{(y+\varepsilon)^3} &\le y^{-1},\\
\frac{\mu^2(y+\varepsilon)}{s^2y^2} &\le C\frac{(1+y)^{2+2\lambda}(y+\varepsilon)}{y^2} \le C(1+y^{1+2\lambda}+y^{-1}+\varepsilon y^{-2}).
\end{aligned}$$

It follows that

$$k_\varepsilon^2 + d_\varepsilon^2 \le C(1+y^{1+2\lambda}+y^{2d}+y^{-1}+\varepsilon y^{-2}). \tag{10.16}$$

uniformly. The inverse moment of order one and all positive moments are available in (10.2), since $\nu > 1$. The last term is handled by (10.3). Consequently,

$$\begin{aligned}
H_\varepsilon(t) \le C(1+\mathcal{R}_\varepsilon(t)), \qquad \int_0^T H_\varepsilon(t)dt \le C,\\
Q_\varepsilon = \iint \frac{e^z+\varepsilon}{s(t,e^z)^2}P_\varepsilon \le s_-^{-2}\mathbb{E}\big[(Y^\varepsilon+\varepsilon)(1+Y^\varepsilon)^{2\lambda}\big] \le Q_*.
\end{aligned} \tag{10.17}$$

The bound on Q_ε is uniform at each time; the bound on H_ε is integrated in time. This distinction is essential.

We next choose the exponents needed to control the conditional weights. For $h \ge 1$ and conjugate exponents $A_h, B_h > 1$, (4.7) gives

$$\int q(1+K_\varepsilon)^h, \quad \iint w_\varepsilon^{\pm h}P_\varepsilon \;\le\; \big(\mathbb{E}v(t,Y_t^\varepsilon)^{hA_h}\big)^{1/A_h}\big(\mathbb{E}v(t,Y_t^\varepsilon)^{-hB_h}\big)^{1/B_h}. \tag{10.18}$$

Bound (10.18) is Proposition 4.1 applied to $V(t,z) = v(t,e^z)$. If $h(\gamma+\omega) < \nu$, the interval $(\omega h/\nu, 1-\gamma h/\nu)$ is nonempty and contained in $[0,1]$. Choose explicitly

$$\vartheta_h = \frac{1}{2}\left(\frac{\omega h}{\nu}+1-\frac{\gamma h}{\nu}\right), \qquad A_h = 1/\vartheta_h, \qquad B_h = 1/(1-\vartheta_h).$$

Then $A_h, B_h > 1$, $A_h^{-1}+B_h^{-1} = 1$ and

$$\omega h A_h < \nu, \qquad \gamma h B_h < \nu.$$

Both factors are uniformly bounded by (10.11). Thus

$$\sup_{\varepsilon,t}\left\{\int q(1+K_\varepsilon)^h + \iint (w_\varepsilon^h + w_\varepsilon^{-h})P_\varepsilon\right\} < \infty \qquad (h \ge 1,\ (\gamma+\omega)h < \nu). \tag{10.19}$$

Since $\nu > \gamma+\omega$, we may fix $h > 1$ with $(\gamma+\omega)h < \nu$. When $\gamma+\omega = 0$, take, for example, $h = 2$. Otherwise, $h = (1+\nu/(\gamma+\omega))/2$ is admissible. Hence

$$\mathcal{B}_{\varepsilon,h} = \int qK_\varepsilon^h \le C_h. \tag{10.20}$$

We can now apply the inherited entropy inequality and obtain bounds at positive times. For each fixed ε, the solution was selected by Proposition 9.1. The constants in (9.4) depend on a, α, T, h, which are common to the family; here $H_x = 0$ because the variance is independent of x. Substituting (10.17) and (10.20) gives (10.12). The passage from $h = 2$ to this $h > 1$ uses the score of order $2h/(h-1)$ established in (3.13); a second moment of K_ε is no longer required. Lemma 9.2 gives

$$e^{2E_\varepsilon} \le Q_\varepsilon\, 4\iint \eta_\varepsilon^2 \left|\frac{\partial}{\partial z}\sqrt{P_\varepsilon}\right|^2 \le Q_*\mathcal{A}_\varepsilon. \tag{10.21}$$

Bound (10.10) implies $E_\varepsilon \geq -\frac{1}{2}\log\pi - \mathbb{E}|Z^\varepsilon|^2 \geq -C$. Apply (9.14) with $c = \alpha/4$, $G_\varepsilon = C(1+t^{-1}) + C\mathcal{R}_\varepsilon$ and $s = t/2$. For almost every $t > 0$,

$$\begin{aligned} E_\varepsilon(t) &\leq C(t/2 + \log 2) + C\int_{t/2}^{t} \mathcal{R}_\varepsilon(u)du - \tfrac{1}{2}\log(\alpha t/(4Q_*)) \\ &\leq C - \tfrac{1}{2}\log t. \end{aligned}$$

The constant is uniform by (10.3). Integration of the entropy inequality between Lebesgue times, using $E_\varepsilon \geq -C$ and $\int_0^T \mathcal{R}_\varepsilon \leq C$, gives (10.13). No uniform pointwise bound on H_ε was used. The initial law may have infinite entropy; neither finite initial entropy nor uniformity in ε of the internal constants in the common theorem was used.

10.2.2 Weak centering without a global Sobolev logarithm

Fix $\varepsilon, \delta > 0$ and write $P = P_\varepsilon$, $R_\pm = \int V^{\pm 1} P\, dz$, $M_\pm = R_\pm/q$, with $M = M_+$. For fixed ε, Theorem 2.1 gives Sobolev roots and moments of $V^{\pm 1}$ of every order. For $1 < r < 2$,

$$\int_\delta^T \iint V^{\pm 1}\left|\frac{\partial P}{\partial x}\right| \leq r\left(\int\left|\frac{\partial P^{1/r}}{\partial x}\right|^r\right)^{1/r}\left(\int V^{\pm r/(r-1)}P\right)^{(r-1)/r} < \infty.$$

Fubini's theorem identifies $\partial R_\pm/\partial x = \int V^{\pm 1}\partial P/\partial x$. The score in Lemma 9.3 also gives $\int R_\pm|S| \leq \left(\int V^{\pm 2}P\right)^{1/2}\left(\int qS^2\right)^{1/2} < \infty$. The local lower bound on q therefore implies, at almost every time,

$$M_\pm \in W^{1,1}_{\mathrm{loc}}(\mathbb{R}), \qquad \frac{\partial M_\pm}{\partial x} = \frac{\partial R_\pm/\partial x - R_\pm S}{q}.$$

Under the probability measure $u = P/q$, the Cauchy–Schwarz inequality gives $M_+M_- \geq 1$. This inequality holds everywhere for the absolutely continuous representatives. Thus M is locally bounded above and away from zero, and $K = M_+M_- - 1$ is locally bounded.

The chain rules on the absolutely continuous sections of P and $\sqrt{wP}$ give, with $f = (\partial P/\partial x)/P - S$ on $\{P > 0\}$ and $m = (\partial M/\partial x)/M$,

$$2\frac{\partial\sqrt{wP}}{\partial x} - S\sqrt{wP} = (f - m)\sqrt{wP}.$$

The weighted fields are extended by zero on $\{P = 0\}$, where derivatives of nonnegative Sobolev functions vanish almost everywhere. Differentiation of the conditional integrals, justified above, gives $\int fu = 0$ and $\int wfu = m$. Hence

$$m^2 \leq KD_x, \qquad D_x = \int w(f-m)^2 u, \qquad K = \int (1-w)^2 u/w. \tag{10.22}$$

The square-root energy establishes finiteness of $\int qD_x$ and gives

$$\mathcal{D}_\varepsilon \leq 8a_+\iint\left|\frac{\partial\sqrt{wP}}{\partial x}\right|^2 + 2a_+\int qS^2 + 4\iint \eta_\varepsilon^2\left|\frac{\partial\sqrt{P}}{\partial z}\right|^2 \leq C\mathcal{A}_\varepsilon + C(1+t^{-1}). \tag{10.23}$$

It remains to check that $\sqrt{P} \in H^1_{\mathrm{loc}}$ at almost every time, as required by Proposition 4.3. Fix such a time. On a compact rectangle I, V is bounded away from zero and M is bounded above, so $w \geq c_I > 0$. By (10.22),

$$\iint_I Pf^2 \leq 2c_I^{-1}\iint_I wP(f-m)^2 + 2\int_{I_x} qm^2 < \infty, \qquad \int_{I_x} qm^2 \leq \sup_{I_x} K\int_{I_x} qD_x.$$

Since $\partial P/\partial x = P(S+f)$ and $\int qS^2 < \infty$, we have $\int_I \mathbf{1}_{\{P>0\}}|\partial P/\partial x|^2/P < \infty$. Applying the chain rule to $\sqrt{P+e}$ and letting $e \downarrow 0$ gives the local L^2 derivative of $\sqrt{P}$ in x. In z, this follows from the energy and the local lower bound on η_ε. The local constants are used only to justify the identities; the uniform bounds below use (10.23) and the conditional moments without these constants.

10.2.3 From entropy estimates to gradient estimates

Set $\beta = r/(2-r)$. The inequality $r < r_{\nu,\gamma,\omega}$ is equivalent to $g_*\beta < \nu$, and hence to the two conditions $\beta < \nu$ and $(\gamma+\omega)\beta < \nu$. The three available moments are

$$\iint \eta_\varepsilon^{-2\beta} P_\varepsilon \le s_-^{-2\beta}\mathbb{E}\big[(Y^\varepsilon+\varepsilon)^\beta (1+Y^\varepsilon)^{2\lambda\beta}\big] \le C,$$
$$\iint w_\varepsilon^{-\beta} P_\varepsilon \le C, \qquad \int q K_\varepsilon^\beta \le C.$$

To check the first bound, use $\varepsilon \le 1$ to obtain

$$(Y^\varepsilon+\varepsilon)^\beta (1+Y^\varepsilon)^{2\lambda\beta} \le (1+Y^\varepsilon)^{(1+2\lambda)\beta}.$$

Its expectation is uniformly bounded by the positive moments already established. The other two bounds follow from (10.19) at $h = \beta > 1$, since $(\gamma+\omega)\beta < \nu$.

The weak identities are those of Section 10.2.2. Proposition 4.3 therefore applies to P_ε with the three moments above. The dissipation is controlled by (10.13) and (10.23), and the scalar score by Lemma 9.3. This gives the two estimates in (10.14). □

10.3 Passage to the limit and conclusion

Proof of Theorem 2.2. Fix the data (H1), (P1)–(P4) and the thresholds (2.19). Proposition 10.1 constructs the approximating solutions and their moments; Proposition 10.2 gives the entropy and gradient bounds. We pass to a common limit, construct the paths, and verify the conclusions of the theorem.

10.3.1 Strong compactness and identification of all fluxes

We first prove that there exist a sequence $\varepsilon_j \downarrow 0$ and a density P such that $P_{\varepsilon_j} \to P$ in $L^1((0,T)\times\mathbb{R}^2)$. With

$$V(t,z) = v(t,e^z), \quad R = \int V P\, dz, \quad M = R/q, \quad w = V/M,$$
$$\eta(t,z) = s(t,e^z)e^{-z/2}, \qquad b(t,z) = e^{-z}\big(\mu(t,e^z) - s(t,e^z)^2/2\big),$$

the moments M are positive and finite almost everywhere, and

$$\begin{aligned} w_\varepsilon P_\varepsilon &\to wP, & \eta_\varepsilon^2 P_\varepsilon &\to \eta^2 P, \\ b_\varepsilon P_\varepsilon &\to bP, & \eta_\varepsilon\sqrt{w_\varepsilon} P_\varepsilon &\to \eta\sqrt{w}P \end{aligned} \quad \text{in } L^1((0,T)\times\mathbb{R}^2) \tag{10.24}$$

along this sequence.

We first estimate the time derivatives through the flux masses. The masses of the logarithmic fluxes satisfy, uniformly in time and ε,

$$\begin{aligned} \iint a w_\varepsilon P_\varepsilon &\le a_+, \\ \iint \eta_\varepsilon^2 P_\varepsilon &\le s_+^2 \mathbb{E}(Y^\varepsilon)^{-1} \le C, \\ \iint |b_\varepsilon| P_\varepsilon &\le C(1+\mathbb{E}(Y^\varepsilon)^{-1}) \le C, \\ \iint |\widehat{\rho}|\eta_\varepsilon\sqrt{a w_\varepsilon} P_\varepsilon &\le \rho_*\sqrt{a_+}\left(\iint \eta_\varepsilon^2 P_\varepsilon\right)^{1/2} \le C. \end{aligned} \tag{10.25}$$

The first bound is calibration, $\int w_\varepsilon P_\varepsilon \, dz = q$. The last is the Cauchy–Schwarz inequality. The weak formulation therefore gives, for every ball D and every cutoff $\chi \in C_c^\infty(D)$,

$$\|\frac{\partial}{\partial t}(\chi P_\varepsilon)\|_{L^\infty(0,T;H^{-4}(D))} \le C_\chi.$$

In dimension two, $H^4(D)$ embeds into $C^2(\overline{D})$, so test-function derivatives of order at most two are controlled by the H^4 norm.

We combine these time estimates with the spatial gradient bounds. For $1 < r < r_{\nu,\gamma,\omega}$, Proposition 10.2 bounds χP_ε in $L^r(\delta, T; W^{1,1}(D))$; the time estimate above controls it in $W^{1,\infty}(\delta, T; H^{-4}(D))$. The application of Simon's criterion in Section 7.1 gives local strong convergence. It requires only $r > 1$.

We next extend the local convergence to the whole space-time domain. Bound (10.10) controls the tails in z. The common marginal q has position moments of every order because its scalar coefficients are bounded; these control the tails in x. Local convergence is therefore global on $(\delta, T) \times \mathbb{R}^2$. Since $\int P_\varepsilon(t) = 1$, the contribution of $(0, \delta)$ to an L^1 difference is at most 2δ. This gives convergence on $(0, T)$ and $\int P(t, x, z)\, dz = q(t, x)$ almost everywhere.

We identify the variance moment before passing to the conditional ratio. Choose $h > 1$ such that $\omega h < \nu$; this is possible since $\nu > \gamma + \omega$. By (10.11) and Lemma 7.1, $VP_\varepsilon \to VP$ in L^1, hence $R_\varepsilon \to R$ in $L^1(dt\, dx)$. Since $V > 0$ and $\int P dz = q > 0$, we have $0 < R < \infty$ almost everywhere. Along a subsequence, the ratio and $w_\varepsilon P_\varepsilon$ converge almost everywhere. The calibrated fluxes have mass one; Section 7.3 gives their strong convergence on $(0, T)$.

We then identify the factor fluxes and the mixed flux. The coefficients (10.4) converge to b, η uniformly on every compact strip in z, uniformly in time, even when μ, s are merely Borel measurable in time. Indeed,

$$|b_\varepsilon - b| = \frac{s(t, e^z)^2 \varepsilon}{2e^z(e^z + \varepsilon)}, \quad |\eta_\varepsilon - \eta| \le s_+ \left|(e^z + \varepsilon)^{-1/2} - e^{-z/2}\right|.$$

Both upper bounds tend to zero uniformly on the strip. For $1 < h < \nu$,

$$|\eta_\varepsilon^2|^h + |b_\varepsilon|^h \le C_h(1 + e^{-hz}).$$

The inverse moments of Proposition 10.1 give uniform integrability of the corresponding fluxes, by the same Hölder estimate on sets A. Almost-everywhere convergence and Vitali's theorem give $\eta_\varepsilon^2 P_\varepsilon \to \eta^2 P$ and $b_\varepsilon P_\varepsilon \to bP$ in L^1. Finally,

$$\eta_\varepsilon \sqrt{w_\varepsilon} P_\varepsilon = \sqrt{\eta_\varepsilon^2 P_\varepsilon}\sqrt{w_\varepsilon P_\varepsilon}.$$

Both factors converge strongly in L^2, since $|\sqrt{s} - \sqrt{t}|^2 \le |s - t|$. Their product therefore converges strongly in L^1. The multipliers a and $\widehat{\rho}\sqrt{a}$ are bounded; mere measurability causes no difficulty for these strong convergences.

10.3.2 Curve of laws, initial trace and stochastic reconstruction

Tightness from (10.10), the scalar moments, and equicontinuity from (10.25) allow us to apply Section 7.5. We obtain a continuous curve μ with trace $\mu_0^{X,\log Y}$, occupation density P, and first marginal $q(t, x)dx$ at every positive time.

Convergence of all fluxes and the common trace give, for $\phi \in C_c^\infty([0, T) \times \mathbb{R}^2)$,

$$\begin{aligned} &- \int P \frac{\partial \phi}{\partial t} - \int \phi(0, x, \log y)\, \mu_0(\, \mathrm{d}x\, \mathrm{d}y) \\ &\quad = \int P \left[\tfrac{1}{2} a w \left(\frac{\partial^2 \phi}{\partial x^2} - \frac{\partial \phi}{\partial x} \right) + \widehat{\rho}\eta\sqrt{aw}\, \frac{\partial^2 \phi}{\partial x \partial z} + \tfrac{1}{2}\eta^2 \frac{\partial^2 \phi}{\partial z^2} + b \frac{\partial \phi}{\partial z} \right]. \end{aligned} \tag{10.26}$$

Integrals without specified limits are over $(0,T)\times\mathbb{R}^2$.

The conditional moment is positive and finite, the fluxes are integrable by (10.25) and their convergence, and μ satisfies (10.26). All the assumptions of Proposition 8.1 therefore hold in (x,z) with coefficients $V,\eta,b,\widehat{\rho}$. It gives a continuous process (X,Z), two Brownian motions with covariation $\widehat{\rho}(t,X_t,Z_t)dt$, identification of the conditional expectation, and the true martingale property of e^X.

Set $Y=e^Z$. Each path of Z is continuous and real-valued on the compact time interval, and hence has a finite minimum. Thus $\inf_{t\le T} Y_t>0$ almost surely, with no assertion of an inverse moment of this infimum. Itô's formula and the expressions for b,η give

$$\begin{aligned} dY &= Yb(t,Z)dt + Y\eta(t,Z)dB + \tfrac{1}{2}Y\eta(t,Z)^2dt \\ &= \left[\mu(t,Y) - s(t,Y)^2/2 + s(t,Y)^2/2\right]dt + s(t,Y)\sqrt{Y}\,dB \\ &= \mu(t,Y)dt + s(t,Y)\sqrt{Y}\,dB. \end{aligned}$$

Disintegration of P gives $M(t,x)=\mathbb{E}[V(t,Z_t)\mid X_t=x]=\mathbb{E}[v(t,Y_t)\mid X_t=x]$ for $dt\,\mathcal{L}(X_t)(dx)$-almost every (t,x). Set $\mathsf{M}=e^{c_0}M$. By (2.13), M is the conditional moment of v and $v/M=\mathsf{v}/\mathsf{M}$. The price equation is therefore exactly (2.20), with no constant added to the denominator.

10.3.3 Regularity of the same solution in the positive coordinate

Strong convergence and (10.14) give, for $1<r<r_{\nu,\gamma,\omega}$, $P^{1/r}\in L^r(\delta,T;W^{1,r}(\mathbb{R}^2))$. Indeed, $|s^{1/r}-t^{1/r}|^r\le|s-t|$ gives strong convergence of the roots; weak compactness of their gradients and identification in distributions give the conclusion. On compact sets, $\eta_\varepsilon\to\eta$ uniformly and is bounded away from zero. Lower semicontinuity of the z part of (10.13) therefore gives

$$\int_\delta^T \iint \eta^2 \left|\frac{\partial}{\partial z}\sqrt{P}\right|^2 < \infty. \tag{10.27}$$

By Fatou's lemma, the limiting inverse moments satisfy $\int_\delta^T\int e^{-pz}P<\infty$ for $p<\nu$.

The density in the positive coordinate is $p_Y(t,x,y)=y^{-1}P(t,x,\log y)$. Set $h=P^{1/r}$ and $y=e^z$. Local chain rules give

$$\begin{aligned} \int\left|\frac{\partial}{\partial x}p_Y^{1/r}\right|^r dt\,dx\,dy &= \int\left|\frac{\partial}{\partial x}h\right|^r dt\,dx\,dz, \\ \int\left|\frac{\partial}{\partial y}p_Y^{1/r}\right|^r dt\,dx\,dy &= \int e^{-rz}\left|\frac{\partial}{\partial z}h - h/r\right|^r dt\,dx\,dz. \end{aligned}$$

Since $\eta(t,z)=s(t,e^z)e^{-z/2}$,

$$\frac{\partial}{\partial z}h = (2/r)P^{1/r-1/2}\frac{\partial}{\partial z}\sqrt{P}.$$

With $\beta=r/(2-r)<\nu$, Hölder's inequality gives

$$\begin{aligned} \int e^{-rz}\left|\frac{\partial}{\partial z}h\right|^r &\le \left(\frac{2}{r}\right)^r \int\left(\eta^2\left|\frac{\partial}{\partial z}\sqrt{P}\right|^2\right)^{r/2} P^{1-r/2}\left(e^z s(t,e^z)^2\right)^{-r/2} \\ &\le \left(\frac{2}{r}\right)^r \left(\int \eta^2\left|\frac{\partial}{\partial z}\sqrt{P}\right|^2\right)^{r/2} \left(\int P\left(e^z s(t,e^z)^2\right)^{-\beta}\right)^{1-r/2} < \infty. \end{aligned}$$

Indeed, the lower bound on s in (2.16) gives

$$\left(ys(t,y)^2\right)^{-\beta} \le s_-^{-2\beta}y^{-\beta}(1+y)^{2\lambda\beta} \le C(y^{-\beta}+1+y^{2\lambda\beta}).$$

The inverse moment is available since $\beta < \nu$; positive moments pass to the limit by Fatou's lemma and are available at every order. The other term is $r^{-r} \int P e^{-rz} < \infty$, since

$$r < \frac{2\nu}{\nu + g_*} \leq \frac{2\nu}{\nu + 1} < \nu \qquad (\nu > 1).$$

These finite integrals allow us to remove the localisations. The mass gives $\int |p_Y^{1/r}|^r = T - \delta$. Finally, the chain rule $\nabla p_Y = r(p_Y^{1/r})^{r-1} \nabla p_Y^{1/r}$ and Hölder's inequality give $p_Y \in L_t^1 W^{1,1}$. All conclusions of Theorem 2.2 therefore hold for the same constructed solution.

10.3.4 Removing the compact-support restriction

For the law in (2.22), clamp X_0 to $[-j, j]$ and Y_0 to $[1/j, j]$, preserving their joint dependence, and apply the preceding construction. For $j \geq 1$,

$$(Y_0^{(j)})^m \leq 1 + Y_0^m, \qquad (Y_0^{(j)})^{-p} \leq 1 + Y_0^{-p}.$$

The generator estimates above thus give uniform positive moments and inverse moments of every order $p < \nu$. The integrated defect uses only $\mathbb{E}(Y_0^{(j)})^{-1}$ and the inverse moment of order $1 + \theta < \nu$; its constant is uniform too. Consequently (10.17), (10.20), entropy entry, and the root bounds have constants independent of j. The scalar marginals q_j have the common score bounds of Proposition 3.5; scalar coupling from $X_0^{(j)} \to X_0$ in L^2 identifies their narrowly uniform limit with the target marginal q.

Apply compactness to the logarithmic occupation densities of these selected solutions. Their factor energies are inherited by lower semicontinuity from the factor-regularised approximations, so (10.27) remains uniform. The root bounds and the flux masses (10.25) give local L^1 compactness; the logarithmic and price moments control the tails. Strong convergence of the joint densities gives strong convergence of q_j. The moment of V of an order greater than one identifies $R = \int VP \, \mathrm{d}z$ and $M = R/q$ almost everywhere. For the calibrated fluxes, their common total mass and almost-everywhere convergence give strong L^1 convergence; the factor fluxes use an inverse moment of order $h \in (1, \nu)$, and the mixed flux is a product of square roots. These are precisely the hypotheses used in (10.26) and reconstruction, now with initial trace $\mu_0^{X, \log Y}$ because the clamped joint laws converge narrowly. The root bounds, factor energy and inverse moments pass to this same limit; the change-of-coordinate estimates above then give the stated global regularity in y. This proves the general initial-law assertion without assuming a Gaussian moment for $\log Y_0$. □

11 Approximation on the whole space

Theorem 11.1 (Diagonal approximation of a calibrated weak solution)**.** *Assume (H1)–(H5). There exist choices*

$$n_j \to \infty, \quad m_j \to \infty, \quad \tau_j \to 0, \quad \kappa_j \to 0$$

and solutions of the corresponding Fourier systems whose piecewise constant densities, extended by zero outside the centred domains, converge in $L^1((0,T) \times \mathbb{R}^2)$ to an occupation density of a calibrated weak solution of the original model. The fluxes also converge in the L^1 spaces of Sections 7–8.

Proof. On each fixed torus, Proposition 5.1 gives a periodic solution p_n as an internal limit, with convergence of the density and all four fluxes. These same solutions satisfy Proposition 6.1; Sections 7–8 provide a subsequence $p_{n_j} \to p$ on the whole space, with the corresponding flux convergence and stochastic reconstruction. For each j, choose an internal approximation p_j^{num} on that torus such that the sum of its L^1 errors relative to p_{n_j} and its four fluxes is less than

$1/j$, while $m_j \geq j$ and $\tau_j, \kappa_j \leq 1/j$. Fixed-torus convergence permits these finitely many requirements. With both densities extended by zero from the same centred domain,

$$\|p_j^{\mathrm{num}} - p\|_1 \leq j^{-1} + \|p_{n_j} - p\|_1 \to 0.$$

The same triangle inequality for each flux proves the claim. □

This result gives no explicit rule relating n, m, τ, κ. The internal constants may deteriorate as the torus grows. The independent limits on a fixed torus therefore do not permit an arbitrary simultaneous limit of all four parameters. Approximate evaluation of q and of the integrals requires further analysis: it must preserve the marginal or control its defect, and satisfy the residual tolerances.

12 Quantitative stability of calibrated fluxes

The estimates in this section concern densities and their calibrated fluxes. Time may be treated as a parameter.

12.1 Different marginals and variances

Let p_1, p_2 be probability densities on $\mathbb{R}^2$, and let v_1, v_2 be strictly positive finite measurable functions. Set

$$q_i(x) = \int_{\mathbb{R}} p_i(x,y)\,\mathrm{d}y, \qquad A_i(x) = \int_{\mathbb{R}} v_i(x,y)p_i(x,y)\,\mathrm{d}y.$$

Assume that $A_i < \infty$ almost everywhere on $\{q_i > 0\}$. Then $A_i > 0$ almost everywhere on this set. Define

$$M_i = A_i/q_i, \qquad F_i = \frac{v_i p_i}{M_i} = q_i \frac{v_i p_i}{A_i} \quad \text{on } \{q_i > 0\}.$$

On $\{q_i = 0\}$, set $F_i = 0$ and $M_i = 1$. Exceptional values do not affect the integrals. In particular,

$$\int F_i(x,y)\,\mathrm{d}y = q_i(x), \qquad \iint F_i = 1.$$

Theorem 12.1 (Stability with different marginals and variances)**.** *Let $\beta > 0$. If*

$$B_\beta = \iint v_1^{-\beta} p_1 < \infty, \qquad d = \iint |v_1 p_1 - v_2 p_2| < \infty,$$

then

$$\|F_1 - F_2\|_{L^1} \leq \|q_1 - q_2\|_{L^1} + 2B_\beta^{1/(\beta+1)} d^{\beta/(\beta+1)}. \tag{12.1}$$

No uniform positive lower bound on M_1 or M_2 is required.

Under these assumptions, suppose that $v_1 = v_2 = v$ and, for some $r > 1$,

$$A_r = \iint v^r (p_1 + p_2) < \infty.$$

With $\epsilon = \|p_1 - p_2\|_1$, we have

$$\|F_1 - F_2\|_1 \leq \epsilon + 2B_\beta^{1/(\beta+1)} A_r^{\beta/[r(\beta+1)]} \epsilon^{\beta(r-1)/[r(\beta+1)]}. \tag{12.2}$$

If $q_1 = q_2$, the term ϵ may be omitted. Moreover, for $H_i = \sqrt{F_i p_i}$,

$$\|H_1 - H_2\|_1 \leq \|F_1 - F_2\|_1^{1/2} + \|p_1 - p_2\|_1^{1/2}. \tag{12.3}$$

Proof. At each x, write $f_i = v_i p_i$ and $d(x) = \int |f_1 - f_2| \, \mathrm{d}y$. If $A_1, A_2 > 0$, then $|A_1 - A_2| \le d(x)$ and

$$\int \left| \frac{f_1}{A_1} - \frac{f_2}{A_2} \right| \, \mathrm{d}y \le \min\{2, 2d(x)/A_1\}.$$

Decomposing $F_1 - F_2 = q_1(f_1/A_1 - f_2/A_2) + (q_1 - q_2)f_2/A_2$ gives

$$\int |F_1 - F_2|(x, y) \, \mathrm{d}y \le |q_1 - q_2|(x) + 2\min\{q_1(x), d(x)/M_1(x)\}. \tag{12.4}$$

If either marginal vanishes, the left side equals the other marginal and the first term suffices, so the bound holds almost everywhere. For $\theta = \beta/(\beta+1)$, $\min(u, v) \le u^{1-\theta} v^{\theta}$ and Hölder yield

$$\int \min(q_1, d/M_1) \, \mathrm{d}x \le d^{\theta} \left(\int q_1 M_1^{-\beta} \, \mathrm{d}x \right)^{1-\theta} \le d^{\theta} B_{\beta}^{1-\theta},$$

where the last step is conditional Jensen for the convex function $z^{-\beta}$. This proves (12.1), also when $d = 0$. If $v_1 = v_2 = v$, Hölder and $|p_1 - p_2| \le p_1 + p_2$ give $\int v|p_1 - p_2| \le A_r^{1/r} \epsilon^{1-1/r}$; also $\|q_1 - q_2\|_1 \le \epsilon$ by integration in y. Substitution proves (12.2), with the marginal term absent when $q_1 = q_2$. Finally,

$$H_1 - H_2 = (\sqrt{F_1} - \sqrt{F_2})\sqrt{p_1} + \sqrt{F_2}(\sqrt{p_1} - \sqrt{p_2}).$$

Cauchy–Schwarz, unit masses and $|\sqrt{u} - \sqrt{v}|^2 \le |u - v|$ give (12.3). □

Remark 12.2 (Time-integrated estimates)**.** The proof of (12.1) also applies with x replaced by (t, x) on $J \times \mathbb{R}$, with all norms and moments integrated over J. In (12.3), the right-hand side is then multiplied by $|J|^{1/2}$ if the densities have mass one at each time. Under uniform bounds on all positive and negative moments of v, estimate (12.2) gives every exponent strictly less than one, with a constant depending on the exponent. It gives neither exponent one nor an estimate of the error in the densities themselves.

12.2 Regularisation of the denominator

Theorem 12.3 (Exact calibration defect)**.** *For a density p, its conditional moment M and its marginal q, define*

$$F = \frac{vp}{M}, \qquad F_\varepsilon = \frac{vp}{M + \varepsilon}, \qquad \varepsilon > 0.$$

For every $0 < s \le 1$,

$$\int F_\varepsilon(x, y) \, \mathrm{d}y = q(x) \frac{M(x)}{M(x) + \varepsilon}, \tag{12.5}$$

$$\|F - F_\varepsilon\|_1 = \int q(x) \frac{\varepsilon}{M(x) + \varepsilon} \, \mathrm{d}x \le \varepsilon^s B_s(p). \tag{12.6}$$

The bound is finite when $B_s(p) = \iint v^{-s} p < \infty$. Under this moment condition, the mixed flux also satisfies

$$\left\| \sqrt{Fp} - \sqrt{F_\varepsilon p} \right\|_1 \le \varepsilon^{s/2} B_s(p)^{1/2}.$$

Independently of the finiteness of $B_s(p)$, let p_ε be another probability density with marginal q_ε and conditional moment M_ε, positive and finite on $\{q_\varepsilon > 0\}$. For $\beta > 0$, $0 < s \le 1$, assume $B_\beta(p) < \infty$, $B_s(p_\varepsilon) < \infty$, and $\iint v|p_\varepsilon - p| < \infty$, and set $F_\varepsilon[p_\varepsilon] = vp_\varepsilon/(M_\varepsilon + \varepsilon)$. Then

$$\begin{aligned} \|F_\varepsilon[p_\varepsilon] - F[p]\|_1 \le{}& \varepsilon^s B_s(p_\varepsilon) + \|q_\varepsilon - q\|_1 \\ &+ 2B_\beta(p)^{1/(\beta+1)} \left(\iint v|p_\varepsilon - p| \right)^{\beta/(\beta+1)}. \end{aligned}$$

Proof. Integration in y gives (12.5); since $0 \le F_\varepsilon \le F$, the L^1 difference equals the mass defect. The inequality $u/(1+u) \le u^s$, $0 < s \le 1$, and conditional Jensen give

$$0 \le q\frac{\varepsilon}{M+\varepsilon} \le \varepsilon^s q M^{-s} \le \varepsilon^s \int v^{-s} p \, \mathrm{d}y. \tag{12.7}$$

Integration proves (12.6), allowing an infinite right side. Cauchy–Schwarz and the square-root inequality used in Theorem 12.1 give $\|\sqrt{Fp} - \sqrt{F_\varepsilon p}\|_1 \le \|F - F_\varepsilon\|_1^{1/2}$. For the last assertion, insert $F[p_\varepsilon]$ in the difference: (12.6) bounds the regularisation term by $\varepsilon^s B_s(p_\varepsilon)$, and (12.1) with $p_1 = p$, $p_2 = p_\varepsilon$ bounds the density term, placing B_β under p as required. □

This estimate does not prove convergence of solutions of the regularised equations: uniform moment bounds and convergence of p_ε remain to be established. It separates the errors that must be controlled.

12.3 Spatial smoothing of a given density

Proposition 12.4. *Fix $J = (\delta, T)$. Let p have mass one at each time, and let $v = v(t,y) > 0$ be independent of x. Suppose that*

$$\frac{\partial}{\partial x} p, \ \frac{\partial}{\partial x}(vp) \in L^1(J \times \mathbb{R}^2), \qquad B_\beta = \int_J \iint v^{-\beta} p < \infty.$$

The derivatives are weak derivatives. Let $K \ge 0$ be a kernel on $\mathbb{R}$, of mass one, with $m_1 = \int |z| K(z) \, \mathrm{d}z < \infty$. For $K_h(z) = h^{-1} K(z/h)$, set $p_h = K_h *_x p$ and form $F[p_h]$ with the same variance v. Then

$$\|F[p_h] - F[p]\|_{L^1(J\times\mathbb{R}^2)} \le h m_1 \left\| \frac{\partial}{\partial x} p \right\|_1 + 2 B_\beta^{1/(\beta+1)} \left(h m_1 \left\| \frac{\partial}{\partial x}(vp) \right\|_1 \right)^{\beta/(\beta+1)}.$$

Proof. For almost every (t,y), the sections $p(t,\cdot,y)$ and $f_{t,y} = v(t,y)p(t,\cdot,y)$ belong to $W^{1,1}(\mathbb{R})$: p is integrable on the section, $v(t,y)$ is a finite constant, and the assumed weak derivatives are integrable there. Absolute continuity and Tonelli give

$$\int |f_{t,y}(x-z) - f_{t,y}(x)| \, \mathrm{d}x \le |z| \int |f'_{t,y}| \, \mathrm{d}x.$$

Integrate this inequality and its version for p in (t,y) and against $K_h(z) \, \mathrm{d}z$. Since multiplication by v commutes with convolution in x,

$$\|p_h - p\|_1 \le h m_1 \|\partial_x p\|_1, \qquad \|vp_h - vp\|_1 \le h m_1 \|\partial_x (vp)\|_1.$$

No global integrability of vp is needed for these nonnegative-integrand applications of Tonelli. To verify finiteness of the conditional moments, use the zero limits at infinity of an integrable $W^{1,1}(\mathbb{R})$ function:

$$\operatorname*{ess\,sup}_x \int v(t,y) p(t,x,y) \, \mathrm{d}y \le \int \|f_{t,y}\|_\infty \, \mathrm{d}y \le \iint |\partial_x (vp)(t,x,y)| \, \mathrm{d}x \, \mathrm{d}y < \infty$$

for almost every t. The same bound holds after convolution, and $v > 0$ gives positivity where the marginal is positive. The time-integrated form of Theorem 12.1 now proves the assertion. □

For the solutions of Theorem 2.1 with v independent of x, these integrability properties follow from the estimates already proved. The displayed smoothing bound uses commutation of multiplication by v with convolution in x; it is not claimed for general $v(t,x,y)$. Indeed, for $1 < r < 2$,

$$\left\| v \frac{\partial}{\partial x} p \right\|_{L^1(J\times\mathbb{R}^2)} \le r \left\| \frac{\partial}{\partial x}(p^{1/r}) \right\|_{L^r} \left(\int_J \iint v^{r/(r-1)} p \right)^{(r-1)/r} < \infty.$$

The Sobolev chain rule gives $\frac{\partial}{\partial x}p = rp^{1-1/r}\frac{\partial}{\partial x}(p^{1/r})$, and $\frac{\partial}{\partial x}(vp) = v\frac{\partial}{\partial x}p$ in distributions because v is independent of x. The preceding integrability bounds justify the product and the use of Fubini's theorem. All positive and negative moments of v are available in this setting. This is a rate for smoothing an already constructed density, not a convergence rate for a calibration algorithm.

13 Vanilla price errors and denominator regularisation

We work on the real line, with zero interest and dividend rates. We first bound price errors in terms of a defect in the marginal equation, and then apply the estimate to regularisation of the calibration denominator. This gives a price estimate corresponding to the flux defect in Section 12.2.

13.1 The marginal equation and its residual

Fix $T_* > 0$ and a Borel coefficient $a : [0, T_*] \times \mathbb{R} \to (0, \infty)$ satisfying

$$0 < a_- \le a(t,x) \le a_+, \qquad \operatorname*{ess\,sup}_{(t,x)} \left(\left| \frac{\partial a}{\partial x} \right| + \left| \frac{\partial^2 a}{\partial x^2} \right| \right) \le K_a. \tag{13.1}$$

The coefficient is Borel measurable in time, and the spatial derivatives are weak derivatives. We choose the spatial $C^{1,1}$ representatives at nonexceptional times and an admissible constant at the remaining times. For a test function ψ, set

$$\mathcal{B}\psi = \frac{\partial^2 \psi}{\partial x^2} - \frac{\partial \psi}{\partial x}, \qquad \mathcal{A}_t \psi = \frac{a(t,\cdot)}{2}\mathcal{B}\psi.$$

Let $\mu, \widehat{\mu}$ be narrowly continuous curves of probability measures on $\mathbb{R}$, and set $\nu_t = \widehat{\mu}_t - \mu_t$. Assume that, for every $\psi \in C_c^\infty(\mathbb{R})$ and every $t \in [0, T_*]$,

$$\begin{aligned} \langle \nu_t, \psi \rangle = \langle \nu_0, \psi \rangle &+ \frac{1}{2}\int_0^t \langle \nu_s, a(s,\cdot)\mathcal{B}\psi \rangle \, \mathrm{d}s \\ &+ \frac{1}{2}\int_0^t \int_{\mathbb{R}} a(s,x)\mathfrak{d}(s,x)\mathcal{B}\psi(x) \, \mathrm{d}x \, \mathrm{d}s + \int_0^t \langle \mathfrak{r}_s, \psi \rangle \, \mathrm{d}s, \end{aligned} \tag{13.2}$$

where $\mathfrak{d} \in L^1((0,T_*) \times \mathbb{R})$ and $\mathfrak{r}_s$ is a measurable family of finite signed measures with $\int_0^{T_*} \|\mathfrak{r}_s\|_{\mathrm{TV}} \, \mathrm{d}s < \infty$. Here $\|\lambda\|_{\mathrm{TV}} = |\lambda|(\mathbb{R})$ denotes the mass of the total variation measure, without a factor $1/2$.

In the LSV application, if $\widehat{p}$ has marginal $\widehat{q}$ and approximate flux $\widehat{F}$, then

$$\mathfrak{d}(t,x) = \int_{\mathbb{R}} \widehat{F}(t,x,y) \, \mathrm{d}y - \widehat{q}(t,x). \tag{13.3}$$

Integrating the joint equation in y gives (13.2) when the target marginal q solves $\frac{\partial}{\partial t}q = \frac{1}{2}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial}{\partial x}\right)(aq)$, with $\mu_t = q(t,x)\,\mathrm{d}x$, provided that the other fluxes vanish under this integration and $\mathfrak{r}$ is the marginal residual. For integrable fluxes, take $\chi \in C_c^\infty(\mathbb{R})$, equal to one on $[-1,1]$, test with $\psi(x)\chi(y/R)$, and let $R \to \infty$. The additional terms contain factors R^{-1} or R^{-2}, so their integrals tend to zero.

13.2 Duality and smooth test functions

Proposition 13.1 (Weak error of the marginal)**.** *Under* (13.1) *and* (13.2)*, for* $\phi \in C_b^2(\mathbb{R})$ *and* $0 \le t \le T_*$*,*

$$\begin{aligned} |\langle \nu_t, \phi \rangle| \le \|\phi\|_\infty &\left(\|\nu_0\|_{\mathrm{TV}} + \int_0^t \|\mathfrak{r}_s\|_{\mathrm{TV}} \, \mathrm{d}s \right) \\ &+ \frac{a_+}{2} \exp(K_a t/2) \|\mathcal{B}\phi\|_\infty \int_0^t \|\mathfrak{d}(s,\cdot)\|_1 \, \mathrm{d}s. \end{aligned} \tag{13.4}$$

Proof. Apply Proposition 3.1 on $[0, T_*]$. The resulting smooth coefficients $a^{(j)}$ satisfy (13.1) with the same constants, and

$$\int_0^{T_*} \sup_{|x|\le L} |a^{(j)}(s,x) - a(s,x)| \, \mathrm{d}s \longrightarrow 0 \quad (L < \infty). \tag{13.5}$$

For fixed t, solve

$$\frac{\partial u_j}{\partial s} + \frac{a^{(j)}}{2} \mathcal{B} u_j = 0, \qquad u_j(t, \cdot) = \phi.$$

The representation $u_j(s,x) = \mathbb{E}[\phi(Z_t^{j,s,x})]$, where $\mathrm{d}Z = -a^{(j)} \, \mathrm{d}s/2 + \sqrt{a^{(j)}} \, \mathrm{d}W$, gives $\|u_j(s,\cdot)\|_\infty \le \|\phi\|_\infty$. For smooth ϕ, spatial differentiation is justified by the first two flow derivatives J, H. With $b = -a^{(j)}/2$, $\sigma = \sqrt{a^{(j)}}$, they satisfy

$$\begin{aligned} \mathrm{d}J &= b_x J \, \mathrm{d}s + \sigma_x J \, \mathrm{d}W, & J_s &= 1, \\ \mathrm{d}H &= (b_x H + b_{xx} J^2) \, \mathrm{d}s + (\sigma_x H + \sigma_{xx} J^2) \, \mathrm{d}W, & H_s &= 0. \end{aligned}$$

The coefficient derivatives along the flow are bounded, so martingale estimates and Gronwall give finite moments for J, H and permit differentiation under the expectation. Set $h_j = \mathcal{B} u_j$. The product rule gives $\mathcal{B}(ah) = a h_{xx} + (2a_x - a) h_x + (a_{xx} - a_x) h$. Consequently,

$$\frac{\partial h_j}{\partial s} + \frac{a^{(j)}}{2} \frac{\partial^2 h_j}{\partial x^2} + \left(\frac{\partial}{\partial x} a^{(j)} - \frac{a^{(j)}}{2} \right) \frac{\partial h_j}{\partial x} + \frac{\frac{\partial^2}{\partial x^2} a^{(j)} - \frac{\partial}{\partial x} a^{(j)}}{2} h_j = 0. \tag{13.6}$$

The drift in this equation is bounded and Lipschitz, and its zeroth-order term is bounded in absolute value by $K_a/2$. The Feynman–Kac formula, or the comparison principle after reversing time, gives

$$\|h_j(s,\cdot)\|_\infty \le \exp(K_a(t-s)/2) \|\mathcal{B}\phi\|_\infty. \tag{13.7}$$

The first derivative of u_j is also bounded, by its differentiated equation or the flow representation above.

Time-dependent test functions are admissible in (13.2). This follows first for $\alpha(s)\beta(x)$, by integration by parts in time, and then by approximation in $C^{1,2}$ on compact sets. To use u_j, multiply it by a cutoff $\chi(x/R)$ equal to one on $[-R, R]$. The terms containing derivatives of χ tend to zero by the bounds on u_j, $\frac{\partial}{\partial x} u_j$, $\frac{\partial^2}{\partial x^2} u_j$, the masses of $\mu_s, \widehat{\mu}_s$, and integrability of $\mathfrak{d}, \mathfrak{r}$. We obtain

$$\begin{aligned} \langle \nu_t, \phi \rangle = \langle \nu_0, u_j(0,\cdot) \rangle + \frac{1}{2} \int_0^t \int a \, \mathfrak{d} \, h_j \, \mathrm{d}x \, \mathrm{d}s + \int_0^t \langle \mathfrak{r}_s, u_j(s,\cdot) \rangle \, \mathrm{d}s \\ + \frac{1}{2} \int_0^t \langle \nu_s, (a - a^{(j)}) h_j \rangle \, \mathrm{d}s. \end{aligned} \tag{13.8}$$

The last term tends to zero. On $[-L, L]$, this follows from (13.5) and $|\nu_s|(\mathbb{R}) \le 2$. Outside this interval, it is bounded by a constant times $\int_0^t (\mu_s + \widehat{\mu}_s)(\{|x| > L\}) \, \mathrm{d}s$, which tends to zero by dominated convergence. The bounds on u_j, h_j, followed by $j \to \infty$, give (13.4). For general $\phi \in C_b^2$, use $\phi_\epsilon = \zeta_\epsilon * \phi$ with a nonnegative smooth unit-mass kernel. Boundedness of ϕ' gives uniform convergence $\phi_\epsilon \to \phi$, while commutation with $\mathcal{B}$ gives $\|\phi_\epsilon\|_\infty \le \|\phi\|_\infty$ and $\|\mathcal{B}\phi_\epsilon\|_\infty \le \|\mathcal{B}\phi\|_\infty$. Apply the smooth estimate and pass to the limit against the finite measures. □

13.3 Uniform bounds for call and put prices

We impose the integrability conditions needed for unbounded payoffs:

$$\begin{aligned} M_* &:= \sup_{t\le T_*} \int_{\mathbb{R}} \exp(x)(\mu_t + \widehat{\mu}_t)(\,\mathrm{d}x) < \infty, \\ &\int_0^{T_*} \int_{\mathbb{R}} \exp(x)|\mathfrak{d}(t,x)|\,\mathrm{d}x\,\mathrm{d}t < \infty, \\ &\int_0^{T_*} \int_{\mathbb{R}} \exp(x)|\mathfrak{r}_t|(\,\mathrm{d}x)\,\mathrm{d}t < \infty. \end{aligned} \tag{13.9}$$

Set

$$\begin{aligned} d_1(t) &= \int_{\mathbb{R}} \exp(x)|\mathfrak{d}(t,x)|\,\mathrm{d}x, \\ E_1(t) &= \int_{\mathbb{R}} \exp(x)|\nu_0|(\,\mathrm{d}x) + \int_0^t \int_{\mathbb{R}} \exp(x)|\mathfrak{r}_s|(\,\mathrm{d}x)\,\mathrm{d}s, \\ \Delta_1(t) &= \int_0^t d_1(s)\,\mathrm{d}s. \end{aligned}$$

The undiscounted prices are

$$C(t,K) = \int (\exp(x) - K)_+ \mu_t(\,\mathrm{d}x), \qquad P(t,K) = \int (K - \exp(x))_+ \mu_t(\,\mathrm{d}x), \qquad K > 0.$$

Write $\widehat{C}, \widehat{P}$ for the same integrals against $\widehat{\mu}_t$, and define

$$\mathcal{E}(t) = \sup_{K>0} \max\{|\widehat{C}(t,K) - C(t,K)|,\, |\widehat{P}(t,K) - P(t,K)|\}.$$

Theorem 13.2 (Vanilla price error from the marginal defect)**.** *Under* (13.1), (13.2) *and* (13.9), *there are constants C_0, C_1, depending only on a_-, a_+, K_a, T_*, such that, for every $0 < t \le T_*$,*

$$\mathcal{E}(t) \le \exp(1) E_1(t) + C_0 \sqrt{M_* \Delta_1(t)}, \tag{13.10}$$

$$\mathcal{E}(t) \le E_1(t) + C_1 \int_0^t \frac{d_1(s)}{\sqrt{t-s}}\,\mathrm{d}s \quad \textit{if this integral is finite.} \tag{13.11}$$

In particular, for $2 < r < \infty$, with $r' = r/(r-1)$,

$$\mathcal{E}(t) \le E_1(t) + C_1 (1 - r'/2)^{-1/r'} t^{1/2 - 1/r} \|d_1\|_{L^r(0,t)}.$$

For $r = \infty$, the last term is $2C_1\sqrt{t}\,\|d_1\|_{L^\infty(0,t)}$. For each $K > 0$, we also have

$$\max\{|\widehat{C}(t,K) - C(t,K)|, |\widehat{P}(t,K) - P(t,K)|\} \le E_1(t) + C_2 K \int_0^t \frac{\|\mathfrak{d}(s,\cdot)\|_1}{\sqrt{t-s}}\,\mathrm{d}s, \tag{13.12}$$

whenever the integral is finite, where C_2 depends on the same constants. Replacing K by $K_{\max}$ gives a uniform bound for $0 < K \le K_{\max}$. If $\nu_0 = 0$, $\mathfrak{r} = 0$ and $\mathfrak{d} \le 0$, then $\widehat{C} \le C$ and $\widehat{P} \le P$.

Proof. Fix a positive function $\varrho \in C_c^\infty((-1,1))$ of mass one. For $0 < h \le 1$, set $\varrho_h(z) = h^{-1}\varrho(z/h)$ and $m_1 = \int |z|\varrho(z)\,\mathrm{d}z \le 1$. For a call, write

$$g_K(x) = (\exp(x) - K)_+, \qquad g_{K,h} = \varrho_h * g_K.$$

The function $z \mapsto (z-K)_+$ is 1-Lipschitz. For $|z| \le h$,

$$|g_K(x-z) - g_K(x)| \le \exp(x)|\exp(-z) - 1| \le \exp(x+h)|z|.$$

Consequently, uniformly in $K > 0$,

$$\begin{gathered} |g_{K,h} - g_K| \le m_1 h \exp(h) \exp(x), \\ 0 \le g_{K,h} \le \exp(h)\exp(x), \qquad 0 \le \frac{\partial g_{K,h}}{\partial x} \le \exp(h)\exp(x). \end{gathered} \tag{13.13}$$

In distributions,

$$\frac{\partial g_K}{\partial x} = \exp(x)\mathbf{1}_{\{x>\log K\}}, \qquad \mathcal{B} g_K = K\delta_{\log K}.$$

Hence

$$\begin{gathered} \mathcal{B} g_{K,h}(x) = K\varrho_h(x - \log K), \\ 0 \le \exp(-x)\mathcal{B} g_{K,h}(x) \le \exp(h)\|\varrho\|_\infty/h, \\ \int \exp(-x)\mathcal{B} g_{K,h}(x)\,\mathrm{d}x \le \exp(h). \end{gathered} \tag{13.14}$$

Solve the backward equation with coefficient $a^{(j)}$ and terminal datum $g_{K,h}$, and denote the solution by $u_{j,h}$. Its stochastic representation is finite: the exponential price of the scalar diffusion is a martingale because its volatility coefficient is bounded. Thus

$$|u_{j,h}(s,x)| \le \exp(h)\exp(x). \tag{13.15}$$

Differentiation through the flow remains justified: the derivatives of $g_{K,h}$ are bounded by $C_h \exp(x)$, and the scalar diffusion has exponential moments of every order on a finite interval. Hölder's inequality and the moments of J, H permit differentiation under the expectation. The same arguments bound both spatial derivatives by $C_h \exp(x)$. To justify the Feynman–Kac representations for their equations, stop the auxiliary diffusions upon leaving $[-R, R]$. Their drift, diffusion and zeroth-order coefficients are bounded. The boundary terms are dominated by a constant times $\exp(\sup_{u \le T_*} |Z_u|)$, which is integrable for these diffusions with bounded coefficients, and tend to zero as $R \to \infty$. This justifies the kernel representations and comparisons below despite the unbounded payoff.

Set $w_{j,h} = \mathcal{B} u_{j,h}$ and $H_{j,h}(s,x) = \exp(-x) w_{j,h}(s,x)$. Substitution of $w_{j,h} = \exp(x) H_{j,h}$ into (13.6) gives

$$\frac{\partial H_{j,h}}{\partial s} + \frac{a^{(j)}}{2}\frac{\partial^2 H_{j,h}}{\partial x^2} + \left(\frac{a^{(j)}}{2} + \frac{\partial}{\partial x} a^{(j)}\right)\frac{\partial H_{j,h}}{\partial x} + \frac{\frac{\partial^2}{\partial x^2} a^{(j)} + \frac{\partial}{\partial x} a^{(j)}}{2} H_{j,h} = 0. \tag{13.16}$$

The comparison principle gives $H_{j,h} \ge 0$ and

$$0 \le w_{j,h}(s,x) \le \exp(x+h)\exp(K_a T_*/2)\|\varrho\|_\infty/h. \tag{13.17}$$

The same transformation of the equation for the first derivative of $u_{j,h}$ gives an equation with no zeroth-order term for $\exp(-x)\frac{\partial}{\partial x} u_{j,h}$. It implies $\left|\frac{\partial}{\partial x} u_{j,h}\right| \le \exp(x+h)$. Since $\frac{\partial^2}{\partial x^2} u_{j,h} = \frac{\partial}{\partial x} u_{j,h} + w_{j,h}$, all required derivatives are bounded by $C_h \exp(x)$, with C_h independent of j, K.

We also have a bound independent of h. Setting $\tau = t - s$ in (13.16) gives exactly

$$\frac{\partial H_{j,h}}{\partial \tau} = \frac{1}{2}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial}{\partial x}\right)(a^{(j)}(t-\tau, x) H_{j,h}).$$

This is the same scalar Fokker–Planck equation, with time reversed in the coefficient. After reversing time, the operator in (13.16) can be written as

$$\frac{\partial}{\partial x}\left(\frac{a^{(j)}}{2}\frac{\partial}{\partial x}\right) + \frac{a^{(j)} + \frac{\partial}{\partial x} a^{(j)}}{2}\frac{\partial}{\partial x} + \frac{\frac{\partial^2}{\partial x^2} a^{(j)} + \frac{\partial}{\partial x} a^{(j)}}{2}.$$

It is uniformly elliptic, and its divergence-form and lower-order coefficients are bounded independently of j. Aronson's Gaussian upper bound [2, 3], in dimension one, therefore gives

$$0 \le \Gamma_j(s,x;t,z) \le \frac{C_\Gamma}{\sqrt{t-s}} \exp\left(-c_\Gamma \frac{|x-z|^2}{t-s}\right).$$

The constants depend only on a_-, a_+, K_a, T_*. They involve neither third spatial derivatives nor time derivatives of $a^{(j)}$. The last bound in (13.14) then gives

$$0 \le w_{j,h}(s,x) \le C_\Gamma \exp(x+h)(t-s)^{-1/2}. \tag{13.18}$$

Identity (13.8) applies to $g_{K,h}$. Indeed, (13.9) and the bounds $C_h \exp(x)$ justify removing the cutoffs $\chi(x/R)$. The coefficient error again tends to zero for fixed h: on $[-L, L]$, use (13.5) with the factor $\exp(L)$; outside this interval, use

$$\int_0^t \int_{|x|>L} \exp(x)(\mu_s + \widehat{\mu}_s)(\,\mathrm{d}x)\,\mathrm{d}s \longrightarrow 0.$$

This convergence follows from space-time integrability, which is ensured by $M_* < \infty$. Thus we first let $j \to \infty$ with h fixed.

Write $m(t) = \int \exp(x)(\mu_t + \widehat{\mu}_t)(\,\mathrm{d}x) \le M_*$. By (13.13), (13.15) and (13.17),

$$\begin{gathered} |\langle \nu_t, g_K \rangle| \le \exp(h) E_1(t) + m_1 h \exp(h) m(t) + B_* \exp(h) \Delta_1(t)/h, \\ B_* = \frac{a_+}{2} \exp(K_a T_*/2) \|\varrho\|_\infty. \end{gathered} \tag{13.19}$$

If $0 < \Delta_1(t) \le M_*$, choose $h = \sqrt{\Delta_1(t)/M_*}$. If $\Delta_1(t) = 0$, let $h \downarrow 0$. If $\Delta_1(t) > M_*$, use $|\langle \nu_t, g_K \rangle| \le m(t) \le M_* \le \sqrt{M_* \Delta_1(t)}$. These three cases prove (13.10), for example with $C_0 = \max\{1, \exp(1)(m_1 + B_*)\}$. Since ϱ is fixed once and for all, this constant does not depend on any approximate data.

For (13.11), use (13.18) in the same identity and let $h \downarrow 0$. We may take $C_1 = a_+ C_\Gamma/2$. Hölder's inequality in time gives the other estimates, since

$$\left(\int_0^t (t-s)^{-r'/2}\,\mathrm{d}s\right)^{1/r'} = (1 - r'/2)^{-1/r'} t^{1/2 - 1/r} \quad (r > 2).$$

For (13.12), apply Aronson's bound directly to (13.6), without multiplying by $\exp(-x)$. After reversing time, the operator is

$$\frac{\partial}{\partial x}\left(\frac{a^{(j)}}{2}\frac{\partial}{\partial x}\right) + \frac{\frac{\partial}{\partial x}a^{(j)} - a^{(j)}}{2}\frac{\partial}{\partial x} + \frac{\frac{\partial^2}{\partial x^2}a^{(j)} - \frac{\partial}{\partial x}a^{(j)}}{2}.$$

The same coefficient assumptions hold, and the terminal datum $\mathcal{B} g_{K,h} = K \varrho_h(\cdot - \log K)$ has integral K. Hence $0 \le w_{j,h}(s,x) \le C'_\Gamma K/\sqrt{t-s}$. The duality identity, followed by $j \to \infty$ and $h \downarrow 0$, gives (13.12) with $C_2 = a_+ C'_\Gamma/2$.

For puts, use the payoff

$$\bar{g}_K(x) = (K - \exp(x))_+ - K = -\min\{K, \exp(x)\}.$$

It satisfies $|\bar{g}_K| \le \exp(x)$ and $\mathcal{B}\bar{g}_K = K \delta_{\log K}$. The smoothing and derivative estimates are the same in absolute value. Since $\nu_t(\mathbb{R}) = 0$, we have $\langle \nu_t, \bar{g}_K \rangle = \widehat{P}(t,K) - P(t,K)$. The same arguments apply uniformly for $K > 0$.

Finally, if $\nu_0 = 0$, $\mathfrak{r} = 0$ and $\mathfrak{d} \le 0$, the term $\int a\,\mathfrak{d}\,w_{j,h}$ in (13.8) is nonpositive. The coefficient error vanishes as $j \to \infty$, and the smoothing error vanishes as $h \downarrow 0$. This proves both comparisons. □

13.4 Application to solutions of the regularised equation

Fix $x_0 \in \mathbb{R}$ and a finite Borel function $v : [0, T_*] \times \mathbb{R}^2 \to (0, \infty)$. The target scalar diffusion is

$$\mathrm{d}Z_t = -\frac{1}{2}a(t, Z_t)\,\mathrm{d}t + \sqrt{a(t, Z_t)}\,\mathrm{d}W_t, \qquad Z_0 = x_0, \qquad \mu_t = \mathcal{L}(Z_t).$$

Proposition 3.2 gives this diffusion and its marginal equation with initial trace δ_{x_0}. The prices C, P in this subsection refer to this target law.

Theorem 13.3 (Bias and vanilla price error of denominator regularisation)**.** *Assume* (13.1)*, and let μ_t be the law of the scalar diffusion with coefficient a, starting from x_0. For each $0 < \varepsilon \le 1$, suppose that an approximate solution is given with joint density p_ε, marginal $\mu_t^\varepsilon = q_\varepsilon(t, x)\,\mathrm{d}x$ for almost every $t > 0$, and a narrowly continuous curve of laws starting from the same x_0. Suppose that its marginal satisfies exactly*

$$\frac{\partial q_\varepsilon}{\partial t} = \frac{1}{2}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial}{\partial x}\right)\left(a q_\varepsilon \frac{M_\varepsilon}{M_\varepsilon + \varepsilon}\right),$$
$$M_\varepsilon(t, x) = \frac{\int v(t, x, y) p_\varepsilon(t, x, y)\,\mathrm{d}y}{q_\varepsilon(t, x)}.$$

Assume $0 < M_\varepsilon < \infty$ where $q_\varepsilon > 0$, and set $M_\varepsilon = 1$ on $\{q_\varepsilon = 0\}$. For $0 < s \le 1$, set

$$B_s^\varepsilon(t) = \iint \exp(x) v(t, x, y)^{-s} p_\varepsilon(t, x, y)\,\mathrm{d}x\,\mathrm{d}y,$$
$$A_s^\varepsilon(t) = \iint v(t, x, y)^{-s} p_\varepsilon(t, x, y)\,\mathrm{d}x\,\mathrm{d}y.$$

Then the prices associated with these laws satisfy

$$C_\varepsilon(t, K) \le C(t, K), \qquad P_\varepsilon(t, K) \le P(t, K).$$

If the integral on the right is finite, then

$$\begin{aligned} 0 \le C(t, K) - C_\varepsilon(t, K) &\le C_1 \varepsilon^s \int_0^t \frac{B_s^\varepsilon(u)}{\sqrt{t - u}}\,\mathrm{d}u, \\ 0 \le P(t, K) - P_\varepsilon(t, K) &\le C_1 \varepsilon^s \int_0^t \frac{B_s^\varepsilon(u)}{\sqrt{t - u}}\,\mathrm{d}u. \end{aligned} \tag{13.20}$$

For each $K > 0$, we also have

$$\max\{C(t, K) - C_\varepsilon(t, K), P(t, K) - P_\varepsilon(t, K)\} \le C_2 K \varepsilon^s \int_0^t \frac{A_s^\varepsilon(u)}{\sqrt{t - u}}\,\mathrm{d}u, \tag{13.21}$$

whenever the integral is finite. In particular, $A_s^* = \sup_{\varepsilon \le 1} \operatorname{ess\,sup}_{u \le T_*} A_s^\varepsilon(u) < \infty$ gives the bound $2C_2 K_{\max} A_s^* \sqrt{t}\,\varepsilon^s$ for $0 < K \le K_{\max}$. If $B_s^* = \sup_{\varepsilon \le 1} \operatorname{ess\,sup}_{u \le T_*} B_s^\varepsilon(u) < \infty$, the error is at most $2C_1 B_s^* \sqrt{t}\,\varepsilon^s$, uniformly in $K > 0$. With only a uniform bound on $\int_0^{T_*} B_s^\varepsilon(u)\,\mathrm{d}u$, estimate (13.10) *gives the rate $\varepsilon^{s/2}$. Existence is an assumption of this theorem; Theorem 13.5 supplies it for a real-valued factor and price-independent variance satisfying* (13.24)*, under its other stated hypotheses.*

Proof. Set $\theta_\varepsilon = M_\varepsilon/(M_\varepsilon + \varepsilon)$, so that $0 \le \theta_\varepsilon \le 1$. We first verify the required moments without assuming convergence of p_ε. For $r > 1$ and $0 < \delta \le 1$, the bounded function $f_\delta(x) = \exp(rx)/(1 + \delta \exp(rx))$ satisfies

$$\mathcal{B} f_\delta = \frac{r \exp(rx)\big((r - 1) - (r + 1)\delta \exp(rx)\big)}{(1 + \delta \exp(rx))^3} \le r(r - 1) f_\delta.$$

Since $0 \le a\theta_\varepsilon \le a_+$, we have $(a\theta_\varepsilon/2)\mathcal{B}f_\delta \le a_+ r(r-1)f_\delta/2$, including where $\mathcal{B}f_\delta < 0$. For fixed δ, the derivatives of f_δ are bounded, so spatial truncation extends the weak formulation to this test function. Gronwall's lemma and monotone convergence as $\delta \downarrow 0$ give

$$\int \exp(rx)\mu_t^\varepsilon(\,\mathrm{d}x) \le \exp\bigl(rx_0 + a_+ r(r-1)t/2\bigr). \tag{13.22}$$

The same bound holds for μ_t. Thus the price moments in (13.9) are uniform.

By (12.5), the defect is

$$\mathfrak{d}_\varepsilon = -q_\varepsilon \frac{\varepsilon}{M_\varepsilon + \varepsilon} \le 0, \qquad \mathfrak{r} = 0, \qquad \nu_0 = 0.$$

In particular, $|\mathfrak{d}_\varepsilon| \le q_\varepsilon$ gives all the basic integrability conditions needed for Theorem 13.2. Multiplying (12.7) by $\exp(x)$ and integrating gives

$$d_1^\varepsilon(t) \le \varepsilon^s B_s^\varepsilon(t).$$

Without this weight, (12.6) gives $\|\mathfrak{d}_\varepsilon(t,\cdot)\|_1 \le \varepsilon^s A_s^\varepsilon(t)$. Theorem 13.2 now applies with $E_1 = 0$. Its kernel and strike estimates give (13.20)–(13.21); its root estimate uses $\Delta_1 \le \varepsilon^s \int_0^t B_s^\varepsilon$. The sign $\mathfrak{d}_\varepsilon \le 0$ gives both price comparisons. Finally, integration of $(t-u)^{-1/2}$ gives the stated bounds under uniform moments. □

The weighted condition on B_s^ε can be checked using the usual factor moments. Indeed, for every $r > 1$, Hölder's inequality and (13.22) give

$$B_s^\varepsilon(t) \le \exp\bigl(x_0 + a_+(r-1)T_*/2\bigr) \left(\iint v^{-sr/(r-1)} p_\varepsilon(t,x,y)\,\mathrm{d}x\,\mathrm{d}y \right)^{(r-1)/r}. \tag{13.23}$$

Thus a uniform bound on a negative moment of order strictly greater than s suffices for the bound uniform over all strikes. On a bounded strike range, (13.21) requires only the negative moment of order s. These moment conditions impose no uniform positive lower bound on M_ε. They must be verified for the regularised family under consideration and do not follow from convergence of the densities alone.

Proposition 13.4 (Conditional linear rate in the two main coefficient classes)**.** *Suppose that the families in Theorem 13.3 are occupation densities of weak joint solutions whose factor has the prescribed initial value and the unchanged factor equation of Theorem 2.1 or Theorem 2.2, respectively. Take $T_* = T$; the factor takes values in $\mathbb{R}$ or $(0,\infty)$, respectively, and in the second case densities are extended by zero outside that state space. Use the normalised variance v and the denominator $M_\varepsilon + \varepsilon$. Under the coefficient assumptions of the corresponding theorem, there is a constant $C_T < \infty$, independent of $\varepsilon \in (0,1]$, $t \in (0,T]$ and $K > 0$, such that*

$$0 \le C(t,K) - C_\varepsilon(t,K) \le C_T\sqrt{t}\,\varepsilon, \qquad 0 \le P(t,K) - P_\varepsilon(t,K) \le C_T\sqrt{t}\,\varepsilon.$$

Theorem 13.5 supplies these solutions in the price-independent real-factor class satisfying (13.24)*; their existence remains an assumption in the other cases of this proposition.*

Proof. In the real-factor case, the factor generator is unchanged by denominator regularisation. Apply the Lyapunov calculation of Section 6.1 on the line with $U = 1 + y^2$, $C_0 = 2C_d + \eta_+^2$ and $\lambda' = -C_0\lambda - 2\eta_+^2\lambda^2$, $\lambda(0) = 1$. Here $L_Y U \le C_0 U$ and $\eta^2|U'|^2 \le 4\eta_+^2 U$, so $(\partial_t + L_Y)e^{\lambda U} \le 0$. Stopping before $|Y| = N$ and then using Fatou, since the assumed continuous solutions exist on $[0,T]$, gives

$$\sup_{\varepsilon, t \le T} \mathbb{E} e^{\lambda(T)|Y_t^\varepsilon|^2} \le e^{1+y_0^2}.$$

Together with (H4), this bounds all inverse-variance moments uniformly in ε. In the positive-factor case, use the comparison function Φ of (10.6). The actual factor diffusion satisfies $s(t,y)^2 y \le s_+^2 y$, so the generator comparison (10.7) remains valid with $s(t,y)^2 y$ in place of $s(t,y)^2 y^2/(y+\varepsilon)$. The stopped comparison and Gamma-integral computation of Section 10.1.2 give

$$\sup_{\varepsilon, t\le T} \mathbb{E}[(Y_t^\varepsilon)^{-u}] < \infty \quad (0<u<\nu), \qquad \sup_{\varepsilon, t\le T} \mathbb{E}[(Y_t^\varepsilon)^{m}] < \infty \quad (m>0).$$

These estimates use no uniqueness of the factor law. Since $\nu > \gamma$, choose $p > 1$ with $\gamma p < \nu$; any $p > 1$ is admissible if $\gamma = 0$. The lower bound on v in (2.17) gives

$$v(t,y)^{-p} \le C_p(1 + y^{-\gamma p} + y^{m-p}),$$

so its expectation is uniformly bounded.

In either case there is an order $p > 1$ with a uniform inverse-variance moment. Take $r = p/(p-1)$ in (13.23) and set $s = 1$. The stock moment bound (13.22) and Hölder's inequality give $\sup_\varepsilon \operatorname{ess\,sup}_{t\le T} B_1^\varepsilon(t) < \infty$. Formula (13.20) with $s = 1$, and $\int_0^t (t-u)^{-1/2}\,\mathrm{d}u = 2\sqrt{t}$, prove the claim. □

Theorem 13.5 (Existence with a regularised denominator)**.** *Assume (H1)–(H3) and (H5), and let $v(t,y) = \exp(\ell(t,y))$, independent of x. Suppose that $\ell(t,\cdot)$ is locally Lipschitz for almost every t, that $\partial_y \ell \in \mathscr{G}_0$, and that $|\ell(t,y)| \le C_e + e y^2$ for every $e > 0$, uniformly in time. Assume in addition that $\ell(\cdot, y)$ is absolutely continuous on $[0,T]$ for almost every y, with a measurable weak derivative satisfying*

$$\begin{aligned} &|\partial_t \ell(t,y)| \le h(t) G(y), \qquad G \ge 0, \quad G \in \mathscr{G}_0, \\ &h \ge 0, \qquad \int_0^T h(t)\log(1+h(t))\,\mathrm{d}t < \infty. \end{aligned} \tag{13.24}$$

Use the continuous temporal representatives determined by these assumptions. Time-independent v is included by taking $h = 0$. Fix $T > 0$, $(x_0, y_0) \in \mathbb{R}^2$ and $\varepsilon > 0$. There exists a continuous weak solution on $[0,T]$ of

$$\begin{aligned} \mathrm{d}X_t^\varepsilon &= -\frac{1}{2} a(t, X_t^\varepsilon) \frac{v(t, Y_t^\varepsilon)}{M_\varepsilon(t, X_t^\varepsilon) + \varepsilon}\,\mathrm{d}t + \sqrt{\frac{a(t, X_t^\varepsilon) v(t, Y_t^\varepsilon)}{M_\varepsilon(t, X_t^\varepsilon) + \varepsilon}}\,\mathrm{d}W_t, \\ \mathrm{d}Y_t^\varepsilon &= b(t, Y_t^\varepsilon)\,\mathrm{d}t + \eta(t, Y_t^\varepsilon)\,\mathrm{d}B_t, \qquad \mathrm{d}\langle W, B\rangle_t = \rho(t, X_t^\varepsilon, Y_t^\varepsilon)\,\mathrm{d}t, \end{aligned} \tag{13.25}$$

starting from (x_0, y_0). Here M_ε is positive, finite and Borel, and

$$M_\varepsilon(t,x) = \mathbb{E}[v(t, Y_t^\varepsilon) \mid X_t^\varepsilon = x]$$

for $\mathrm{d}t\,\mathcal{L}(X_t^\varepsilon)(\mathrm{d}x)$-almost every (t,x). The occupation measure has a density p_ε, and, for every $0 < \delta < T$ and $1 < r < 2$,

$$p_\varepsilon^{1/r} \in L^r(\delta, T; W^{1,r}(\mathbb{R}^2)), \qquad p_\varepsilon \in L^1(\delta, T; W^{1,1}(\mathbb{R}^2)).$$

The process $\exp(X^\varepsilon)$ is a true martingale. The marginal $q_\varepsilon = \int p_\varepsilon\,\mathrm{d}y$ satisfies

$$\partial_t q_\varepsilon = \frac{1}{2}(\partial_x^2 + \partial_x)\left(a q_\varepsilon \frac{M_\varepsilon}{M_\varepsilon + \varepsilon}\right), \qquad q_\varepsilon\big|_{t=0} = \delta_{x_0}, \tag{13.26}$$

in the measure-valued weak sense. The point initial condition may be replaced by any μ_0 satisfying (2.8); then $q_\varepsilon|_{t=0} = \mu_0^X$ and $\mathbb{E}e^{X_t^\varepsilon} = \int e^x \mu_0^X(dx)$. Constants in the density estimates may depend on ε. The proof is given in Appendix D.

Corollary 13.6 (Uniform linear price bias)**.** *Under the assumptions of Theorem 13.5, choose such a solution for every $0 < \varepsilon \le 1$. Let $C_\varepsilon, P_\varepsilon$ be its call and put prices, and let C, P be the prices of the target scalar diffusion with the same initial price law (δ_{x_0} or μ_0^X). There is a constant $C_T < \infty$, independent of the chosen solutions, $\varepsilon \in (0,1]$, $t \in (0,T]$ and $K > 0$, such that*

$$0 \le C(t,K) - C_\varepsilon(t,K) \le C_T \varepsilon \sqrt{t}, \qquad 0 \le P(t,K) - P_\varepsilon(t,K) \le C_T \varepsilon \sqrt{t}.$$

Proof. Theorem 13.5 supplies the weak joint solutions required in the real-valued factor case of Proposition 13.4. Its proof gives inverse-variance moments of every finite order uniformly in ε, because the factor equation is unchanged and $v^{\pm 1} \in \mathscr{G}_0$. Together with (13.22) and (13.23), this gives a uniform bound on B_1^ε. For a point initial condition apply (13.20) with $s = 1$. For a general initial law, the Gaussian factor bound starts from (3.5), and the stock estimate has initial factor $\int e^{rx}\, \mu_0^X(\,\mathrm{d}x)$. Hölder with any inverse-variance moment of order greater than one gives the same uniform bound on B_1^ε. Theorem 13.2 applies with equal initial marginals, zero residual and defect $-q_\varepsilon \varepsilon/(M_\varepsilon + \varepsilon)$, giving the same estimate. Finally $\int_0^t (t-u)^{-1/2}\,\mathrm{d}u = 2\sqrt{t}$. □

The spatial representatives and exceptional sections in Theorem 13.5 follow the conventions of Section 2; changes on those space-time null sets are occupation-null. No time derivative of a, b, η, ρ or spatial derivative of b is required. The variance $v(t,y)$ is the one used with the fixed additive parameter ε. For a family regularised with $\mathsf{M}_\varepsilon + \varepsilon$, identity (2.14) replaces the regularisation parameter at time u by $\varepsilon e^{-c_0(u)}$. Applying (12.7) at each time gives the same bounds with $A_s^\varepsilon(u)$ and $B_s^\varepsilon(u)$ replaced by $e^{-sc_0(u)} A_s^\varepsilon(u)$ and $e^{-sc_0(u)} B_s^\varepsilon(u)$, computed for that family. The temporal integrability or uniform bounds required by the chosen price estimate must hold for these weighted moments. A lower bound on c_0 suffices when the corresponding bounds on A_s^ε or B_s^ε are available for that family; no such lower bound follows from (H4).

There is also a choice of regularisation that respects the chosen normalisation without any bound on c_0. Replace the denominator in the original variables by

$$\mathsf{M}_\varepsilon(t,x) + \varepsilon \exp(c_0(t)).$$

For each given joint law, (2.14) then gives exactly

$$\frac{\mathsf{v}(t,y)}{\mathsf{M}_\varepsilon(t,x) + \varepsilon \exp(c_0(t))} = \frac{v(t,y)}{M_\varepsilon(t,x) + \varepsilon}.$$

Consequently, any existing solution with this time-dependent regularisation satisfies the estimates for the normalised denominator, including Proposition 13.4 when its assumptions hold. If the normalised variance satisfies (13.24) and the factor is real-valued, Theorem 13.5 supplies existence for this choice under its hypotheses. The added term is positive, finite and Borel measurable; no time derivative of c_0 is used. This is a different regularisation from adding a constant ε to M_ε; it does not remove the weighted-moment requirement for that constant choice.

13.5 The price equation and the scope of the estimates

To express the defect in the equation for the price surface, suppose that the marginals have densities and set $U(t,k) = \widehat{C}(t, \exp(k)) - C(t, \exp(k))$. Derivatives are understood in distributions. We have

$$\left(\frac{\partial^2}{\partial k^2} - \frac{\partial}{\partial k} \right) U(t,k) = \exp(k)(\widehat{q} - q)(t,k).$$

For $\zeta \in C_c^\infty(\mathbb{R})$, the function $\Psi(x) = \int (\exp(x) - \exp(k))_+ \zeta(k)\,\mathrm{d}k$ satisfies $\mathcal{B}\Psi(x) = \exp(x)\zeta(x)$. The integrability conditions (13.9) permit the use of Ψ in (13.2) and the application of Fubini's theorem. It follows that

$$\frac{\partial U}{\partial t} = \frac{a(t,k)}{2} \left(\frac{\partial^2 U}{\partial k^2} - \frac{\partial U}{\partial k} \right) + \frac{a(t,k)}{2} \exp(k) \mathfrak{d}(t,k) + \int (\exp(x) - \exp(k))_+ \, \mathfrak{r}_t(\,\mathrm{d}x).$$

This is a linear equation with a source term; the preceding duality proofs treat the nonsmooth terminal payoff.

The exponential weight reflects payoff growth and is not controlled by unweighted L^1 convergence. The residual here must be a time-integrable signed measure with the stated moments. Proposition 5.3 controls only the projected residual on a fixed torus: its marginal is zero, and consistency and boundary terms remain to be estimated before applying the whole-space price bounds.

14 Scope of the results

The existence theorems establish neither uniqueness of the coupled system nor strong existence. For a positive factor, $\nu > \max\{1, \gamma+\omega\}$ ensures control of the integrated defect and a conditional moment of order greater than one; the critical CIR case $\nu = 1$ is not covered. Correlations of modulus one on a positive-time interval and endpoint regularity exponents also remain outside the results proved here. Theorem B.1 treats a vanishing or singular variance separately for a Brownian factor. No optimality of the sufficient conditions is asserted.

Theorem 13.5 treats the denominator $M + \varepsilon$ for a fixed $\varepsilon > 0$, a real-valued factor and price-independent $v(t, y)$ satisfying (13.24). It does not prove regularised existence for arbitrary time-dependent variance or Heston, exact calibration for $\varepsilon > 0$, or convergence of the joint laws as $\varepsilon \downarrow 0$. Corollary 13.6 concerns vanilla prices and uses moment bounds uniform in ε; the density estimates in the existence proof may depend on ε.

The stability and regularisation estimates in Section 12 concern densities and fluxes; they do not give a convergence rate for the density of a numerical solution. The Fourier construction establishes neither propagation of chaos for a particle system nor quantitative stability of prices of path-dependent products. In particular, a Hölder estimate for the fluxes does not give a Gronwall inequality for the difference of two solutions.

A Three extensions of the assumptions on the line

A.1 Borel variance with zero correlation

Theorem A.1 (Variance without a factor derivative)**.** *Retain (H1)–(H3). Suppose that $\rho = 0$ and replace (H4) by the following condition alone: $\mathsf{v}(t, x, y) = e^{c_0(t)} v(t, x, y) > 0$ is finite and Borel measurable, c_0 is finite and Borel measurable, $\log v$ is locally Lipschitz in x for almost every (t, y), $\partial_x \log v \in \mathscr{G}_0$, and*

$$\forall e > 0 \quad \exists C_e < \infty : \qquad |\log v(t, x, y)| \le C_e + e y^2.$$

No continuity or derivative of v in y is required. All conclusions of Theorem 2.1 remain valid.

Proof. Use the approximations of Section 3.4 with $\ell = \log v$: cutoff in both spatial coordinates, positive convolution, and exponentiation. Subquadratic growth of $|\ell|$ preserves the uniform growth bounds for $V_n^{\pm 1}$ and their almost-everywhere convergence; these operations require no derivative of ℓ. For fixed n, the periodic construction of Section 5 applies with $\rho_n = 0$.

Indeed,

$$k_{\rho_n} = \rho_n \partial_y \eta_n + \eta_n \partial_y \rho_n + \tfrac{1}{2} \rho_n \eta_n \partial_y \log V_n = 0,$$

because both ρ_n and $\partial_y \rho_n$ vanish. Thus the only entropy term containing a y-derivative of $\log V_n$ vanishes before any uniform estimate is taken. The original sources are

$$H_n = \iint p_n \left(b_n/\eta_n - \frac{\partial \eta_n}{\partial y} \right)^2, \qquad J_n = \int q_n \frac{(\partial a_n/\partial x + a_n + a_n S_n)^2}{a_n} K_n.$$

The Gaussian moments, conditional moments and score control these two terms. The additional source H_x of Section 4 is controlled by $\partial_x \log v \in \mathscr{G}_0$. The centering and root identities in that section use no y-derivative of v when $\rho = 0$. Coercivity and the resulting gradient estimate therefore give the same root bounds.

The compactness argument in Section 7.1 applies. Lemma 7.1, with moments of V_n of order greater than one, gives $V_n p_n \to vp$ in L^1 and identifies the ratio. The fluxes and reconstruction follow from Sections 7.3–8. Normalisation is as in (2.13).

□

A.2 Bounds local in strictly positive time

Theorem A.2 (Degeneracy at the initial time)**.** *Retain (H1), (H3), the global upper bound $0 < \eta(t,y) \le \eta_+$ for $t > 0$, and the global bound $|\ell(t,x,y)| \le C_e + ey^2$ from (H4). Assume the same local spatial regularity as in (H2), (H4), (H5), but only the following bounds: for every $\delta > 0$ and $e > 0$, there exist $C_{\delta,e} < \infty$ and $\rho_\delta < 1$ such that, on $[\delta, T]$,*

$$\eta^{-1} + \left|\frac{\partial \eta}{\partial y}\right| + |\partial_x \ell| + \left|\frac{\partial \ell}{\partial y}\right| + \left|\frac{\partial \rho}{\partial y}\right| \le C_{\delta,e} e^{ey^2}, \qquad |\rho| \le \rho_\delta. \tag{A.1}$$

The values at $t = 0$ may be chosen arbitrarily, finite and Borel measurable. The conclusions of Theorem 2.1 remain valid, with regularity constants depending on δ. No uniform rate as $\delta \downarrow 0$ is asserted.

Proof. Choose $s_N \downarrow 0$ in a full-measure set of admissible time sections and freeze (η, ℓ, ρ) at s_N on $[0, s_N)$, retaining a, b. For each N, smooth and periodise these coefficients as in Section 3.4, and apply Proposition 5.1. The common upper bound on η, the one-sided drift bound and the global subquadratic bound on ℓ give uniform Gaussian factor moments and all conditional-weight moments, independently of N and the torus. The scalar score bounds are also uniform.

Fix $\delta > 0$. For $s_N < \delta/8$ and convolution radius less than $\delta/8$, all coefficients used on $[\delta/2, T]$ satisfy the common bounds in (A.1) (using their constant admissible extensions at the upper endpoint). In particular the original entropy sources and the additional source H_x in Section 4 are uniformly bounded there, as is $\int \eta^{-2} p$. Periodic entropy coercivity and comparison between $\delta/2$ and δ give

$$\operatorname*{ess\,sup}_{[\delta,T]} E + \int_\delta^T D \, dt \le C_\delta.$$

Here D is the centred dissipation from that section. Its weak centering identities hold on every fixed torus, so the weighted gradient estimates give the same uniform root bounds for every $1 < r < 2$. This argument uses the periodic inequalities before the external limit and does not require passage of the entropy inequality to the whole space at this stage.

Choose a diagonal sequence of these periodic solutions with $N \to \infty$, radii tending to infinity and smoothing radii tending to zero. The coefficients converge locally almost everywhere: on every positive-time compact cylinder the unfrozen coefficients eventually equal the target coefficients. The flux masses remain bounded up to time zero by

$$\iint awp \le a_+, \qquad \iint \eta^2 p \le \eta_+^2, \qquad \iint |b| p \le C_T, \qquad \iint |\rho| \eta \sqrt{aw}\, p \le \eta_+ \sqrt{a_+}.$$

On each cylinder $(\delta, T) \times [-L, L]^2$, the gradient estimate and the bound for the time derivative in H^{-4} give strong L^1 compactness as in Section 7.1. Extract over integer L and a sequence $\delta \downarrow 0$. The common position moments make the spatial tails uniformly small, and the interval $(0, \delta)$ contributes at most 2δ to the L^1 distance. Thus the convergence holds on $(0, T) \times \mathbb{R}^2$. For every smooth compactly supported spatial test ϕ, the displayed flux bounds also give

$$|\langle \mu_t^n - \mu_s^n, \phi \rangle| \le C_\phi |t - s|,$$

with a constant independent of both approximation indices. This estimate and convergence of the initial laws preserve the initial trace in the narrowly continuous limiting curve. Weighted-product convergence identifies the conditional moment; conservation of mass identifies the calibrated flux, as in Section 7.3. Stochastic reconstruction proves the assertion, with constants depending on δ. □

A.3 Exponential growth and two drift conditions

Proposition A.3 (Entropy source involving the derivative of the drift)**.** *In the smooth periodic setting of Proposition 4.2, set*

$$H_{\partial b}(p) = \iint p\left(k_\rho^2 + \left|\frac{\partial}{\partial y}\eta\right|^2\right), \qquad B_-(p) = \iint p\left(\frac{\partial}{\partial y}b\right)_-.$$

Then

$$\langle \mathcal{F}_t(p), \log(p/q)\rangle + c_\alpha \mathcal{D}(p) \le \frac{2}{\alpha}H_{\partial b}(p) + B_-(p) + \frac{J(p)+H_x(p)}{2\alpha}. \tag{A.2}$$

The corresponding time-dependent inequality holds for the same periodic solution constructed in Proposition 5.1.

Proof. Periodic integration by parts in the factor flux gives

$$\iint bpg = \iint b\,\frac{\partial}{\partial y}p = -\iint \left(\frac{\partial}{\partial y}b\right)p.$$

With the notation $A = \sqrt{aw}(f-m)$, $B = \eta g$, the sum of the three fluxes is therefore

$$\begin{aligned} &-\frac{1}{2}\iint p(A^2 + 2\rho AB + B^2) + \iint p\left(-Ak_\rho - B\frac{\partial \eta}{\partial y}\right) \\ &\quad - \iint b_y p - \frac{1}{2}\int \mathcal{B}m - \frac{1}{2}\iint awp(f-m)(L-n). \end{aligned}$$

The quadratic form is bounded below by $\alpha(A^2+B^2)$. The Cauchy–Schwarz and Young inequalities give

$$\sqrt{\mathcal{D}H_{\partial b}} \le \frac{\alpha}{8}\mathcal{D} + \frac{2}{\alpha}H_{\partial b}, \qquad \tfrac{1}{2}\sqrt{\mathcal{D}J} \le \frac{\alpha}{8}\mathcal{D} + \frac{J}{2\alpha}, \qquad -\iint \left(\frac{\partial}{\partial y}b\right)p \le B_-(p).$$

The remaining covariance is bounded by $\alpha\mathcal{D}/8 + H_x/(2\alpha)$ as in Proposition 4.2; it vanishes when V is independent of x. This proves (A.2) with the same choice of c_α. At each implicit step, the same entropy test gives this inequality with the coefficients of that step. On a fixed torus, $\frac{\partial}{\partial y}b$ is bounded and the discrete coefficients converge uniformly; strong convergence of the densities passes B_- and $H_{\partial b}$ to the limit. Strong convergence of the entropy and lower semicontinuity of the dissipation are as in the proof of Proposition 5.1. The same solution therefore inherits both entropy inequalities, with no additional regularity assumption on the solution. □

Proposition A.4 (A free parameter in the Gaussian moment bound)**.** *For the approximations of Section 6.1, set*

$$\Gamma_T = \frac{c_* C_*}{2\eta_+^2(e^{C_* T}-1)}, \qquad c_* = 4/\pi^2, \tag{A.3}$$

where C_* is the constant from Section 6.1. For every $0 < \Gamma < \Gamma_T$, there exists $C_{\Gamma,T} < \infty$ such that

$$\sup_n \sup_{t\le T} \int e^{\Gamma y^2}\, d\mu_t^n \le C_{\Gamma,T}. \tag{A.4}$$

This assertion uses only the upper bound on η, the one-sided drift bound, and Gaussian initial data with variance $\varepsilon_n \downarrow 0$.

Proof. The calculation of Section 6.1 applies with

$$\lambda_{\lambda_0}(t) = \frac{\lambda_0 e^{-C_* t}}{1 + (2\eta_+^2 \lambda_0 / C_*)(1 - e^{-C_* t})}, \qquad \lambda_0 > 0.$$

For $\varepsilon_n \le (4\lambda_0)^{-1}$, the initial integral is at most $\sqrt{2}\exp\{\lambda_0(1 + 2y_0^2)\}$. There are only finitely many remaining indices, and their tori are compact. Dividing the numerator and denominator in $\lambda_{\lambda_0}(T)$ by λ_0 shows that

$$c_* \lambda_{\lambda_0}(T) = \frac{c_* e^{-C_* T}}{\lambda_0^{-1} + (2\eta_+^2 / C_*)(1 - e^{-C_* T})} \uparrow \frac{c_* C_*}{2\eta_+^2 (e^{C_* T} - 1)} = \Gamma_T.$$

Given $\Gamma < \Gamma_T$, choose a finite λ_0 for which $\Gamma < c_* \lambda_{\lambda_0}(T)$. The Lyapunov estimate then controls $e^{\Gamma y^2}$ for every sufficiently large n. Increasing the constant to include the finitely many remaining compact tori proves (A.4). No moment at Γ_T is asserted. □

Theorem A.5 (Controlled quadratic growth)**.** *Retain (H1), $0 < \eta \le \eta_+$, $yb(t,y) \le C_d(1 + y^2)$, $|\rho| \le \rho_* < 1$, and the local spatial regularity of η, ℓ, ρ in (H2), (H4), (H5). Assume also $\partial_x \ell \in \mathscr{G}_0$, uniformly in (t,x); this is automatic when v is independent of x. Suppose, with nonnegative rates and constants uniform in the other variables, that*

$$v \le Ce^{L_+ y^2}, \qquad v^{-1} \le Ce^{L_- y^2}, \qquad \eta^{-1} \le Ce^{L_\eta y^2},$$
$$|b| \le Ce^{L_b y^2}, \qquad \left|\frac{\partial}{\partial y}\eta\right| \le Ce^{L_d y^2}, \qquad \left|\frac{\partial}{\partial y}\ell\right| \le Ce^{L_\ell y^2}, \qquad \left|\frac{\partial}{\partial y}\rho\right| \le Ce^{L_\rho y^2}.$$

The normalisation $\mathsf{v} = e^{c_0(t)} v$ is allowed as in (H4). Set

$$L = L_+ + L_-, \qquad B = \max\{L, 2L_\eta\}, \qquad Q_0 = \max\{2(L_b + L_\eta), 2L_d, 2L_\ell, 2L_\rho\}.$$

The constant Γ_T is defined by (A.3)*, with the constant D_0 from* (3.9)*. Assume one of the following two conditions:*

(E1) $\max\{B, Q_0\} < \Gamma_T$*;*

(E2) *$b(t,\cdot)$ is locally absolutely continuous for almost every t, its weak derivative satisfies $\left(\frac{\partial}{\partial y} b\right)_- \le Ce^{D_b y^2}$, and, with*

$$Q_1 = \max\{L_b, D_b, 2L_d, 2L_\ell, 2L_\rho\}, \qquad \max\{B, Q_1\} < \Gamma_T.$$

Then there exists a calibrated weak solution starting from (x_0, y_0), with the marginal, occupation-density and martingale conclusions of Theorem 2.1. For this same solution,

$$p^{1/r} \in L^r(\delta, T; W^{1,r}(\mathbb{R}^2)), \qquad p \in L^1(\delta, T; W^{1,1}(\mathbb{R}^2)) \quad \left(0 < \delta < T,\ 1 < r < \frac{2\Gamma_T}{\Gamma_T + B}\right). \tag{A.5}$$

When $B = 0$, the upper endpoint of the interval is 2*.*

Proof. We begin with the periodic approximations and keep track of the available growth margins. For fixed n, the smoothed and periodised coefficients are strictly positive where required and satisfy the assumptions of Proposition 5.1. The solutions p_n therefore exist before the following estimates are made. Proposition A.4 still applies: its proof requires no subquadratic growth bound on v or its derivatives. It gives (A.4) for every $\Gamma < \Gamma_T$.

Smoothing preserves the rates up to an arbitrarily small loss. Indeed, for $|h| \le 1$ and $\epsilon > 0$,

$$(y - h)^2 \le (1 + \epsilon) y^2 + 1 + \epsilon^{-1}, \qquad |y|^m \le C_{m,\epsilon} e^{\epsilon y^2}.$$

For η^{-1}, Jensen's inequality gives $(\eta * \varphi_n)^{-1} \le \eta^{-1} * \varphi_n$; joining to η_+ does not decrease η_n. For ℓ_n, the bounds $-\log C - L_- y^2 \le \ell \le \log C + L_+ y^2$ pass through the cutoff and convolution. Derivatives of the cutoffs add only polynomial factors. Each loss is chosen after fixing the exponents needed below; strict inequalities make these choices possible.

We first choose a conditional-moment exponent strictly greater than one. For $h \ge 1$ and $A^{-1} + (A')^{-1} = 1$, Proposition 4.1 gives

$$\int q_n (1 + K_n)^h, \quad \iint w_n^{\pm h} p_n \le \left(\iint V_n^{hA} p_n \right)^{1/A} \left(\iint V_n^{-hA'} p_n \right)^{1/A'}.$$

These moments are bounded whenever $hL < \Gamma < \Gamma_T$: choose

$$\frac{hL_+}{\Gamma} < \frac{1}{A} < 1 - \frac{hL_-}{\Gamma}.$$

The interval is nonempty, and the choice can be made in $(0, 1)$, including when one rate is zero. Fix $\Gamma_0 \in (\max\{B, Q_i\}, \Gamma_T)$, where $i = 0$ in (E1) and $i = 1$ in (E2). Choose $h > 1$ with $hL < \Gamma_0$; if $L = 0$, take $h = 2$. With $h' = h/(h-1)$ and the score of order $2h'$ from Proposition 3.5,

$$\begin{aligned} J_n(t) &\le C \left(\int q_n K_n^h \right)^{1/h} \left(\int q_n (1 + S_n^2)^{h'} \right)^{1/h'} \\ &\le C_h (1 + t^{-h'})^{1/h'} \le C_h (1 + t^{-1}). \end{aligned} \tag{A.6}$$

No second moment of K_n is required.

We now estimate the entropy sources under each of the two drift assumptions. Under (E1), the squares of $k_{\rho,n}$ and $b_n/\eta_n - \frac{\partial}{\partial y}\eta_n$ are bounded by $C_\epsilon e^{(Q_0+\epsilon)y^2}$. Under (E2), use (A.2). The weak derivative of b is locally integrable in (t, y): on a compact spatial interval, its total variation is bounded by its negative variation and two values of b, all controlled uniformly in time by the growth bounds. For the cutoff $\chi_n(y) = \chi(y/n)$ and the positive kernel,

$$\begin{aligned} \left(\frac{\partial}{\partial y} b_n \right)_- &\le \left[\chi_n \left(\frac{\partial}{\partial y} b \right)_- + \left| \frac{\partial}{\partial y} \chi_n \right| |b| \right] * \varphi_n \\ &\le C_\epsilon \exp\{(\max\{D_b, L_b\} + \epsilon) y^2\}. \end{aligned}$$

Zero extension in time introduces no additional spatial derivative. The other squares in the source have rates at most $\max\{2L_d, 2L_\ell, 2L_\rho\}$, up to the same loss. In both cases, (A.4) at Γ_0 controls the source and $\iint \eta_n^{-2} p_n$, since $2L_\eta < \Gamma_0$.

For price-dependent variance use Proposition 5.1 with the dissipation of Section 4. Its additional source satisfies, with the exponent $h > 1$ chosen above and $h' = h/(h-1)$,

$$H_x \le a_+ \left(\iint w_n^h p_n \right)^{1/h} \left(\iint |\partial_x \log V_n|^{2h'} p_n \right)^{1/h'} \le C.$$

The first factor is controlled by the conditional moments already proved; the second by the Gaussian moment and $\partial_x \ell \in \mathscr{G}_0$. The cutoff derivative in x adds only a polynomial in y, so smoothing preserves this bound. The drift integration by parts in (E2) is unaffected, with the same additional H_x term. Thus the strict growth budgets require no extra exponential rate for this derivative; the entropy dissipation coefficient can be reduced to $\alpha/8$. Coercivity (6.11), the entropy lower bound from the y^2 moment, and the comparison in Section 6.4 yield

$$\operatorname*{ess\,sup}_{t \in [\delta, T]} E_n(t) + \int_\delta^T \mathcal{D}_n(t)\, dt \le C_\delta.$$

In the price-dependent case, $\mathcal{D}_n$ here denotes the dissipation of Section 4. These estimates apply directly to the periodic solutions. Passage of the entropy inequality to the whole space is not used here.

We choose the Sobolev exponent using the remaining moment bounds. Fix $1 < r < 2\Gamma_T/(\Gamma_T + B)$ and $\beta = r/(2-r)$. Then $\beta B < \Gamma_T$. Choose $\Gamma_r \in (\max\{\Gamma_0, \beta B\}, \Gamma_T)$. The preceding Gaussian and conditional moments control

$$\iint p_n w_n^{-\beta}, \qquad \int q_n K_n^{\beta}, \qquad \iint p_n \eta_n^{-2\beta},$$

since $\beta L < \Gamma_r$ and $2\beta L_\eta < \Gamma_r$. Proposition 4.3 gives bounds for the roots and gradients of p_n on (δ, T).

We finally identify the limiting fluxes and reconstruct the stochastic process. In both cases, $L_b < \Gamma_0$ and $L_+ < \Gamma_0$. There therefore exist $s_b, s_v > 1$ such that $s_b L_b < \Gamma_0$ and $s_v L_+ < \Gamma_0$. The corresponding moments control $b_n p_n$ and $V_n p_n$ by Lemma 7.1. The flux masses are uniform up to $t = 0$: the calibrated flux has mass one, η_n is bounded, the mixed flux is controlled by the Cauchy–Schwarz inequality, and the drift by its moment of order s_b.

The gradients and these time estimates give the compactness of Section 7.1. Position moments control the tails. The variance moment converges strongly and identifies M; conservation of mass in Section 7.3 gives the calibrated and mixed fluxes. The initial trace and continuous laws are as in Section 7.5. Proposition 8.1 applies. Choose a countable family of cylinders and an increasing sequence of exponents r. Any further extraction of gradients concerns the same limiting density; interpolation and Hölder's inequality extend the bounds to every exponent in (A.5). No moment at Γ_T or regularity at the endpoint exponent has been used. □

Remark A.6. Condition (E2) is an alternative to (E1): it requires a weak spatial derivative of b, but replaces control of b^2/η^2 by control of $\left(\frac{\partial}{\partial y} b\right)_-$. Condition (E1) remains available without a derivative of b. The constant Γ_T depends on the time horizon and coefficients; no optimality is asserted.

B Brownian factor and variance vanishing or singular at a point

Theorem B.1 (Powers of the factor and compensated correlation)**.** *Let a satisfy (H1), $(x_0, y_0) \in \mathbb{R}^2$, $-1 < \theta < 1$, and let $\varrho : [0,T] \times \mathbb{R} \to \mathbb{R}$ be Borel measurable with $|\varrho| \le \rho_* < 1$. Define*

$$v_\theta(y) = |y|^\theta \quad (y \neq 0), \qquad v_\theta(0) = \begin{cases} 0, & \theta > 0, \\ 1, & \theta \le 0, \end{cases} \qquad \rho(t,x,y) = \varrho(t,x)\frac{y}{\sqrt{1+y^2}}.$$

There exists a continuous weak solution, starting from (x_0, y_0), of

$$\begin{aligned} dX_t &= -\tfrac{1}{2} a(t, X_t)\frac{v_\theta(Y_t)}{M(t, X_t)}dt + \sqrt{a(t, X_t)\frac{v_\theta(Y_t)}{M(t, X_t)}}\, dW_t, \\ dY_t &= dB_t, \qquad d\langle W, B\rangle_t = \rho(t, X_t, Y_t)dt, \\ M(t,x) &= \mathbb{E}[v_\theta(Y_t) \mid X_t = x] \quad \text{for } dt\,\mathcal{L}(X_t)(dx)\text{-almost every } (t,x). \end{aligned} \tag{B.1}$$

The conditional moment has a positive finite Borel version; its value at $t = 0$ may be chosen arbitrarily positive. At every $t > 0$, X_t has density $q(t,\cdot)$ from (2.10), and Y_t has law $\mathcal{N}(y_0, t)$. The price e^X is a true martingale. The occupation density p satisfies

$$p^{1/r} \in L^r(\delta, T; W^{1,r}(\mathbb{R}^2)), \qquad p \in L^1(\delta, T; W^{1,1}(\mathbb{R}^2)) \quad \left(0 < \delta < T,\ 1 < r < \frac{2}{1+|\theta|}\right). \tag{B.2}$$

Multiplying v_θ by $e^{c_0(t)}$, with c_0 finite and Borel measurable, preserves these conclusions after normalisation.

Proof. We approximate the variance and correlation together so that their compensation is preserved. Take $R_n = 8(n+1)$, $0 < \varepsilon_n \le 1$ tending to zero, and the coefficients a_n and marginals q_n from Section 3.4. Set

$$\omega_n = \frac{\pi}{2R_n}, \qquad F_n(y) = \frac{\sin(\omega_n y)}{\omega_n}, \qquad V_n(y) = (\varepsilon_n + F_n(y)^2)^{\theta/2}, \qquad b_n = 0, \quad \eta_n = 1.$$

Let $\varrho_n(t,x)$ be a positive convolution of ϱ, extended by zero outside the time interval, then multiplied by an x-cutoff equal to one on $[-R_n/2, R_n/2]$ and zero near the join. Along a subsequence, $\varrho_n \to \varrho$ almost everywhere, and $|\varrho_n| \le \rho_*$. Define

$$\rho_n(t,x,y) = \varrho_n(t,x)\frac{F_n(y)F_n'(y)}{\sqrt{1+F_n(y)^2}}.$$

Although F_n is antiperiodic with period $2R_n$, F_n^2 and F_nF_n' are periodic. All coefficients are therefore smooth on the torus, with $V_n > 0$ and $|\rho_n| \le \rho_*$. Proposition 5.1, with periodised Gaussian initial data, constructs p_n and its curve of laws μ^n. For $y \neq 0$, $V_n(y) \to v_\theta(y)$ and $\rho_n(t,x,y) \to \rho(t,x,y)$ almost everywhere. The line $y = 0$ is Lebesgue-null.

We first identify the factor marginal and its moments. Integration of the joint equation in x cancels all x-fluxes, including the mixed flux. Thus $r_n(t,y) = \int p_n(t,x,y)dx$ solves $\frac{\partial}{\partial t} r_n = \frac{1}{2}\frac{\partial^2}{\partial y^2} r_n$, with initial data the periodised Gaussian of mean y_0 and variance ε_n. Periodic scalar uniqueness identifies r_n. Its Fourier series gives, for $t > 0$,

$$\begin{aligned}\|r_n(t)\|_\infty &\le \frac{1}{2R_n}\left(1 + 2\sum_{k\ge 1} e^{-k^2\pi^2(t+\varepsilon_n)/(2R_n^2)}\right)\\ &\le \frac{1}{2R_n} + \frac{1}{\sqrt{2\pi(t+\varepsilon_n)}}.\end{aligned} \tag{B.3}$$

The last inequality uses the decreasing-integrand bound $\sum_{k\ge1} e^{-ck^2} \le \int_0^\infty e^{-cz^2}\,\mathrm{d}z = \sqrt{\pi}/(2\sqrt{c})$. The marginal is that of the centred projection of $y_0 + \sqrt{t+\varepsilon_n}G$, which decreases absolute value; its polynomial moments are therefore uniform.

On $[-R_n, R_n]$,

$$c_* y^2 \le F_n(y)^2 \le y^2, \qquad c_* = 4/\pi^2.$$

Write $\vartheta = |\theta|$. If $s\vartheta < 1$, the singular weight is bounded by $c_*^{-s\vartheta/2}|y|^{-s\vartheta}$, and the growing weight by $(1+y^2)^{s\vartheta/2}$. Using (B.3) on $|y| < 1$ and unit mass elsewhere gives

$$\sup_n \sup_{\delta\le t\le T} \iint (V_n^s + V_n^{-s})p_n < \infty \qquad (\delta > 0,\ s\vartheta < 1). \tag{B.4}$$

Indeed, the singular integral equals $\int_{-1}^1 |y|^{-s\vartheta}dy = 2/(1-s\vartheta)$. The nonsingular weight has moments of every order. For $\theta = 0$, both weights equal one.

For $h\vartheta < 1$, Proposition 4.1 applies with a conjugate exponent close to one for the singular weight: $1 < A' < 1/(h\theta)$ if $\theta > 0$, or $1 < A < 1/(h|\theta|)$ if $\theta < 0$. The other weight has moments of every order. For $\theta = 0$, take $A = A' = 2$. Thus

$$\sup_n \sup_{\delta\le t\le T} \left\{\int q_n(1+K_n)^h + \iint (w_n^h + w_n^{-h})p_n\right\} < \infty, \qquad h\vartheta < 1. \tag{B.5}$$

In particular, such an $h > 1$ exists since $\vartheta < 1$.

We now verify the uniform bound resulting from the compensation. Since $\eta_n = 1$ and $b_n = 0$, we have $d_n = 0$ and $k_n = \frac{\partial}{\partial y}\rho_n + \rho_n \frac{\partial}{\partial y}\log V_n/2$. The identities $F_n'' = -\omega_n^2 F_n$, $|F_n'| \le 1$ give

$$\begin{aligned}\frac{\partial}{\partial y}\rho_n &= \varrho_n\left(\frac{(F_n')^2}{(1+F_n^2)^{3/2}} - \frac{\omega_n^2 F_n^2}{\sqrt{1+F_n^2}}\right),\\ \rho_n\frac{\partial}{\partial y}\log V_n &= \varrho_n\theta\frac{F_n^2(F_n')^2}{\sqrt{1+F_n^2}(\varepsilon_n + F_n^2)}.\end{aligned} \tag{B.6}$$

Since $\omega_n^2|F_n| \le \omega_n \le \pi/2$, these expressions are uniformly bounded. Consequently, $H_n = \iint k_n^2 p_n \le C$. The bound $|\rho_n| \le \rho_* < 1$ preserves coercivity of the covariance. The derivative of $\log V_n$ alone is not estimated uniformly.

These moment and source bounds give entropy and gradient estimates away from time zero. Fix $\delta > 0$ and $h > 1$ with $h\vartheta < 1$. Hölder estimate (A.6), applied on $[\delta/4, T]$ with (B.5), gives $J_n \le C_\delta$. Periodic coercivity (6.11) applies with $Q_n = \iint \eta_n^{-2} p_n = 1$. The quadratic moment of the factor gives a uniform lower bound on E_n. The comparison in Section 6.4, first between $\delta/4$ and $\delta/2$, followed by integration of the entropy inequality, gives

$$\operatorname*{ess\,sup}_{t\in[\delta,T]} E_n(t) + \int_\delta^T \mathcal{D}_n(t)dt \le C_\delta.$$

For $1 < r < 2/(1+\vartheta)$, $\beta = r/(2-r)$ satisfies $\beta\vartheta < 1$. Proposition 4.3 applies to the periodic solutions with (B.5) at $h = \beta$ and $\iint \eta_n^{-2\beta} p_n = 1$. It supplies the gradient bounds needed for compactness. No uniform moment at $t = 0$ of singular V_n^{-s} or V_n^s has been assumed.

We use these estimates to pass to the limit and retain the initial trace. The flux masses satisfy, on all of $[0, T]$,

$$\iint a_n w_n p_n \le a_+, \qquad \iint \eta_n^2 p_n = 1, \qquad \iint |\rho_n| \sqrt{a_n w_n}\, p_n \le \rho_* \sqrt{a_+}.$$

The marginal moments control the tails. Sections 7.1–7.5 give strong compactness, a continuous curve of laws, and its trace $\delta_{(x_0,y_0)}$. The first marginal is q by (3.11); the second is $\mathcal{N}(y_0, t)$ by convergence of the projected Gaussians.

On $[\delta, T]$, choose $1 < s < 1/\vartheta$ (any $s > 1$ if $\vartheta = 0$). Lemma 7.1 and (B.4) give $V_n p_n \to v_\theta p$ in L^1. The conditional moment is positive and finite since $q > 0$ and $v_\theta > 0$ almost everywhere in y. Conservation of mass in Section 7.3 then gives $w_n p_n \to wp$ in L^1 on $(0, T)$; products of square roots give the mixed flux. The gradients pass to the same density by lower semicontinuity and give (B.2).

We finish by checking that the value assigned to the variance at zero does not affect reconstruction. To apply Proposition 8.1, temporarily replace $v_\theta(0)$ by 1 when $\theta > 0$. This changes neither the conditional moment nor the fluxes, since the occupation measure has a density. After reconstruction, $Y = y_0 + B$, so Tonelli's theorem gives

$$\mathbb{E}\int_0^T \mathbf{1}_{\{Y_t=0\}} dt = \int_0^T \mathbb{P}(Y_t = 0)dt = 0.$$

Returning to $v_\theta(0) = 0$ changes neither the drift integrals nor the stochastic integrals: the bracket of their difference is zero. When $\theta < 0$, the finite value assigned at zero is likewise immaterial. The two Brownian motions and their covariation are those of the reconstruction proposition. Calibration and the martingale property follow, and (2.13) handles $e^{c_0(t)}$. □

Remark B.2. For $\theta > 0$, the variance vanishes at a state the factor may visit; for $\theta < 0$, it is unbounded in every neighbourhood of that state. The correlation may be nonzero away from zero and merely Borel measurable in (t, x). Here $|\theta| < 1$ ensures a conditional moment of order greater than one; the cases $|\theta| = 1$ are not covered. Compensation (B.6) does not justify a nonzero constant correlation near the singularity.

C Factors on an interval and changes of variable

Theorem C.1 (Transferring the existence result)**.** *Let I be an open interval and $F : I \to \mathbb{R}$ a strictly increasing C^2 bijection with $F' > 0$. Write $G = F^{-1}$. For Borel coefficients*

$b(t,y), \eta(t,y), \mathsf{v}(t,x,y), \rho(t,x,y)$ with $y \in I$, with $\eta, \mathsf{v} > 0$, define, for $y = G(z)$,

$$\begin{aligned} \widetilde{\eta}(t,z) &= F'(y)\eta(t,y), \\ \widetilde{b}(t,z) &= F'(y)b(t,y) + \tfrac{1}{2}F''(y)\eta(t,y)^2, \\ \widetilde{\mathsf{v}}(t,x,z) &= \mathsf{v}(t,x,y), \qquad \widetilde{\rho}(t,x,z) = \rho(t,x,y). \end{aligned} \tag{C.1}$$

Suppose that the transformed coefficients satisfy (H1)–(H5), with initial point $(x_0, F(y_0))$, $y_0 \in I$. Then there exists a weak solution of model (2.11) on $\mathbb{R}\times I$, with factor $dY = b(t,Y)dt + \eta(t,Y)dB$ remaining in I on $[0,T]$. Calibration to q and the martingale property are preserved. If P is the density constructed in (x,z), the density in (x,y) is

$$p_Y(t,x,y) = P(t,x,F(y))F'(y). \tag{C.2}$$

For every $0 < \delta < T$ and $1 < r < 2$, its roots belong to

$$L^r(\delta, T; W^{1,r}_{\mathrm{loc}}(\mathbb{R}\times I)).$$

The same regularity holds globally in space, on each interval (δ, T), if, with $J = G'$,

$$J^{-1}, \quad J'/J^2 \in \mathscr{G}_0 \quad \text{as functions of } z. \tag{C.3}$$

Proof. Theorem 2.1 gives (X, Z, W, B) with the transformed coefficients. Set $Y = G(Z)$. The compact range of each continuous path of Z maps into a compact subset of I, so Y reaches neither boundary on $[0,T]$. Since $G' = 1/F'$ and $G'' = -F''/(F')^3$, localised Itô's formula and exhaustion give

$$G'\widetilde{b} + \tfrac{1}{2}G''\widetilde{\eta}^2 = b + \frac{F''\eta^2}{2F'} - \frac{F''\eta^2}{2F'} = b, \qquad G'\widetilde{\eta} = \eta.$$

Thus the required factor equation and covariation hold. The identity $\widetilde{\mathsf{v}}(t,X,Z) = \mathsf{v}(t,X,Y)$ preserves the conditional expectation and price ratio; X, its marginal and its martingale property are unchanged.

The change of variable $z = F(y)$ in the occupation integral gives (C.2). On compact subsets of I, $F', 1/F', F''$ are bounded, so the Sobolev composition and product rules applied to $p_Y^{1/r} = P^{1/r}(t,x,F(y))(F'(y))^{1/r}$ give the stated local regularity on (δ,T). To prove global regularity, we retain the weights introduced by the change of variable. With $h = P^{1/r}$ and $y = G(z)$, we have $p_Y^{1/r}(x,G(z)) = h(x,z)J(z)^{-1/r}$. Thus

$$\int \left|\frac{\partial}{\partial x} p_Y^{1/r}\right|^r dx\, dy = \int \left|\frac{\partial}{\partial x} h\right|^r dx\, dz, \tag{C.4}$$

$$\int \left|\frac{\partial}{\partial y} p_Y^{1/r}\right|^r dx\, dy = \int J^{-r} \left|\frac{\partial}{\partial z} h - \frac{J'}{rJ} h\right|^r dx\, dz. \tag{C.5}$$

Fix $r < s < 2$. The chain rule $\frac{\partial}{\partial z}P^{1/r} = (s/r)P^{1/r-1/s}\frac{\partial}{\partial z}P^{1/s}$ and Hölder's inequality in space-time give

$$\int J^{-r}\left|\frac{\partial}{\partial z}P^{1/r}\right|^r \le (s/r)^r \left(\int \left|\frac{\partial}{\partial z}P^{1/s}\right|^s\right)^{r/s} \left(\int P J^{-rs/(s-r)}\right)^{1-r/s}.$$

The first factor is finite by Theorem 2.1. The second is finite by (C.3) and the Gaussian moment. The remaining term in (C.5) is at most $r^{-r}\int P|J'/J^2|^r$, which is also finite. The mass gives the L^r norm of the root. Finally, $\nabla p_Y = r(p_Y^{1/r})^{r-1}\nabla p_Y^{1/r}$ and Hölder's inequality give $p_Y \in L^1_t W^{1,1}$ globally. □

Remark C.2 (Transfer of other classes)**.** The stochastic calculation (C.1) and density formula (C.2) also transfer Theorems A.1, A.2, A.5. Local regularity retains the range of exponents in the theorem used. For a global conclusion, the moments of the weights in (C.5) must be checked; it does not follow from the C^2 regularity of the change of variable alone.

D Existence with a regularised denominator

We prove Theorem 13.5 under (13.24), with initial law (2.8). The time-independent case is obtained by setting $h = 0$. On a bounded interval in y, the temporal derivative bound gives

$$\sup_{|y|\le R} |\ell(t,y) - \ell(s,y)| \le C_R \int_s^t h(u)\,\mathrm{d}u.$$

This holds first for almost every y and admissible times. The uniformly locally Lipschitz spatial representatives extend it in y; completion in time supplies the continuous temporal representatives and endpoint traces. The budget $\int h\log(1+h) < \infty$ implies $h \in L^1(0,T)$. Fix $\varepsilon > 0$, suppress its superscript on the process and density, and set $\alpha = 1 - \rho_* > 0$. All spatial representatives are understood with the almost-everywhere conventions following that theorem.

D.1 The entropy identity

Initially work with smooth positive coefficients and a smooth positive probability density p on $\mathbb{T}_{L_x} \times \mathbb{T}_{L_y}$, with $v = v(t,y)$. Let

$$q = \int p\,\mathrm{d}y, \qquad M = \frac{\int vp\,\mathrm{d}y}{q}, \qquad D = M + \varepsilon,$$
$$h = \partial_x \log p, \qquad g = \partial_y \log p, \qquad S = \partial_x \log q.$$

The nonlinear forward operator is

$$\mathcal{F}_\varepsilon(p) = \tfrac{1}{2}(\partial_x^2 + \partial_x)(avp/D) + \partial_x\partial_y(\rho\eta\sqrt{av/D}\,p) + \tfrac{1}{2}\partial_y^2(\eta^2 p) - \partial_y(bp). \tag{D.1}$$

Define the convex functional and its first variation by

$$\mathcal{H}_\varepsilon(p) = \iint \frac{p\log p}{v}\,\mathrm{d}x\,\mathrm{d}y + \frac{1}{\varepsilon}\int q\log q\,\mathrm{d}x, \qquad z_p = \frac{\log p + 1}{v} + \frac{\log q + 1}{\varepsilon}. \tag{D.2}$$

The notation $\mathcal{H}_{\varepsilon t}$ records its time dependence when needed. For smooth coefficients,

$$\frac{d}{dt}\mathcal{H}_{\varepsilon t}(p_t) = \langle \mathcal{F}_{\varepsilon t}(p_t), z_{p_t}\rangle + \int p_t \log p_t\, \partial_t(v^{-1}). \tag{D.3}$$

The following identity concerns its spatial first variation at fixed time. Set

$$A = \sqrt{a/D}\,h, \qquad B = \eta g/\sqrt{v}, \qquad \mathcal{I}_\varepsilon(p) = \iint p(A^2 + B^2),$$
$$k = \rho\partial_y\eta + \eta\partial_y\rho + \tfrac{1}{2}\rho\eta\partial_y\ell, \qquad d = b/\eta - \partial_y\eta.$$

Lemma D.1 (Cancellation of the derivative of the conditional moment)**.** *In the smooth periodic setting above,*

$$\begin{aligned}\langle \mathcal{F}_\varepsilon(p), z_p\rangle = &-\tfrac{1}{2}\iint p(A^2 + 2\rho AB + B^2) - \frac{1}{2\varepsilon}\int \frac{aMq}{D}S^2\,\mathrm{d}x \\ &+ \frac{1}{2\varepsilon}\int(\partial_x^2 a + \partial_x a)q\,\mathrm{d}x + \iint p\,\mathcal{R}, \\ \mathcal{R} = &-\frac{Ak}{\sqrt{v}} + B\left[\frac{d}{\sqrt{v}} + \frac{\eta\partial_y\ell}{2\sqrt{v}}(\log p + 1)\right] - \frac{\eta d\partial_y\ell}{v}(\log p + 1).\end{aligned} \tag{D.4}$$

Proof. Write $m = (\partial_x M)/D$. Differentiation under the integral gives

$$\int ph\,\mathrm{d}y = qS, \qquad \int vph\,\mathrm{d}y = q(\partial_x M + MS), \qquad \partial_x z_p = h/v + S/\varepsilon.$$

One integration by parts in the price flux gives

$$I_x = -\frac{1}{2}\iint \left(\frac{h}{v}+\frac{S}{\varepsilon}\right)\frac{avp}{D}(h-m) - \frac{1}{2}\iint \left(\frac{h}{v}+\frac{S}{\varepsilon}\right)\frac{(\partial_x a + a)vp}{D}.$$

At each fixed x, the first integral has the factor

$$\begin{aligned}\int p(h-m)h\,\mathrm{d}y + \frac{S}{\varepsilon}\int vp(h-m)\,\mathrm{d}y &= \int ph^2\,\mathrm{d}y - mqS + \frac{qS}{\varepsilon}(Dm + MS - Mm)\\ &= \int ph^2\,\mathrm{d}y + \frac{qM}{\varepsilon}S^2.\end{aligned}$$

Indeed, $\partial_x M = Dm$, so

$$\int vp(h-m)\ \mathrm{d}y = q(Dm + MS - Mm) = q(\varepsilon m + MS).$$

After multiplication by S/ε, its term qmS cancels the term $-mqS$ from the first integral. This is precisely where $D - M = \varepsilon$ is used. The lower-order term simplifies because

$$\frac{1}{D}\left(\int ph\,\mathrm{d}y + \frac{S}{\varepsilon}\int vp\,\mathrm{d}y\right) = \frac{qS}{\varepsilon} = \frac{\partial_x q}{\varepsilon}.$$

Its periodic integration by parts yields $\int(\partial_x^2 a + \partial_x a)q/(2\varepsilon)$.

For the factor flux, use

$$\partial_y z_p = \frac{g - (\partial_y \ell)(\log p + 1)}{v}, \qquad \tfrac{1}{2}\partial_y(\eta^2 p) - bp = p(\tfrac{1}{2}\eta^2 g - \eta d).$$

Multiplication after one integration by parts gives

$$I_y = -\tfrac{1}{2}\iint pB^2 + \iint pB\left[\frac{d}{\sqrt{v}} + \frac{\eta\partial_y\ell}{2\sqrt{v}}(\log p + 1)\right] - \iint p\frac{\eta d\partial_y\ell}{v}(\log p + 1).$$

For $C = \rho\eta\sqrt{av/D}$, integration in x first gives $I_{xy} = -\iint(\partial_x z_p)\partial_y(Cp)$. The contribution of S/ε vanishes upon integration in y. Moreover, $\partial_y(Cp) = \sqrt{av/D}\,p(\rho\eta g + k)$, so

$$I_{xy} = -\iint p\left(\rho AB + \frac{Ak}{\sqrt{v}}\right).$$

Adding the three identities proves the formula. No derivative of ρ in x and no second derivative of v has been used. □

Define a nonnegative weight

$$\Lambda = 1 + \frac{k^2 + d^2 + \eta^2|\partial_y\ell|^2}{v} + \frac{|\eta d\partial_y\ell|}{v}. \tag{D.5}$$

Since $A^2 + 2\rho AB + B^2 \geq \alpha(A^2 + B^2)$, Young's inequality in (D.4) gives

$$\langle \mathcal{F}_\varepsilon(p), z_p\rangle + \frac{\alpha}{4}\mathcal{I}_\varepsilon(p) \leq \frac{K_a}{2\varepsilon} + C_\alpha\iint p\Lambda(1 + |\log p|^2). \tag{D.6}$$

The nonnegative term involving S^2 has been discarded. This bound requires no separately prescribed scalar score, unlike the conditional entropy estimate of Section 4.3.

D.2 Controlling the logarithms

The following estimates explain why the source in (D.6) can be absorbed. The constants below are uniform on square tori of side length at least one whenever the displayed moment bounds are uniform. Integrals on such tori use centred representatives.

Lemma D.2 (Moment bounds imply entropy coercivity)**.** *Suppose $\int p = 1$, and that, for every finite j, the moments of $|x|^j$, $|y|^j$, $v^{\pm j}$, η^{-j} and Λ^j under p are bounded by constants independent of the torus. Suppose also $a \geq a_- > 0$ and $v^{\pm 1}(y) \leq C_e \exp(ey^2)$ uniformly in the approximating coefficients. Then, with constants that may depend on ε,*

$$\|\nabla p\|_1 \leq C\mathcal{I}_\varepsilon(p)^{1/2}, \qquad \int p^2 \leq C(1 + \mathcal{I}_\varepsilon(p)), \tag{D.7}$$

$$\int p|\log p|^j \leq C_j\{1 + [\log(2 + \mathcal{I}_\varepsilon(p))]^j\}, \tag{D.8}$$

$$-C_\varepsilon \leq \mathcal{H}_\varepsilon(p) \leq C_\varepsilon\{1 + \log(2 + \mathcal{I}_\varepsilon(p))\}. \tag{D.9}$$

For every $\xi > 0$,

$$\iint p\Lambda(1 + |\log p|^2) \leq \xi\mathcal{I}_\varepsilon(p) + C_\xi. \tag{D.10}$$

Proof. Cauchy–Schwarz gives

$$\int |\partial_x p| \leq \left(\int \frac{a|\partial_x p|^2}{Dp}\right)^{1/2} \left(\int \frac{Dp}{a}\right)^{1/2}, \qquad \int |\partial_y p| \leq \left(\int \frac{\eta^2|\partial_y p|^2}{vp}\right)^{1/2} \left(\int \frac{vp}{\eta^2}\right)^{1/2}.$$

The remaining factors are bounded by the assumed moments, since

$$\int Dp/a \leq a_-^{-1}(\int vp + \varepsilon), \qquad \int vp/\eta^2 \leq (\int v^2 p)^{1/2}(\int \eta^{-4}p)^{1/2}.$$

The two-dimensional inequality $\|p\|_2 \leq C(\|\nabla p\|_1 + L^{-1}\|p\|_1)$ on a square of period L has a constant independent of $L \geq 1$, by rescaling the unit torus. This proves (D.7).

Under the probability measure $p\,\mathrm{d}x\,\mathrm{d}y$, put $U = \log^+ p$ and $A_2 = \int p^2$. For $s \geq 0$, Markov's inequality gives $\mathbb{P}_p(U > s) \leq \min(1, A_2 e^{-s})$. Integrating $js^{j-1}\mathbb{P}_p(U > s)$ and splitting at $\log(1 + A_2)$ gives

$$\int p(\log^+ p)^j \leq C_j[1 + \log^j(1 + A_2)].$$

For the negative part set $W = 1 + x^2 + y^2$. On $\{e^{-W} \leq p < 1\}$, $\log^- p \leq W$. On $\{p < e^{-W}\}$, writing $p = e^{-u}$ gives $p(\log^- p)^j = e^{-u}u^j \leq C_j e^{-W/2}$. The integral of $e^{-W/2}$ on any centred domain is bounded by its integral on $\mathbb{R}^2$. The position moments bound $\int pW^j$. This proves (D.8).

The positive part of the weighted joint entropy satisfies

$$\int \frac{p}{v}\log^+ p \leq (\int pv^{-2})^{1/2}(\int p(\log^+ p)^2)^{1/2} \leq C[1 + \log(2 + \mathcal{I}_\varepsilon)].$$

Its negative part is bounded by the same argument with $\log^- p$. For the marginal, $\|\partial_x q\|_1 \leq \|\partial_x p\|_1$, whence $\|q\|_\infty \leq L^{-1} + \|\partial_x q\|_1 \leq C(1 + \sqrt{\mathcal{I}_\varepsilon})$. This bounds $\int q\log^+ q$. The preceding negative-log argument in one dimension, using the moments of x, bounds $\int q\log^- q$. These estimates prove (D.9).

Finally,

$$\int p\Lambda(1 + |\log p|^2) \leq (\int p\Lambda^2)^{1/2}(\int p(1 + |\log p|^2)^2)^{1/2} \leq C[1 + \log^2(2 + \mathcal{I}_\varepsilon)].$$

Since $\log^2(2+s) = o(s)$, the last expression is at most $\xi\mathcal{I}_\varepsilon + C_\xi$ for any $\xi > 0$, proving (D.10). □

On a fixed torus with smooth strictly positive coefficients, the lemma applies to every probability density: all coefficient and position moments are bounded by the corresponding suprema. Its constants may then depend on that torus. Combining (D.6) and (D.10) gives

$$\langle \mathcal{F}_\varepsilon(p), z_p \rangle + c\mathcal{I}_\varepsilon(p) \le C_\varepsilon, \qquad c = \alpha/8. \tag{D.11}$$

D.3 Construction on a fixed torus

The marginal is an unknown here, so the parametrisation in Section 5.1 must be replaced by the following entropy variables.

Lemma D.3 (Mass-constrained entropy variables)**.** *For every smooth function $z(x, y)$ on the fixed torus there is a unique positive probability density $p[z]$ satisfying*

$$\frac{\log p[z] + 1}{v} + \frac{\log q[z] + 1}{\varepsilon} = z - \lambda, \qquad q[z] = \int p[z] \, \mathrm{d}y, \tag{D.12}$$

for a scalar λ. It is smooth. On a fixed finite-dimensional space of smooth functions, its dependence on z is continuous (in fact smooth).

Proof. For a fixed x, z, λ, the marginal is the unique positive solution of

$$q = \int \exp\{v(y)(z(x, y) - \lambda) - 1 - v(y)/\varepsilon\} q^{-v(y)/\varepsilon} \, \mathrm{d}y.$$

The right side is positive, continuous and strictly decreasing in q, with limits $+\infty$ at zero and zero at infinity. The left side is increasing. To make the implicit differentiation explicit, write the right-hand side as $R(q, \lambda; x)$ and denote its positive integrand by $P(q, \lambda; x, y)$. Then

$$\partial_q(q - R) = 1 + \int \frac{v(y)}{\varepsilon q} P \, \mathrm{d}y > 0, \qquad \partial_\lambda(q - R) = \int v(y) P \, \mathrm{d}y > 0.$$

The implicit function theorem gives a smooth local solution, and uniqueness joins these local solutions into the positive solution $q(x; \lambda)$. Moreover,

$$\partial_\lambda q = -\frac{\int vP \, \mathrm{d}y}{1 + \int vP/(\varepsilon q) \, \mathrm{d}y} < 0.$$

Thus the solution is strictly decreasing in λ. As λ tends to $+\infty$ it tends uniformly to zero, and as λ tends to $-\infty$ it tends uniformly to infinity; these statements follow by comparison with any fixed level of q and the positive minimum of v. Consequently exactly one λ gives $\int q \, \mathrm{d}x = 1$. Its derivative with respect to λ is strictly negative, so the implicit function theorem also gives continuity and smooth dependence of λ. Recover p from the integrand in the displayed formula. It satisfies (D.12), and all the claims follow. □

Use the real Fourier space X_m of frequencies at most m, with zero total mean (not zero y-mean). Equip it with the H^4 inner product. At a time step $t_k = k\tau$, evaluate all smooth coefficients at t_k, use the inverse entropy map $p_{t_k}[z]$, and solve

$$\kappa(z^k, \phi)_{H^4} + \left\langle \frac{p_{t_k}[z^k] - p^{k-1}}{\tau} - \mathcal{F}^k_\varepsilon(p_{t_k}[z^k]), \phi \right\rangle = 0, \qquad \phi \in X_m. \tag{D.13}$$

Here p^0 is smooth, positive, and has mass one, and $0 < \kappa \le 1$.

Proposition D.4 (Periodic weak existence)**.** *Each step of* (D.13) *has a solution. As $m \to \infty$, $\tau \to 0$ and $\kappa \to 0$, a subsequence of the piecewise constant densities converges strongly in space-time L^1 to a periodic weak solution. It has a narrowly continuous curve of probability laws with initial density p^0. The solution can be selected to satisfy, in distributions in time,*

$$(\mathcal{H}_\varepsilon(p))' + \frac{\alpha}{4}\mathcal{I}_\varepsilon(p) \le \frac{K_a}{2\varepsilon} + C_\alpha \int p\Lambda(1 + |\log p|^2) + \int p \log p \, \partial_t(v^{-1}). \tag{D.14}$$

Proof. The Hessian of the entropy on positive densities has quadratic form

$$\delta^2\mathcal{H}_\varepsilon(p)[f,f] = \int \frac{f^2}{vp} + \frac{1}{\varepsilon}\int \frac{(\int f\,\mathrm{d}y)^2}{q} \geq 0.$$

Convexity and equal masses imply

$$\mathcal{H}_{\varepsilon t_k}(p_{t_k}[z]) - \mathcal{H}_{\varepsilon t_k}(p^{k-1}) \leq \langle p_{t_k}[z] - p^{k-1}, z\rangle,$$

because the scalar λ in (D.12) cancels. Likewise, $\langle \mathcal{F}^k_\varepsilon(p[z]), z\rangle = \langle \mathcal{F}^k_\varepsilon(p[z]), z_{p[z]}\rangle$, since $\mathcal{F}^k_\varepsilon$ has zero total mass. The continuous residual map has

$$(\Phi(z), z)_{H^4} \geq \kappa\|z\|^2_{H^4} + \frac{\mathcal{H}_{\varepsilon t_k}(p[z]) - \mathcal{H}_{\varepsilon t_k}(p^{k-1})}{\tau} + c\mathcal{I}_\varepsilon(p[z]) - C_\varepsilon.$$

The entropy has a finite lower bound over probability densities on the fixed torus (apply $s\log s \geq -1/e$ to the two integrals). Thus the right side is positive on a sufficiently large sphere. Brouwer's argument gives a zero. Here and below $p[z]$ in a single-step calculation means $p_{t_k}[z]$. Set $H_k = \mathcal{H}_{\varepsilon t_k}(p^k)$ and $I_k = \mathcal{I}_{\varepsilon t_k}(p^k)$. Testing with z^k gives

$$H_k - H_{k-1} + c\tau I_k + \tau\kappa\|z^k\|^2_{H^4} \leq C\tau + E_k, \quad E_k = \int p^{k-1}\log p^{k-1}[v(t_k)^{-1} - v(t_{k-1})^{-1}].$$

On this fixed smooth cylinder $\|\partial_t(v^{-1})\|_\infty < \infty$. Lemma D.2 and absorption of a logarithm give, for every $\xi > 0$,

$$|E_k| \leq C\tau\int p^{k-1}|\log p^{k-1}| \leq \tau(\xi I_{k-1} + C_\xi), \qquad I_0 < \infty.$$

Since $\sum_{k=1}^l \tau I_{k-1} = \tau I_0 + \sum_{k=1}^{l-1}\tau I_k$, choose $\xi < c/2$ and absorb the common terms. Summation yields

$$\sup_k \mathcal{H}_\varepsilon(p^k) + \sum_k \tau\mathcal{I}_\varepsilon(p^k) + \kappa\sum_k \tau\|z^k\|^2_{H^4} \leq C. \tag{D.15}$$

All constants in this paragraph concern the fixed torus and initial density. On this fixed torus, write $a_* = \min a > 0$, $\eta_* = \min\eta > 0$ and $v^* = \max v < \infty$. Since $M \leq v^*$, the dissipation satisfies

$$\mathcal{I}_\varepsilon(p) = 4\int \frac{a}{M+\varepsilon}|\partial_x\sqrt{p}|^2 + 4\int\frac{\eta^2}{v}|\partial_y\sqrt{p}|^2 \geq 4\min\left\{\frac{a_*}{v^*+\varepsilon}, \frac{\eta_*^2}{v^*}\right\}\int|\nabla\sqrt{p}|^2.$$

Thus the dissipation controls the full gradient of $\sqrt{p}$, with a constant that is allowed to depend on this torus. Hence

$$\sum_k \tau\|\sqrt{p^k}\|^2_{H^1} + \sum_k \tau(\|p^k\|^2_{W^{1,1}} + \|p^k\|_2^2) \leq C.$$

All fluxes have bounded L^1 norm on the fixed torus, since $v/(M+\varepsilon) \leq v_+/\varepsilon$.

We use the compactness argument of Section 5.3, including the projected time derivatives and the untested Fourier modes. The constant mode is controlled by mass conservation. For the affine interpolant $\widetilde{p}$ and the Fourier projection P_m,

$$\|\partial_t P_m\widetilde{p}\|_{L^2_t H^{-4}} \leq C, \qquad \|P_m(\widetilde{p} - \bar{p})\|_{L^2_t H^{-4}} \leq C\tau.$$

The penalisation is bounded here by $\kappa^2\sum\tau\|z^k\|^2_{H^4} \leq C\kappa$. For every probability measure on this two-dimensional torus, $\|(I-P_m)\mu\|_{H^{-4}} \leq C(1+m)^{-3}$, by summing the spectral tail. Thus time translations in $L^1_t H^{-4}$ are at most $C(h + \tau + (1+m)^{-3})$. The compact interpolation inequality

(5.18) between $W^{1,1}$, L^1 and H^{-4} gives small time translations in L^1; spatial translations follow from the gradient bound. The Fréchet–Kolmogorov criterion gives strong L^1 convergence and, after extraction, almost-everywhere convergence.

The marginals and the v-weighted marginals converge in L^1. On $\{q > 0\}$ their quotient converges almost everywhere. On $\{q = 0\}$ the limiting density vanishes almost everywhere in y, and the bound $vp/(M + \varepsilon) \le v_+ p/\varepsilon$ handles this set. Uniform integrability of the densities therefore gives strong convergence of the price flux; the mixed flux follows by taking products of square roots. The other fluxes converge directly. Summation by parts in time in (D.13) now gives the weak equation. The penalty tends to zero when tested against a smooth function, by $\sqrt{\kappa}(\kappa \sum \tau \|z^k\|_{H^4}^2)^{1/2}$. The projected time estimates and compactness of the torus give a narrowly continuous curve and its initial trace, as in Section 5.3.

To retain (D.14), apply (D.6) at each step and multiply the discrete inequality by a nonnegative smooth time test. Write p_j for a convergent sequence of the discrete densities on the fixed space-time cylinder Q, and put $\varphi(s) = s|\log s|^2$, with $\varphi(0) = 0$. The $L^2(Q)$ bound implies uniform integrability of $\varphi(p_j)$: for every measurable $E \subset Q$ and every $R > 1$,

$$\int_E \varphi(p_j) \le \max_{0 \le s \le R} \varphi(s)\, |E| + \sup_{s > R} \frac{|\log s|^2}{s} \int_Q p_j^2.$$

First choose R large and then $|E|$ small. Almost-everywhere convergence and Vitali's theorem therefore give $p_j |\log p_j|^2 \to p|\log p|^2$ in $L^1(Q)$. The same argument applies to $p_j \log p_j$. Moreover, $\int q_j^2 \le L_y \int p_j^2$, and the strong convergence of the marginals gives, after extraction, $q_j \log q_j \to q \log q$ in L^1. On this fixed smooth cylinder the weights are bounded and their time-step approximations converge uniformly. Consequently the entropy and the spatial source converge in L^1 in time. For the temporal term let p_j^- be the preceding-step interpolant. The time-translation estimate implies $p_j^- \to p$ in L^1; the first cell contributes at most $2\tau_j$. Also p_j^- is bounded in $L^2(Q)$, since its squared norm differs from the current-step bound by at most $\tau_j \|p^0\|_2^2$. Vitali's theorem therefore gives $p_j^- \log p_j^- \to p \log p$ in $L^1(Q)$. The smooth-coefficient difference quotients

$$\frac{v(t_k, y)^{-1} - v(t_{k-1}, y)^{-1}}{\tau_j} \longrightarrow \partial_t (v^{-1})(t, y)$$

converge uniformly. Hence the sums of E_k against a smooth time test converge to the temporal source in (D.14).

For the dissipation put

$$c_{x,j} = \frac{(M_j + \varepsilon) p_j}{a_j}, \qquad c_{y,j} = \frac{v p_j}{\eta_j^2}.$$

Both coefficients are bounded by a fixed multiple of p_j. On $\{q > 0\}$, the convergence of the two marginals gives $M_j \to M$ almost everywhere after extraction. On $\{q = 0\}$, we have $p = 0$ for almost every y, so both coefficients tend to zero by these bounds, whether or not M_j converges there. Strong L^1 convergence of p_j gives uniform integrability; hence

$$c_{x,j} \longrightarrow \frac{(M + \varepsilon) p}{a}, \qquad c_{y,j} \longrightarrow \frac{vp}{\eta^2} \quad \text{in } L^1(Q),$$

where $M = 1$ may be chosen on the exceptional set. For $i = x, y$, the perspective identity is

$$\int \frac{|\partial_i p|^2}{c_i} = \sup_{\phi \in C^\infty_{\mathrm{per}}} \left\{ -2 \int p \partial_i \phi - \int c_i \phi^2 \right\}. \tag{D.16}$$

For space-time integrals, the tests are also smooth and compactly supported in time. The value is zero when numerator and denominator vanish, and infinite when the denominator vanishes

but the numerator does not. Completing the square and approximating in $L^2(c_i\,\mathrm{d}t\,\mathrm{d}x\,\mathrm{d}y)$ proves the identity; if the distributional derivative has no such representation, the supremum is infinite. Each expression on the right is continuous under the strong convergence just proved, so the supremum is lower semicontinuous. The same argument allows a nonnegative smooth time weight in both integrals. Summation by parts in the discrete entropy inequality now proves (D.14). □

D.4 Uniform bounds on expanding tori

Use Proposition 3.4 and the construction in Section 3.4. In particular, take $R_n = 2n + 2$ and $v_n = \exp((\chi(\cdot/n)\ell) * \vartheta_{1/n})$ with a product mollifier in time and factor space. Extend ℓ by its constant temporal traces. The variance remains independent of x. The other coefficients are smoothed and made periodic as there. The resulting coefficients converge locally almost everywhere, satisfy common $\mathscr{G}_0$ bounds for $v_n^{\pm1}, \eta_n^{-1}, b_n, \partial_y\eta_n, \partial_y\rho_n, \partial_y \log v_n$, and obey

$$a_- \le a_n \le a_+, \qquad |\partial_x a_n| + |\partial_x^2 a_n| \le C_a,$$
$$0 < \eta_n \le \eta_+, \qquad |\rho_n| \le \rho_*, \qquad \operatorname{sgn}(y) b_n \le C(1 + |y|).$$

Their strictly positive minima may depend on n. The proof uses only their common growth bounds, not these minima, in the external limit. Apply Proposition D.4 with the joint initial densities of Section 3.3; write p_n for the chosen solutions.

D.4.1 Moments independent of the approximation

The y-marginal solves the factor equation on the circle, independently of the price marginal. Use U_R, C_* and λ from (6.3) and (6.1). Their generator bounds depend only on the factor coefficients, which are unchanged here; hence the test $e^{\lambda(t)U_R}$ and initial bound (3.5) give

$$\sup_{n,t\le T} \int \exp(\kappa_T y^2)\,\mathrm{d}\mu_t^n \le C_T, \qquad \kappa_T = c_*\lambda(T) > 0. \tag{D.17}$$

The x-marginal is not the fixed target. Instead, it solves

$$\partial_t q_n = \tfrac{1}{2}(\partial_x^2 + \partial_x)(a_n\theta_n q_n), \qquad \theta_n = M_n/(M_n + \varepsilon) \in [0, 1].$$

For an integer $j \ge 2$, derivatives of $U_R^{j/2}$ satisfy $|(U_R^{j/2})'| + |(U_R^{j/2})''| \le C_j U_R^{j/2}$ uniformly in $R \ge 1$. Testing the marginal equation with this periodic function and using $0 \le a_n\theta_n \le a_+$ gives Gronwall's bound

$$\sup_{n,t\le T} \int |x|^j\,\mathrm{d}\mu_t^n \le C_{j,T}. \tag{D.18}$$

The initial bounds follow from (2.8): convolution with the Gaussian and centred periodisation preserve uniform moments of each fixed order. The weak equation permits these smooth time-independent tests at every time. No regularity of θ_n is needed.

Every fixed power of each coefficient in the common $\mathscr{G}_0$ bounds is integrable uniformly by choosing its growth exponent smaller than κ_T divided by that power. Products are handled by Hölder. In particular, the weight Λ_n in (D.5) has moments of every finite order uniformly in n, t. All the assumptions of Lemma D.2 are therefore uniform for this family.

D.4.2 Temporal source and entropy entry

The Gaussian moment and the common coefficient envelopes also bound every moment of G_n/v_n. The identity $\partial_t(v^{-1}) = -v^{-1}\partial_t\ell$ gives

$$\left|\int p \log p\,\partial_t(v^{-1})\right| \le h(t)\int p\frac{G}{v}|\log p| \le h(t)\left(\int p(G/v)^2\right)^{1/2}\left(\int p|\log p|^2\right)^{1/2}.$$

The Gaussian moment bounds the first factor uniformly, and Lemma D.2 controls the second. The additional entropy source is therefore bounded, for almost every t, by

$$\left| \int p \log p \, \partial_t(v^{-1}) \right| \le Ch(t)[1 + \log(2 + \mathcal{I}_{\varepsilon t}(p))].$$

For each $\xi > 0$, elementary maximisation in $I \ge 0$ gives

$$Ch[1 + \log(2 + I)] \le \xi I + C_\xi[1 + h \log(1 + h)].$$

Indeed for $I \ge 0$ the derivative of $Ch \log(2 + I) - \xi I$ is $Ch/(2 + I) - \xi$, so its maximum is attained either at zero or at $I = Ch/\xi - 2$; substitution yields the stated bound. Thus L^1 control of h alone is not what this absorption argument proves: the sufficient temporal budget here is $L \log L$.

The product convolution described above satisfies

$$|\partial_t \ell_n(t, y)| \le h_n(t) G_n(y), \qquad h_n = h * \vartheta_n,$$

where h is extended by zero and G_n has common $\mathscr{G}_0$ bounds. For $\Phi(u) = u \log(1 + u)$, convexity and positivity of the kernel give

$$\int_{\mathbb{R}} \Phi(h_n(t)) \, \mathrm{d}t \le \int_0^T \Phi(h(t)) \, \mathrm{d}t.$$

After constructing the fixed-torus solutions, combine (D.14) with Lemma D.2 and the preceding absorption to obtain

$$\begin{aligned} H_n' + c\mathcal{I}_{\varepsilon n} &\le g_n(t), \qquad g_n = C_\varepsilon[1 + \Phi(h_n)], \\ -C_\varepsilon \le H_n &\le C_\varepsilon[1 + \log(2 + \mathcal{I}_{\varepsilon n})], \qquad \sup_n \int_0^T g_n < \infty. \end{aligned} \tag{D.19}$$

Here $H_n = \mathcal{H}_{\varepsilon,n,t}(p_n)$ and all constants are independent of n; dependence on ε is allowed. Inverting the upper bound gives $H_n' + c_1 e^{H_n/c_2} \le g_n + C_1$. The following comparison accounts for the singular part of the temporal derivative.

Lemma D.5 (BV entropy comparison)**.** *Let $H, e^{H/B} \in L^1_{\mathrm{loc}}(0, T)$, with $A, B > 0$ and $C \ge 0$, and suppose that*

$$DH + Ae^{H/B} \, \mathrm{d}t \le C \, \mathrm{d}t$$

in distributions on $(0, T)$. Then H has a right-continuous BV_{loc} representative satisfying

$$H(t) \le Ct + B \log \frac{B}{At}, \qquad 0 < t < T.$$

Proof. Set $F(t) = H(t) - Ct$. The distribution DF is a nonpositive measure, so F has a right-continuous nonincreasing representative and belongs to $BV_{\mathrm{loc}}(0, T)$. Choose this representative and set $H = F + Ct$. Its absolutely continuous derivative satisfies $F_a' \le -Ae^{H/B}$ almost everywhere, and the singular part of DF is nonpositive. The function $U = e^{-F/B}$ is nondecreasing and belongs to BV_{loc}. The BV chain rule gives

$$U_a' = -B^{-1} e^{-F/B} F_a' \ge (A/B) e^{Ct/B} \ge A/B.$$

The continuous singular part is $-B^{-1} e^{-F/B} D^c F$, a nonnegative measure, and every jump of U is nonnegative because every jump of F is downward. Thus $DU \ge (A/B) \, \mathrm{d}t$, including its singular part. For $0 < s < t < T$, right continuity gives

$$U(t) - U(s) \ge A(t - s)/B.$$

Since $U(s) > 0$, it follows that $H(t) \le Ct + B \log(B/[A(t - s)])$. Let $s \downarrow 0$. No value of the initial entropy is used. □

To use this argument with a time-dependent source, fix $0 < s < t < T$ and set $F_n(u) = H_n(u) - \int_s^u (g_n + C_1)$. Its derivative has nonpositive singular part, and $F'_{n,a} \le -c_1 e^{F_n/c_2}$ because $g_n + C_1 \ge 0$. The same BV chain rule gives

$$H_n(t) \le \int_s^t (g_n + C_1)\, \mathrm{d}u - c_2 \log(c_1(t-s)/c_2).$$

Taking $s = t/2$, using the common L^1 bound for g_n, and integrating (D.19) from a Lebesgue time in $(\delta/2, \delta)$ gives

$$H_n(t) \le C_{\varepsilon,T}(1 + |\log t|), \qquad \int_\delta^T \mathcal{I}_{\varepsilon n}(t)\, \mathrm{d}t \le C_{\varepsilon,\delta,T}. \tag{D.20}$$

The lower entropy bound controls the terminal value. No initial entropy bound enters these estimates.

D.4.3 Roots and flux masses

Let $1 < r < 2$ and $\beta = r/(2-r) > 1$. Jensen's inequality on each conditional law gives $\int p_n M_n^\beta \le \int p_n v_n^\beta$. Thus

$$\sup_{n,t} \int p_n (D_n/a_n)^\beta < \infty, \qquad \sup_{n,t} \int p_n (v_n/\eta_n^2)^\beta < \infty.$$

The weighted Hölder inequality and the Sobolev chain rule give

$$\int |\partial_x p_n^{1/r}|^r \le r^{-r} \left(\int \frac{a_n |\partial_x p_n|^2}{D_n p_n} \right)^{r/2} \left(\int p_n (D_n/a_n)^\beta \right)^{1-r/2},$$
$$\int |\partial_y p_n^{1/r}|^r \le r^{-r} \left(\int \frac{\eta_n^2 |\partial_y p_n|^2}{v_n p_n} \right)^{r/2} \left(\int p_n (v_n/\eta_n^2)^\beta \right)^{1-r/2}.$$

For weak densities these identities follow by applying the chain rule to $(p+s)^{1/r} - s^{1/r}$ and letting $s \downarrow 0$; derivatives vanish on the zero set. This gives uniform $L^r_t W^{1,r}$ root bounds and $L^r_t W^{1,1}$ bounds for p_n on (δ, T).

The four flux masses are bounded on the entire interval, including near zero:

$$\int a_n v_n p_n / D_n = \int a_n \theta_n q_n \le a_+, \qquad \int \eta_n^2 p_n \le \eta_+^2,$$
$$\int |b_n| p_n \le C_T, \qquad \int |\rho_n| \eta_n \sqrt{a_n v_n / D_n}\, p_n \le \rho_* \eta_+ \sqrt{a_+}.$$

The last estimate is Cauchy–Schwarz and $\int (v_n/D_n) p_n \le 1$.

D.5 External limit and stochastic reconstruction

The root bounds give local compactness in space. The weak equation and the flux mass bounds give $\partial_t(\chi p_n)$ bounded in $L^\infty_t H^{-4}$ for each smooth compactly supported cutoff χ. Simon's compactness criterion [14], or the space-time translation argument above, gives strong local L^1 convergence on (δ, T). Bounds (D.17)–(D.18) control the tails. Since every density has mass one, the contribution of $(0, \delta)$ to the L^1 distance is at most 2δ. After a diagonal extraction, zero extensions from the centred domains satisfy

$$p_n \longrightarrow p \quad \text{in } L^1((0,T) \times \mathbb{R}^2) \quad \text{and almost everywhere.} \tag{D.21}$$

No derivative is taken across the boundary of a zero extension.

By the weighted-product argument of Lemma 7.1, uniform moments of v_n of order greater than one imply $v_n p_n \to vp$ in L^1. The same argument applies to $b_n p_n$. Integrating in y gives strong convergence of q_n and $A_n = \int v_n p_n \,\mathrm{d}y$ to q and $A = \int vp \,\mathrm{d}y$.

On $\{q > 0\}$, positivity of v implies $0 < A < \infty$ almost everywhere, so $M = A/q$ is positive and finite there. Complete it by one on the remaining set. After extraction $M_n \to M$ almost everywhere on $\{q > 0\}$. Define $F_n = v_n p_n/(M_n + \varepsilon)$ and $F = vp/(M + \varepsilon)$. Then $F_n \to F$ almost everywhere: on $\{q = 0\}$, $p = 0$ almost everywhere in y, and $F_n \le v_n p_n/\varepsilon \to 0$ there. The same domination and the strong convergence of $v_n p_n$ give uniform integrability and tightness of F_n; equivalently dominated convergence with varying L^1 majorants gives

$$F_n \to F \quad \text{in } L^1((0,T) \times \mathbb{R}^2). \tag{D.22}$$

Here division by the fixed ε is legitimate; the estimate is not uniform as it tends to zero. The products $\sqrt{F_n}\sqrt{p_n}$ converge in L^1, because their factors converge in L^2. Bounded, almost-everywhere convergent multipliers $a_n, \eta_n^2, \rho_n \eta_n \sqrt{a_n}$ therefore identify all four fluxes.

The moment bounds and the weak equations give tightness and equicontinuity against smooth compactly supported spatial tests, uniformly up to time zero. Extraction on a countable family of tests yields a narrowly continuous probability curve μ_t with initial value μ_0. Strong convergence identifies $\mathrm{d}t\,\mu_t = p \,\mathrm{d}t \,\mathrm{d}x \,\mathrm{d}y$. Passing to the limit in compactly supported space-time tests gives the equation with operator (D.1) and this initial trace. Integration in y gives (13.26); cutoffs in y are removed using the integrable fluxes. The limiting root bounds follow as in Section 7.4, by strong convergence of the roots, weak compactness of their gradients on each compact set, and spatial exhaustion.

We apply the reconstruction argument of Section 8.1 directly to this probability curve; Proposition 8.1, whose statement prescribes the calibrated marginal, is not needed. Set $w = v/(M + \varepsilon)$. The drift and covariance are

$$\begin{pmatrix} -aw/2 \\ b \end{pmatrix}, \qquad \begin{pmatrix} aw & \rho\eta\sqrt{aw} \\ \rho\eta\sqrt{aw} & \eta^2 \end{pmatrix}.$$

The covariance is positive semidefinite, and its entries and the drift are integrable against $\mathrm{d}t\,\mu_t$ by the flux bounds. Trevisan's superposition theorem [15, Theorem 2.5] therefore gives a continuous martingale problem with these marginals. Dividing the compensated coordinate martingales by $\sqrt{aw}$ and η, respectively, gives two continuous local martingales of bracket t, with cross bracket $\int \rho \,\mathrm{d}t$. Lévy's characterisation gives W, B. The denominators are positive and finite off an occupation-null set, and their stochastic integrals have bracket t, so no uniform lower bound is required. Disintegration identifies M as the conditional moment.

To check the price martingale, put $\theta = M/(M + \varepsilon)$, with arbitrary values on the occupation-null exceptional set. Then $0 \le a\theta \le a_+$. Apply (13.22), with its initial value replaced by $\int e^{rx}\,\mu_0^X(\,\mathrm{d}x)$:

$$\int e^{rx}\,\mathrm{d}\mu_t^X \le e^{a_+ r(r-1)t/2} \int e^{rx}\,\mu_0^X(\,\mathrm{d}x), \qquad r > 1.$$

The proof there uses only the marginal equation and $0 \le a\theta \le a_+$, both available here. Now test with $e^x \chi(x/R)$. Since

$$(\partial_x^2 - \partial_x)(e^x \chi(x/R)) = e^x \{R^{-2}\chi''(x/R) + R^{-1}\chi'(x/R)\},$$

the generator integral tends to zero by the uniform exponential moment and $a\theta \le a_+$. Thus $\mathbb{E}e^{X_t} = \int e^x\,\mu_0^X(\,\mathrm{d}x)$ at every time. Itô's formula makes e^X a positive local martingale; constant expectation makes it a true martingale. This proves Theorem 13.5 under its temporal and initial-law assumptions.

The temporal entropy source was passed to the limit on each fixed smooth torus. Its external limit is not needed: the uniform energy bounds suffice for flux compactness and reconstruction.

The finite systems (D.13) prescribe total mass and use the entropy variable (D.12). Their convergence is subsequential for fixed ε and exact spatial integrals; no convergence rate or convergence guarantee for an implemented solver follows.

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